Syllabus 2026

GEOGRAPHY (50)
CLASS 10
H.C.G.-Paper-2

*There will be **one** paper of **two** hours duration carrying 80 marks and Internal Assessment of 20 marks.*

Candidates will be expected to make the fullest use of sketches, diagrams, graphs and charts in their answers.

Questions set may require answers involving the interpretation of photographs of geographical interest.

PART – I
MAP WORK

1. Interpretation of Topographical Maps

(a) *Locating features with the help of a four figure or a six-figure grid reference.*

(b) *Definition of contour and contour interval. Identification of landforms marked by contours (steep slope, gentle slope, hill, valley, ridge / water divide, escarpment), triangulated height, spot height, bench mark, relative height/ depth.*

(c) *Interpretation of colour tints and conventional symbols used on a topographical survey of India map.*

(d) *Identification and definition of types of scale given on the map.*

Measuring distances and calculating area using the scale given therein.

(e) *Marking directions between different locations, using eight cardinal points.*

(f) *Identify: Site of prominent villages and/or towns, types of land use / land cover and means of communication with the help of the index given at the bottom of the sheet.*

(g) *Identification of drainage (direction of flow and pattern) and settlement patterns.*

(h) *Identification of natural and man-made features.*

2. Map of India

On an outline map of India, candidates will be required to locate, mark and name the following:

***Mountains, Peaks and Plateaus:** Himalayas, Karakoram, Aravali, Vindhyas, Satpura, Western and Eastern Ghats, Nilgiris, Garo, Khasi, Jaintia, Mount Godwin Austin (K2), Mount Kanchenjunga. Deccan Plateau, Chota Nagpur Plateau.*

***Plains:** Gangetic Plains and Coastal plains – (Konkan, Kanara, Malabar, Coromandel, Northern Circars).*

***Desert:** Thar (The Great Indian Desert)*

***Rivers:** Indus, Ravi, Beas, Chenab, Jhelum, Satluj, Ganga, Yamuna, Ghaghra, Gomti, Gandak, Kosi, Chambal, Betwa, Son, Damodar, Brahmaputra, Narmada, Tapti, Mahanadi, Godavari, Krishna, Cauveri, Tungabhadra.*

***Water Bodies:** Gulf of Kutch, Gulf of Khambhat, Gulf of Mannar, Palk Strait, Andaman Sea, Chilka Lake, Wular Lake.*

***Passes:** Karakoram, Nathu-La Passes.*

Latitude and Longitudes: Tropic of Cancer, Standard Meridian (82° 30′E).

***Direction of Winds:** South West Monsoons (Arabian Sea and Bay of Bengal Branches), North East Monsoons and Western Disturbances.*

***Distribution of Minerals:** Oil - Mumbai High (Offshore Oil Field) and Digboi. Iron – Singhbhum, Coal – Jharia.*

Soil Distribution – Alluvial, Laterite, Black and Red Soil.

***Cities** - Delhi, Mumbai, Kolkata, Chennai, Hyderabad, Bengaluru, Kochi, Chandigarh, Srinagar, Vishakhapatnam, Allahabad.*

***Population** - Distribution of Population (Dense and sparse).*

PART - II
GEOGRAPHY OF INDIA

3. Location, Extent and Physical features

- *Position and Extent of India. (through Map only).*
- *The physical features of India – mountains, plateaus, plains and rivers. (through Map only).*

4. Climate

Distribution of Temperature, Rainfall, winds in Summer and Winter and factors affecting the climate of the area. Monsoon and its mechanism. Seasons: March to May – Summer; June to September – Monsoon; October to November - Retreating Monsoon. December to February – Winter.

5. Soil Resources

- *Types of soil (alluvial, black, red and laterite) distribution, composition and characteristics such as colour, texture, minerals and crops associated.*
- *Soil Erosion – causes, prevention and conservation.*

6. Natural Vegetation

- *Importance of forests.*
- *Types of vegetation (tropical evergreen, tropical deciduous, tropical desert, littoral and mountain), distribution and correlation with their environment.*
- *Forest conservation.*

Syllabus

7. **Water Resources**
   - *Sources (Surface water and ground water).*
   - *Need for conservation and conservation practices (Rain water harvesting and its importance).*
   - *Irrigation: Importance and methods.*
8. **Mineral and Energy Resources**
   - *Iron ore, Manganese, Copper – uses and their distribution.*
   - *Conventional Sources: Coal, Petroleum, Natural gas (distribution, advantages and disadvantages).*

     Hydel power (Bhakra Nangal Dam and Hirakud).
   - *Non-conventional Sources: Solar, wind, tidal, geo-thermal, nuclear and bio-gas (generation and advantages).*
9. **Agriculture**
   - *Indian Agriculture – importance, problems and reforms.*
   - *Types of farming in India: subsistence and commercial: shifting, intensive, extensive, plantation and mixed.*
   - *Agricultural seasons (rabi, kharif, zayad).*
   - *Climatic conditions, soil requirements, methods of cultivation, processing and distribution of the following crops:*
     - *rice, wheat, millets and pulses.*
     - *sugarcane, oilseeds (groundnut mustard and soya bean).*
     - *cotton, jute, tea and coffee.*
10. **Manufacturing Industries**

    Importance and classification
    - *Agro based Industry - Sugar, Textile (Cotton and Silk).*
    - *Mineral based Industry – Iron & Steel (TISCO, Bhilai, Rourkela, Vishakhapatnam) Petro Chemical and Electronics.*
11. **Transport**

    Importance and Modes – Roadways, Railways, Airways and Waterways – Advantages and disadvantages.
12. **Waste Management**
    - *Impact of waste accumulation - spoilage of landscape, pollution, health hazards, effect on terrestrial, aquatic (fresh water and marine) life.*
    - *Need for waste management.*
    - *Methods of safe disposal - segregation, dumping and composting.*
    - *Need and methods for reducing, reusing and recycling waste.*

INTERNAL ASSESSMENT
PRACTICAL / PROJECT WORK

Candidates will be required to prepare a project report on any **one** topic. The topics for assignments may be selected from the list of suggested assignments given below. Candidates can also take up an assignment of their choice under any of the broad areas given below.

Suggested list of assignments:

1. Local Geography:
   (a) Land use pattern in different regions of India– a comparative analysis.
   (b) The survey of a local market on the types of shops and services offered.
2. Environment:
   Wildlife conservation efforts in India.
3. Current Geographical Issues:
   Development of tourism in India.
4. Transport in India:
   Development of Road, Rail, Water and Air routes.
5. List different type of industries in the States and collect information about the types of raw materials used, modes of their procurement and disposal of wastes generated. Classify these industries as polluting or environment friendly and suggest possible ways of reducing pollution caused by these units.
6. Need for industrialization in India, the latest trends and its impact on economy of India.
7. Visit a water treatment plant, sewage treatment plant or garbage dumping or vermicomposting sites in the locality and study their working.

EVALUATION

The assignments/project work is to be evaluated by the subject teacher and by an External Examiner. (The External Examiner may be a teacher nominated by the Head of the school, who could be from the faculty, **but not teaching the subject in the section/class.** For example, a teacher of Geography of Class VIII may be deputed to be an External Examiner for Class X, Geography projects.)

The Internal Examiner and the External Examiner will assess the assignments independently.

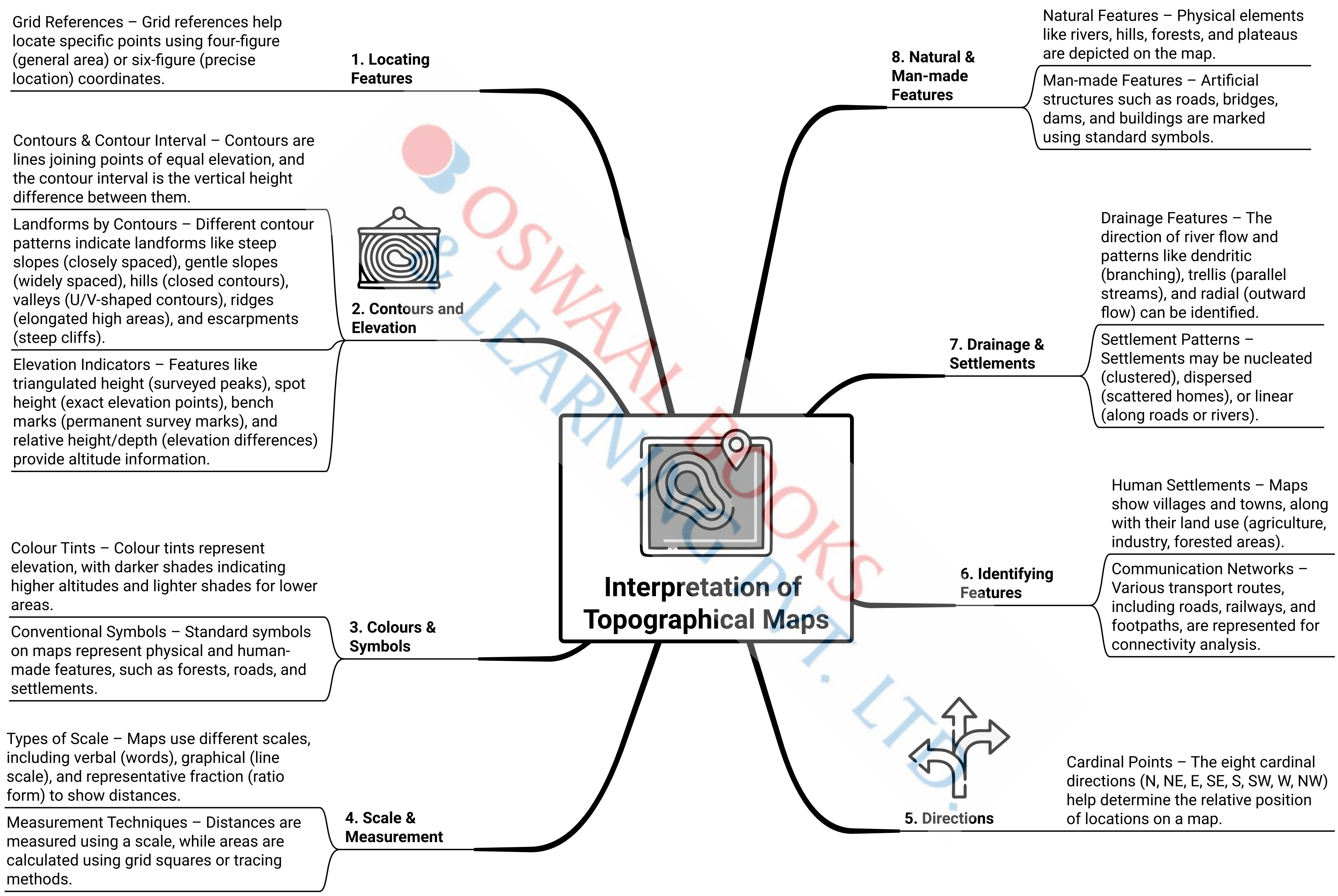
Award of Marks	**(20 Marks)**
Subject Teacher (Internal Examiner)	10 marks
External Examiner	10 marks

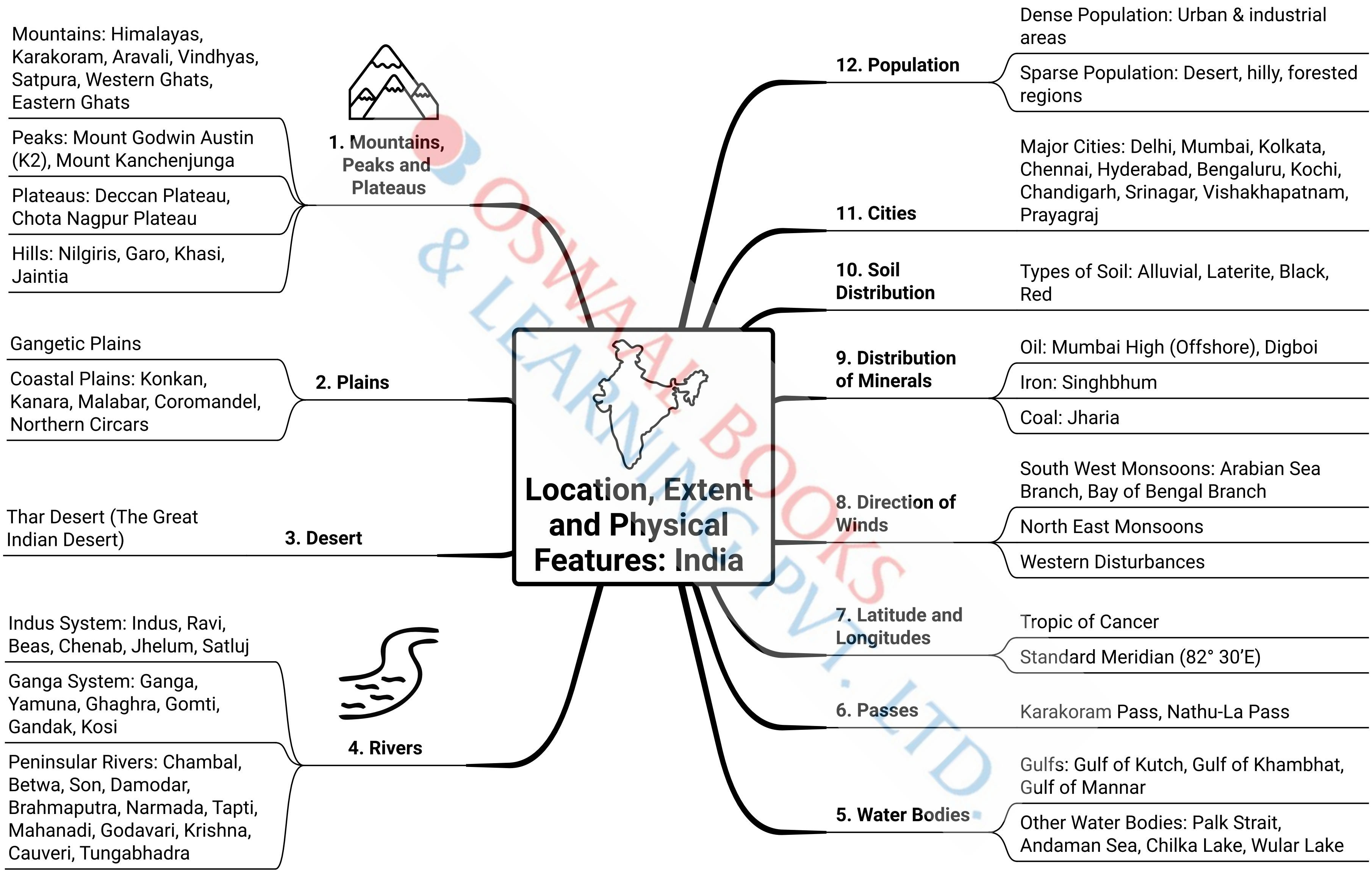
The total marks obtained out of 20 are to be sent to CISCE by the Head of the school.

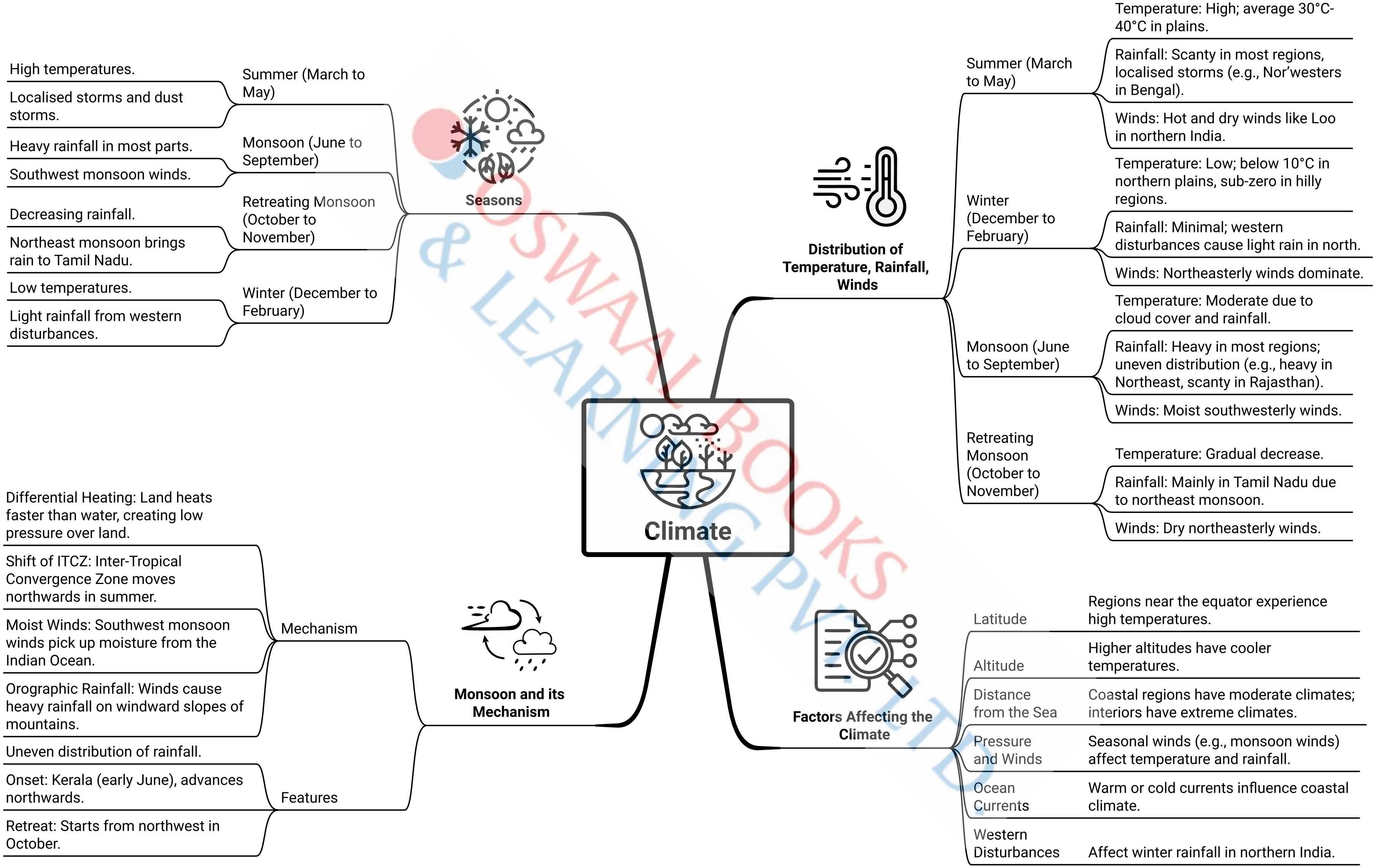
The Head of the school will be responsible for the online entry of marks on CISCE's CAREERS portal by the due date.

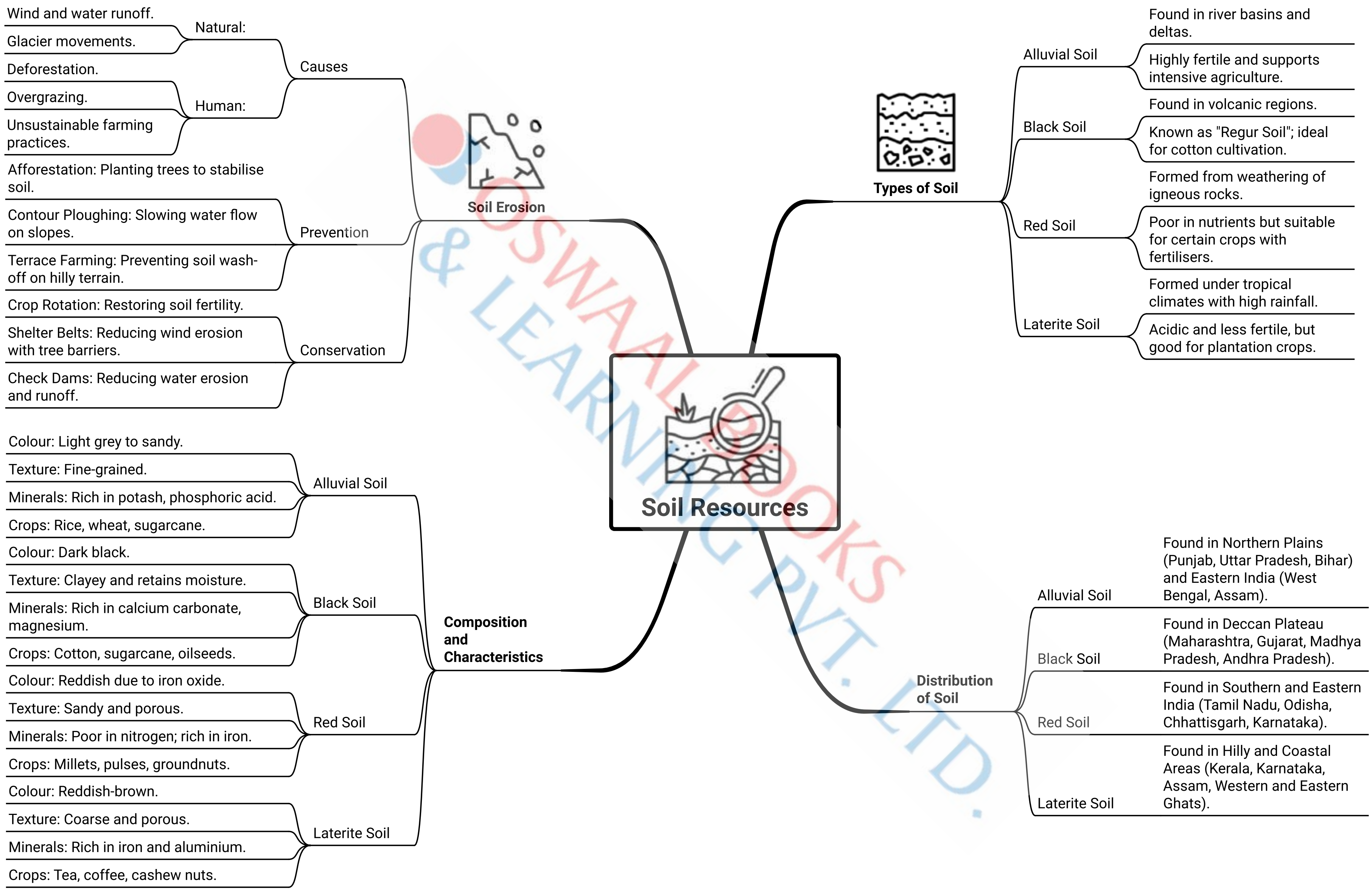
INTERNAL ASSESSMENT IN GEOGRAPHY - GUIDELINES FOR MARKING WITH GRADES

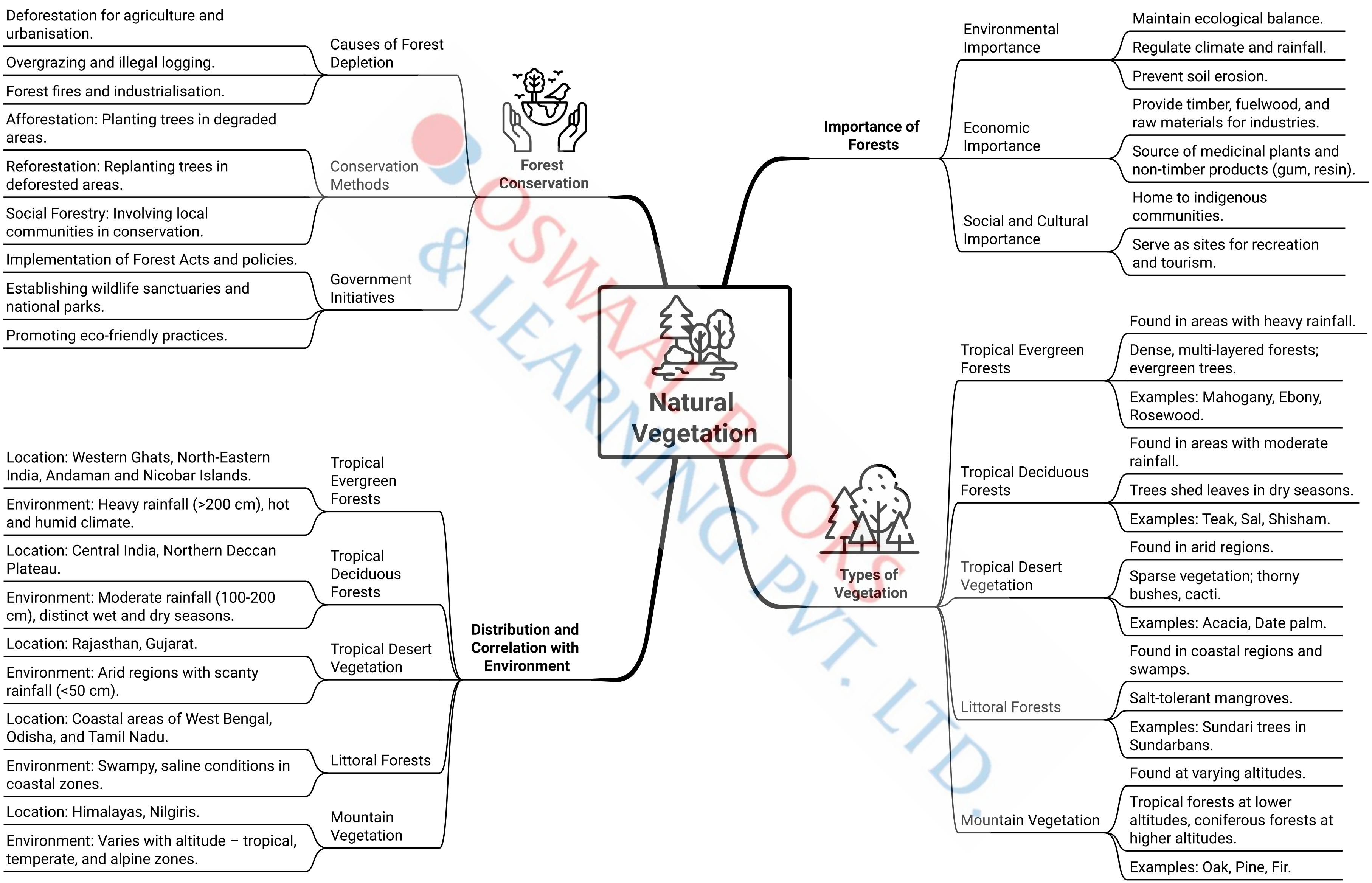
Criteria	Preparation	Procedure/ Testing	Observation	Inference/Results	Presentation
Grade I (4 marks)	Gives complete theoretical information using relevant geographical terms	States the objectives and defines the aspects to be studied.	Studies text and source material and makes a list.	States theoretical information in a coherent and concise manner using geographical terminology. Uses a variety of techniques. Shows resourcefulness. Supports investigation with relevant evidence.	Neatly and correctly stated statement of intent and conclusion matches with objectives.
Grade II (3 marks)	Provides adequate information using appropriate terms.	States objectives but not the limitations of the study.	Makes a limited list of source material only from secondary sources.	Uses sound methodology-using methods suggested. Makes a valid statement about the data collected. Attempts to develop explanations using available information.	Limited use of reference material and a presentation, which is routine.
Grade III (2 marks)	States objectives using some geographical terms but mostly in descriptive terms.	Only lists the aspects to be studied.	References are minimal.	Uses methodology in which selective techniques are applied correctly. Makes descriptive statement. Analysis is limited. Relates and describes systematically the data collected. Tries to relate conclusion to original aim.	Simple and neat with correct placement of references, acknowledgements, contents, maps and diagrams.
Grade IV (1 mark)	States intent without using relevant geographical terms but explaining them correctly.	Shows evidence of what to look for and how to record the same.	Uses methodology with some techniques but is unable to systematically record data and collect information.	Makes few relevant statements. Does analyse data that is not presented or tends to copy analysis available from other sources. Makes superficial conclusions. Link between the original aim and conclusion is not clear.	Neat but lacking in correct placement of table of contents, maps, diagrams and pictures.
Grade V (0 marks)	Does not make any use of geographical terms.	Has not collected any relevant data and has not presented sources correctly.	Does not use any logical technique and does not follow the methodology suggested.	Does not analyse data. Does not use the suggested methods. Makes conclusions but does not relate them to the original aim.	Presents the report without reference.

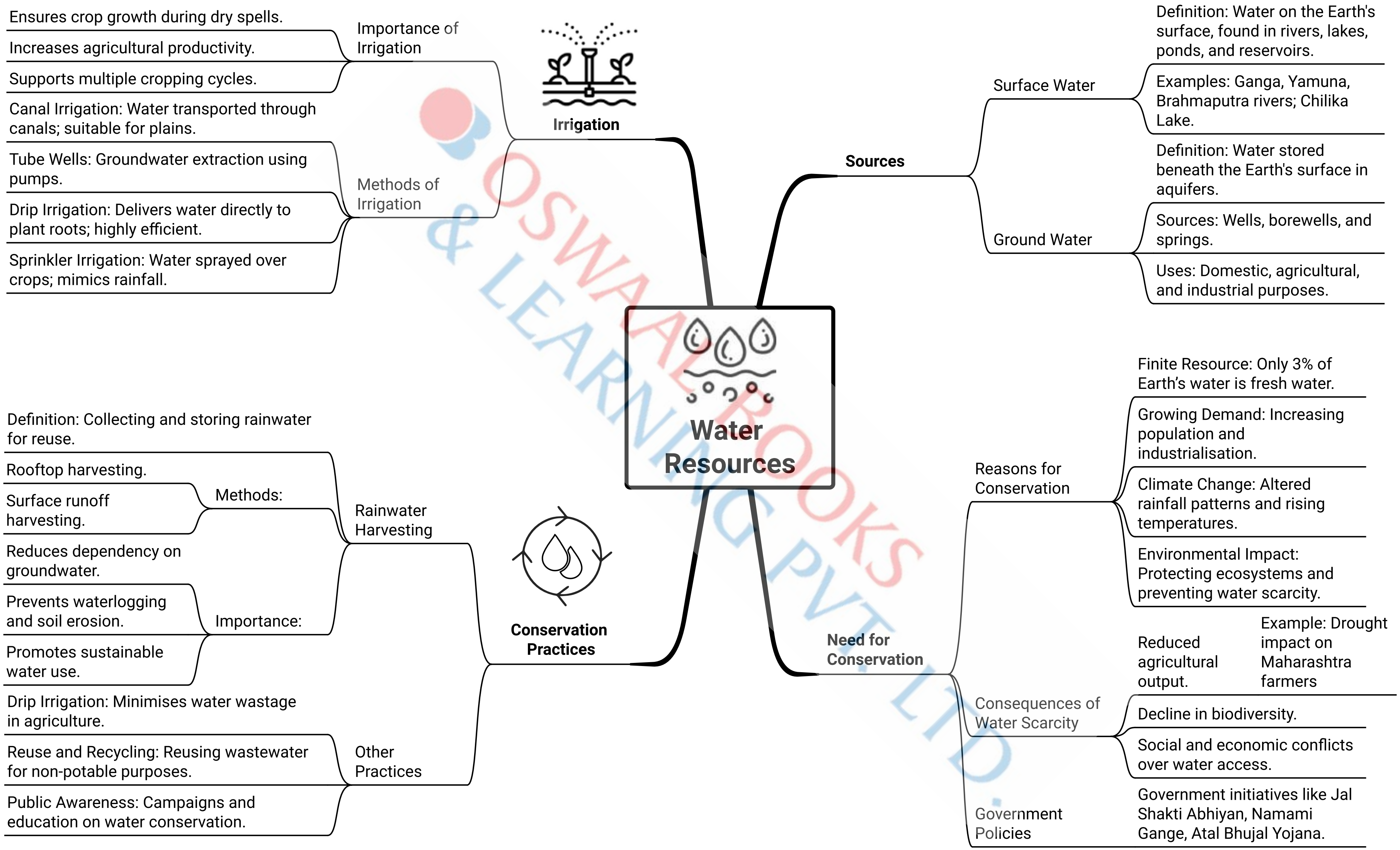

Interpretation of Topographical Maps
1. Locating Features
Grid References – Grid references help locate specific points using four-figure (general area) or six-figure (precise location) coordinates.
2. Contours and Elevation
Contours & Contour Interval – Contours are lines joining points of equal elevation, and the contour interval is the vertical height difference between them.
Landforms by Contours – Different contour patterns indicate landforms like steep slopes (closely spaced), gentle slopes (widely spaced), hills (closed contours), valleys (U/V-shaped contours), ridges (elongated high areas), and escarpments (steep cliffs).
Elevation Indicators – Features like triangulated height (surveyed peaks), spot height (exact elevation points), bench marks (permanent survey marks), and relative height/depth (elevation differences) provide altitude information.
3. Colours & Symbols
Colour Tints – Colour tints represent elevation, with darker shades indicating higher altitudes and lighter shades for lower areas.
Conventional Symbols – Standard symbols on maps represent physical and human-made features, such as forests, roads, and settlements.
4. Scale & Measurement
Types of Scale – Maps use different scales, including verbal (words), graphical (line scale), and representative fraction (ratio form) to show distances.
Measurement Techniques – Distances are measured using a scale, while areas are calculated using grid squares or tracing methods.
5. Directions
Cardinal Points – The eight cardinal directions (N, NE, E, SE, S, SW, W, NW) help determine the relative position of locations on a map.
6. Identifying Features
Human Settlements – Maps show villages and towns, along with their land use (agriculture, industry, forested areas).
Communication Networks – Various transport routes, including roads, railways, and footpaths, are represented for connectivity analysis.
7. Drainage & Settlements
Drainage Features – The direction of river flow and patterns like dendritic (branching), trellis (parallel streams), and radial (outward flow) can be identified.
Settlement Patterns – Settlements may be nucleated (clustered), dispersed (scattered homes), or linear (along roads or rivers).
8. Natural & Man-made Features
Natural Features – Physical elements like rivers, hills, forests, and plateaus are depicted on the map.
Man-made Features – Artificial structures such as roads, bridges, dams, and buildings are marked using standard symbols.

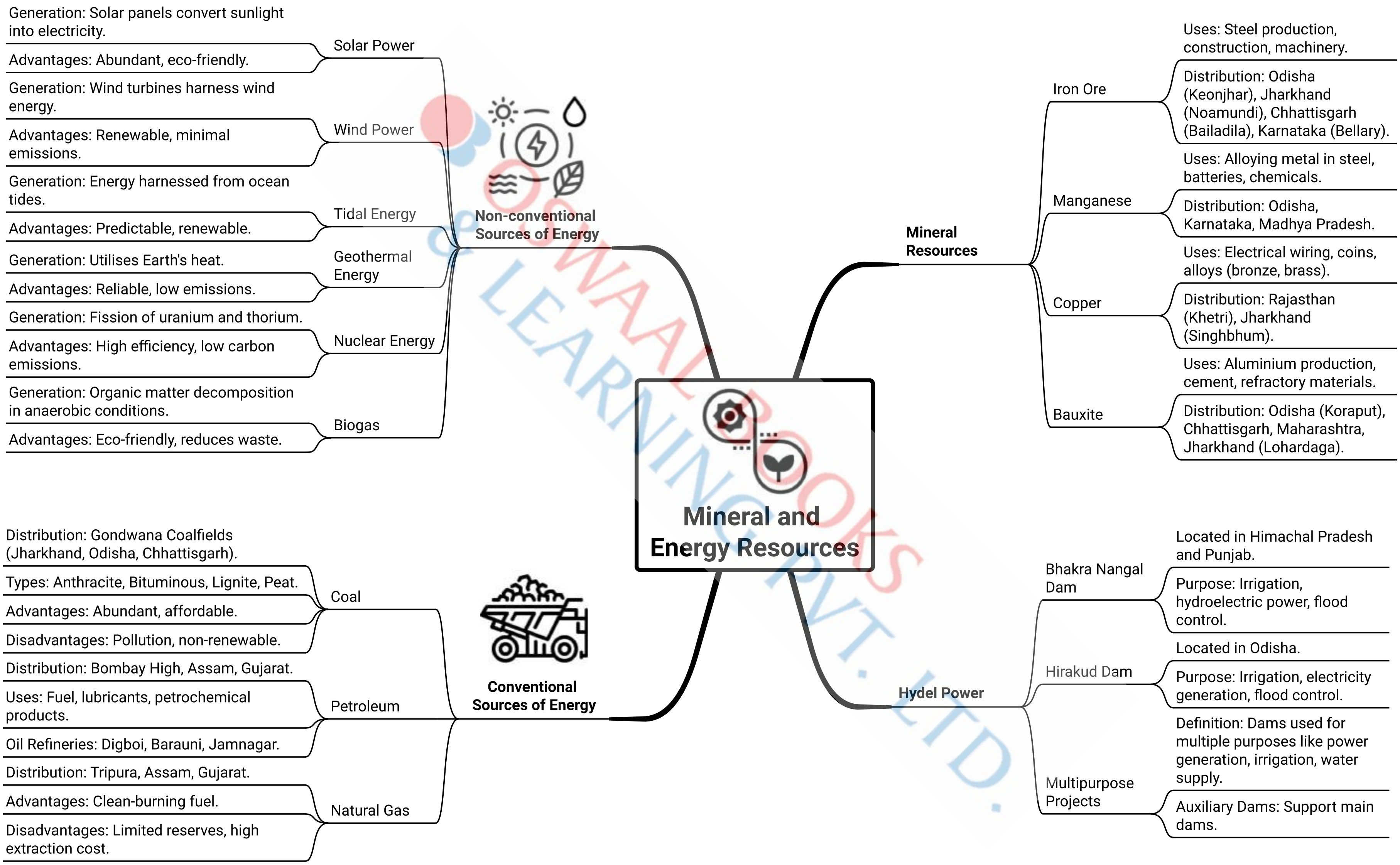

Location, Extent and Physical Features: India
1. Mountains, Peaks and Plateaus
Mountains: Himalayas, Karakoram, Aravali, Vindhyas, Satpura, Western Ghats, Eastern Ghats
Peaks: Mount Godwin Austin (K2), Mount Kanchenjunga
Plateaus: Deccan Plateau, Chota Nagpur Plateau
Hills: Nilgiris, Garo, Khasi, Jaintia
2. Plains
Gangetic Plains
Coastal Plains: Konkan, Kanara, Malabar, Coromandel, Northern Circars
3. Desert
Thar Desert (The Great Indian Desert)
4. Rivers
Indus System: Indus, Ravi, Beas, Chenab, Jhelum, Satluj
Ganga System: Ganga, Yamuna, Ghaghra, Gomti, Gandak, Kosi
Peninsular Rivers: Chambal, Betwa, Son, Damodar, Brahmaputra, Narmada, Tapti, Mahanadi, Godavari, Krishna, Cauveri, Tungabhadra
5. Water Bodies
Gulfs: Gulf of Kutch, Gulf of Khambhat, Gulf of Mannar
Other Water Bodies: Palk Strait, Andaman Sea, Chilka Lake, Wular Lake
6. Passes
Karakoram Pass, Nathu-La Pass
7. Latitude and Longitudes
Tropic of Cancer
Standard Meridian (82° 30′E)
8. Direction of Winds
South West Monsoons: Arabian Sea Branch, Bay of Bengal Branch
North East Monsoons
Western Disturbances
9. Distribution of Minerals
Oil: Mumbai High (Offshore), Digboi
Iron: Singhbhum
Coal: Jharia
10. Soil Distribution
Types of Soil: Alluvial, Laterite, Black, Red
11. Cities
Major Cities: Delhi, Mumbai, Kolkata, Chennai, Hyderabad, Bengaluru, Kochi, Chandigarh, Srinagar, Vishakhapatnam, Prayagraj
12. Population
Dense Population: Urban & industrial areas
Sparse Population: Desert, hilly, forested regions


Climate
Seasons
Summer (March to May)
High temperatures.
Localised storms and dust storms.
Monsoon (June to September)
Heavy rainfall in most parts.
Southwest monsoon winds.
Retreating Monsoon (October to November)
Decreasing rainfall.
Northeast monsoon brings rain to Tamil Nadu.
Winter (December to February)
Low temperatures.
Light rainfall from western disturbances.
Distribution of Temperature, Rainfall, Winds
Summer (March to May)
Temperature: High; average 30°C-40°C in plains.
Rainfall: Scanty in most regions, localised storms (e.g., Nor'westers in Bengal).
Winds: Hot and dry winds like Loo in northern India.
Winter (December to February)
Temperature: Low; below 10°C in northern plains, sub-zero in hilly regions.
Rainfall: Minimal; western disturbances cause light rain in north.
Winds: Northeasterly winds dominate.
Monsoon (June to September)
Temperature: Moderate due to cloud cover and rainfall.
Rainfall: Heavy in most regions; uneven distribution (e.g., heavy in Northeast, scanty in Rajasthan).
Winds: Moist southwesterly winds.
Retreating Monsoon (October to November)
Temperature: Gradual decrease.
Rainfall: Mainly in Tamil Nadu due to northeast monsoon.
Winds: Dry northeasterly winds.
Monsoon and its Mechanism
Mechanism
Differential Heating: Land heats faster than water, creating low pressure over land.
Shift of ITCZ: Inter-Tropical Convergence Zone moves northwards in summer.
Moist Winds: Southwest monsoon winds pick up moisture from the Indian Ocean.
Orographic Rainfall: Winds cause heavy rainfall on windward slopes of mountains.
Features
Uneven distribution of rainfall.
Onset: Kerala (early June), advances northwards.
Retreat: Starts from northwest in October.
Factors Affecting the Climate
Latitude
Regions near the equator experience high temperatures.
Altitude
Higher altitudes have cooler temperatures.
Distance from the Sea
Coastal regions have moderate climates; interiors have extreme climates.
Pressure and Winds
Seasonal winds (e.g., monsoon winds) affect temperature and rainfall.
Ocean Currents
Warm or cold currents influence coastal climate.
Western Disturbances
Affect winter rainfall in northern India.


Soil Resources
Types of Soil
Alluvial Soil
Found in river basins and deltas.
Highly fertile and supports intensive agriculture.
Black Soil
Found in volcanic regions.
Known as "Regur Soil"; ideal for cotton cultivation.
Red Soil
Formed from weathering of igneous rocks.
Poor in nutrients but suitable for certain crops with fertilisers.
Laterite Soil
Formed under tropical climates with high rainfall.
Acidic and less fertile, but good for plantation crops.
Distribution of Soil
Alluvial Soil
Found in Northern Plains (Punjab, Uttar Pradesh, Bihar) and Eastern India (West Bengal, Assam).
Black Soil
Found in Deccan Plateau (Maharashtra, Gujarat, Madhya Pradesh, Andhra Pradesh).
Red Soil
Found in Southern and Eastern India (Tamil Nadu, Odisha, Chhattisgarh, Karnataka).
Laterite Soil
Found in Hilly and Coastal Areas (Kerala, Karnataka, Assam, Western and Eastern Ghats).
Soil Erosion
Causes
Natural:
Wind and water runoff.
Glacier movements.
Human:
Deforestation.
Overgrazing.
Unsustainable farming practices.
Prevention
Afforestation: Planting trees to stabilise soil.
Contour Ploughing: Slowing water flow on slopes.
Terrace Farming: Preventing soil wash-off on hilly terrain.
Conservation
Crop Rotation: Restoring soil fertility.
Shelter Belts: Reducing wind erosion with tree barriers.
Check Dams: Reducing water erosion and runoff.
Composition and Characteristics
Alluvial Soil
Colour: Light grey to sandy.
Texture: Fine-grained.
Minerals: Rich in potash, phosphoric acid.
Crops: Rice, wheat, sugarcane.
Black Soil
Colour: Dark black.
Texture: Clayey and retains moisture.
Minerals: Rich in calcium carbonate, magnesium.
Crops: Cotton, sugarcane, oilseeds.
Red Soil
Colour: Reddish due to iron oxide.
Texture: Sandy and porous.
Minerals: Poor in nitrogen; rich in iron.
Crops: Millets, pulses, groundnuts.
Laterite Soil
Colour: Reddish-brown.
Texture: Coarse and porous.
Minerals: Rich in iron and aluminium.
Crops: Tea, coffee, cashew nuts.

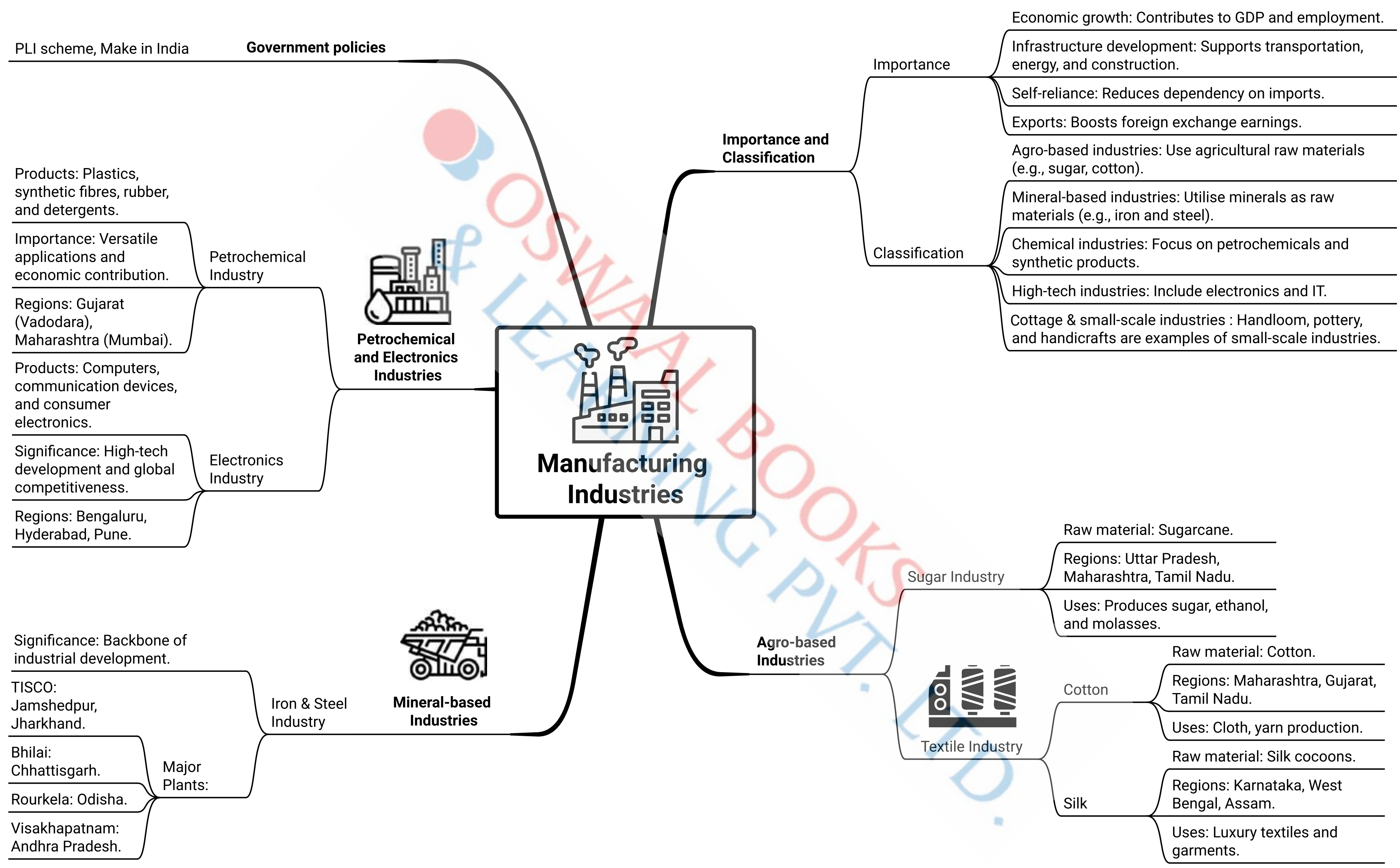

Natural Vegetation
Forest Conservation
Causes of Forest Depletion
Deforestation for agriculture and urbanisation.
Overgrazing and illegal logging.
Forest fires and industrialisation.
Conservation Methods
Afforestation: Planting trees in degraded areas.
Reforestation: Replanting trees in deforested areas.
Social Forestry: Involving local communities in conservation.
Government Initiatives
Implementation of Forest Acts and policies.
Establishing wildlife sanctuaries and national parks.
Promoting eco-friendly practices.
Importance of Forests
Environmental Importance
Maintain ecological balance.
Regulate climate and rainfall.
Prevent soil erosion.
Economic Importance
Provide timber, fuelwood, and raw materials for industries.
Source of medicinal plants and non-timber products (gum, resin).
Social and Cultural Importance
Home to indigenous communities.
Serve as sites for recreation and tourism.
Types of Vegetation
Tropical Evergreen Forests
Found in areas with heavy rainfall.
Dense, multi-layered forests; evergreen trees.
Examples: Mahogany, Ebony, Rosewood.
Tropical Deciduous Forests
Found in areas with moderate rainfall.
Trees shed leaves in dry seasons.
Examples: Teak, Sal, Shisham.
Tropical Desert Vegetation
Found in arid regions.
Sparse vegetation; thorny bushes, cacti.
Examples: Acacia, Date palm.
Littoral Forests
Found in coastal regions and swamps.
Salt-tolerant mangroves.
Examples: Sundari trees in Sundarbans.
Mountain Vegetation
Found at varying altitudes.
Tropical forests at lower altitudes, coniferous forests at higher altitudes.
Examples: Oak, Pine, Fir.
Distribution and Correlation with Environment
Tropical Evergreen Forests
Location: Western Ghats, North-Eastern India, Andaman and Nicobar Islands.
Environment: Heavy rainfall (>200 cm), hot and humid climate.
Tropical Deciduous Forests
Location: Central India, Northern Deccan Plateau.
Environment: Moderate rainfall (100-200 cm), distinct wet and dry seasons.
Tropical Desert Vegetation
Location: Rajasthan, Gujarat.
Environment: Arid regions with scanty rainfall (<50 cm).
Littoral Forests
Location: Coastal areas of West Bengal, Odisha, and Tamil Nadu.
Environment: Swampy, saline conditions in coastal zones.
Mountain Vegetation
Location: Himalayas, Nilgiris.
Environment: Varies with altitude – tropical, temperate, and alpine zones.

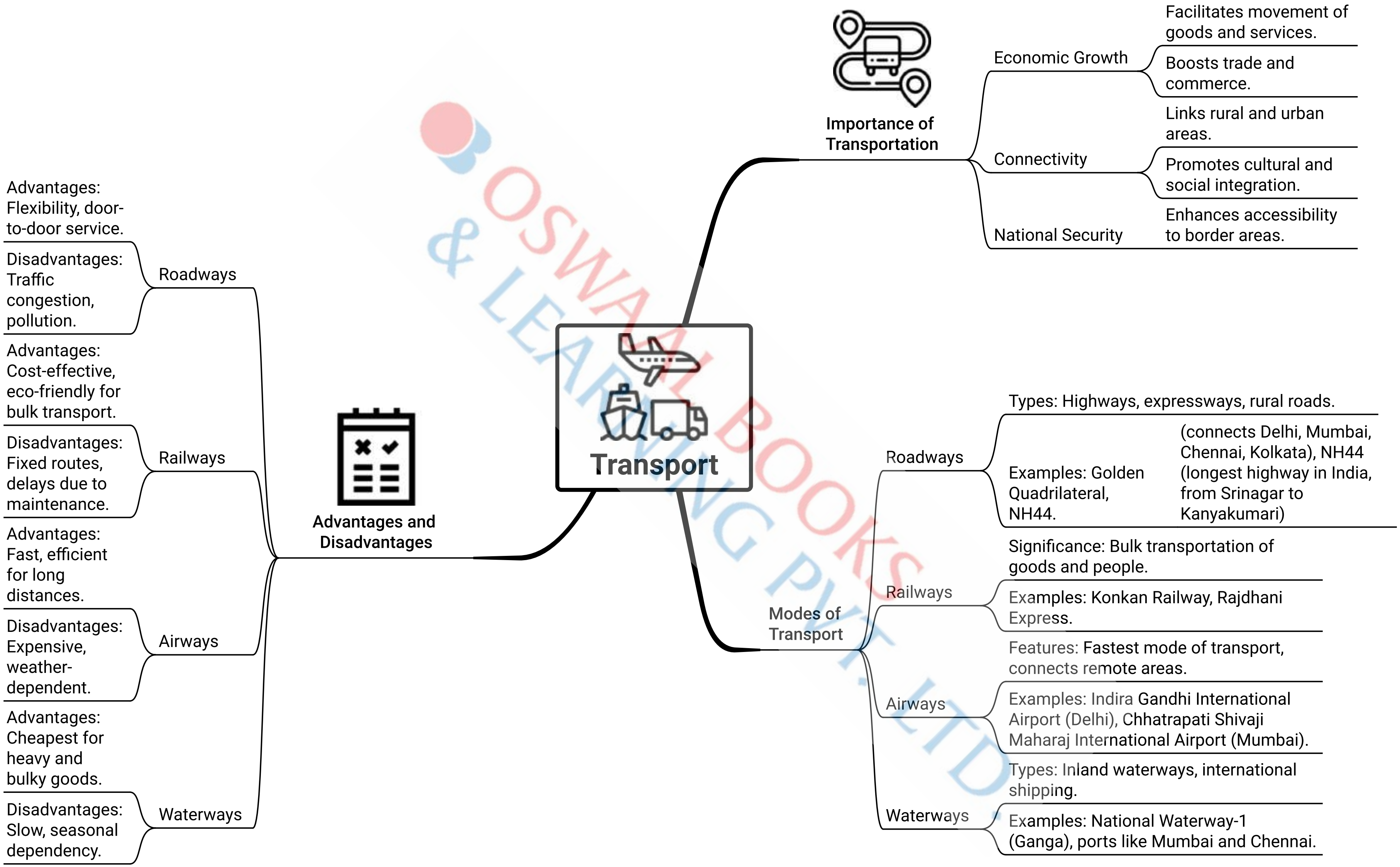

Water Resources
Sources
Surface Water
Definition: Water on the Earth's surface, found in rivers, lakes, ponds, and reservoirs.
Examples: Ganga, Yamuna, Brahmaputra rivers; Chilika Lake.
Ground Water
Definition: Water stored beneath the Earth's surface in aquifers.
Sources: Wells, borewells, and springs.
Uses: Domestic, agricultural, and industrial purposes.
Need for Conservation
Reasons for Conservation
Finite Resource: Only 3% of Earth's water is fresh water.
Growing Demand: Increasing population and industrialisation.
Climate Change: Altered rainfall patterns and rising temperatures.
Environmental Impact: Protecting ecosystems and preventing water scarcity.
Consequences of Water Scarcity
Reduced agricultural output.
Example: Drought impact on Maharashtra farmers
Decline in biodiversity.
Social and economic conflicts over water access.
Government Policies
Government initiatives like Jal Shakti Abhiyan, Namami Gange, Atal Bhujal Yojana.
Irrigation
Importance of Irrigation
Ensures crop growth during dry spells.
Increases agricultural productivity.
Supports multiple cropping cycles.
Methods of Irrigation
Canal Irrigation: Water transported through canals; suitable for plains.
Tube Wells: Groundwater extraction using pumps.
Drip Irrigation: Delivers water directly to plant roots; highly efficient.
Sprinkler Irrigation: Water sprayed over crops; mimics rainfall.
Conservation Practices
Rainwater Harvesting
Definition: Collecting and storing rainwater for reuse.
Methods:
Rooftop harvesting.
Surface runoff harvesting.
Importance:
Reduces dependency on groundwater.
Prevents waterlogging and soil erosion.
Promotes sustainable water use.
Other Practices
Drip Irrigation: Minimises water wastage in agriculture.
Reuse and Recycling: Reusing wastewater for non-potable purposes.
Public Awareness: Campaigns and education on water conservation.

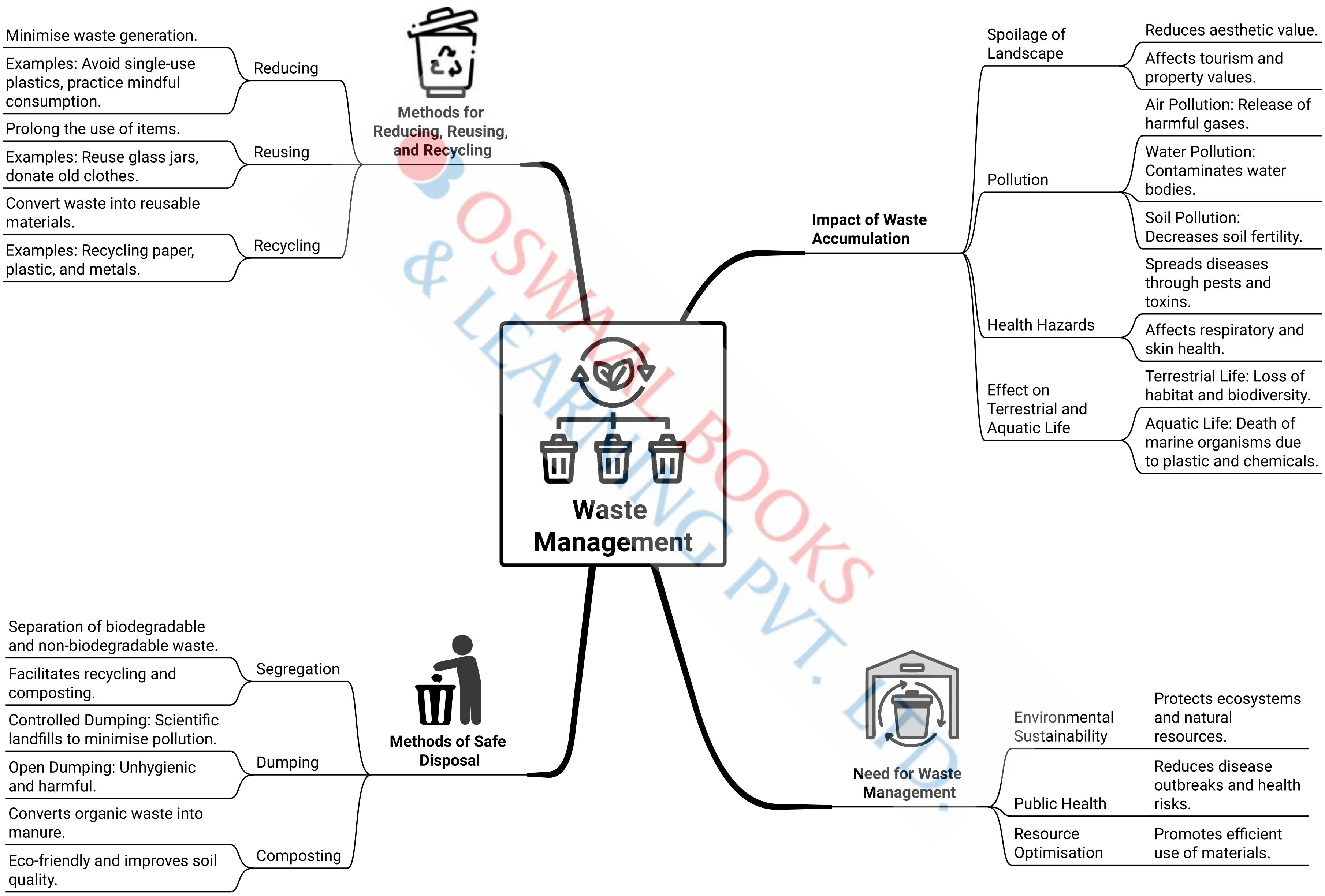

Mineral and Energy Resources
Mineral Resources
Iron Ore
Uses: Steel production, construction, machinery.
Distribution: Odisha (Keonjhar), Jharkhand (Noamundi), Chhattisgarh (Bailadila), Karnataka (Bellary).
Manganese
Uses: Alloying metal in steel, batteries, chemicals.
Distribution: Odisha, Karnataka, Madhya Pradesh.
Copper
Uses: Electrical wiring, coins, alloys (bronze, brass).
Distribution: Rajasthan (Khetri), Jharkhand (Singhbhum).
Bauxite
Uses: Aluminium production, cement, refractory materials.
Distribution: Odisha (Koraput), Chhattisgarh, Maharashtra, Jharkhand (Lohardaga).
Hydel Power
Bhakra Nangal Dam
Located in Himachal Pradesh and Punjab.
Purpose: Irrigation, hydroelectric power, flood control.
Hirakud Dam
Located in Odisha.
Purpose: Irrigation, electricity generation, flood control.
Multipurpose Projects
Definition: Dams used for multiple purposes like power generation, irrigation, water supply.
Auxiliary Dams: Support main dams.
Non-conventional Sources of Energy
Solar Power
Generation: Solar panels convert sunlight into electricity.
Advantages: Abundant, eco-friendly.
Wind Power
Generation: Wind turbines harness wind energy.
Advantages: Renewable, minimal emissions.
Tidal Energy
Generation: Energy harnessed from ocean tides.
Advantages: Predictable, renewable.
Geothermal Energy
Generation: Utilises Earth's heat.
Advantages: Reliable, low emissions.
Nuclear Energy
Generation: Fission of uranium and thorium.
Advantages: High efficiency, low carbon emissions.
Biogas
Generation: Organic matter decomposition in anaerobic conditions.
Advantages: Eco-friendly, reduces waste.
Conventional Sources of Energy
Coal
Distribution: Gondwana Coalfields (Jharkhand, Odisha, Chhattisgarh).
Types: Anthracite, Bituminous, Lignite, Peat.
Advantages: Abundant, affordable.
Disadvantages: Pollution, non-renewable.
Petroleum
Distribution: Bombay High, Assam, Gujarat.
Uses: Fuel, lubricants, petrochemical products.
Oil Refineries: Digboi, Barauni, Jamnagar.
Natural Gas
Distribution: Tripura, Assam, Gujarat.
Advantages: Clean-burning fuel.
Disadvantages: Limited reserves, high extraction cost.

Agriculture

- **Indian Agriculture**
  - Importance
    - Food security: Provides sustenance for the population.
    - Economic contribution: Major contributor to GDP and employment.
    - Raw materials: Supplies industries like textiles and sugar.
  - Problems
    - Small landholdings: Fragmented farms hinder productivity.
    - Dependency on monsoons: Irregular rainfall affects output.
    - Low mechanisation: Limited use of modern technology.
    - Soil degradation: Overuse of chemicals and poor management.
  - Reforms
    - Irrigation projects: Dams and canals for water availability.
    - Green Revolution: High-yield varieties, fertilisers, mechanisation.
    - Minimum Support Price (MSP): Ensures fair price for crops.
    - Crop diversification: Encourages growth of alternative crops.
- **Types of Farming in India**
  - Subsistence Farming
    - Definition: Produces food for family consumption.
    - Types:
      - Shifting: Slash-and-burn method (e.g., North-East India).Shifting cultivation in Meghalaya is called Jhum farming.
      - Intensive: High input on small plots for maximum yield.
  - Commercial Farming
    - Definition: Produces crops for sale in markets.
    - Types:
      - Extensive: Large-scale, mechanised farming (e.g., wheat in Punjab).
      - Plantation: Single crop (e.g., tea in Assam, coffee in Karnataka).
      - Mixed: Crops and livestock on the same farm.
- **Agricultural Seasons**
  - Rabi (Winter)
    - Sowing: October-November.
    - Harvesting: March-April.
    - Crops: Wheat, barley, mustard.
  - Kharif (Monsoon)
    - Sowing: June-July.
    - Harvesting: October-November.
    - Crops: Rice, maize, millets.
  - Zayad (Summer)
    - Sowing: March-April.
    - Harvesting: June-July.
    - Crops: Watermelon, cucumber, vegetables.
- **Major Crops**
  - Food Crops
    - Rice:
      - Climatic Conditions: High temperature, heavy rainfall.
      - Soil: Alluvial soil.
      - Distribution: West Bengal, Uttar Pradesh.
    - Wheat:
      - Climatic Conditions: Moderate temperature, light rainfall.
      - Soil: Well-drained loamy soil.
      - Distribution: Punjab, Haryana.
    - Millets:
      - Crops: Jowar, bajra, ragi.
      - Conditions: Drought-resistant.
      - Distribution: Maharashtra, Karnataka.
    - Pulses:
      - Crops: Gram, lentil.
      - Conditions: Low rainfall.
      - Distribution: Madhya Pradesh, Rajasthan.
  - Cash Crops
    - Sugarcane:
      - Conditions: Tropical, abundant water.
      - Soil: Fertile alluvial soil.
      - Distribution: Uttar Pradesh, Maharashtra.
    - Oilseeds:
      - Groundnut: Gujarat, Tamil Nadu.
      - Mustard: Rajasthan, Haryana.
      - Soybean: Madhya Pradesh, Maharashtra.
  - Fibre Crops
    - Cotton:
      - Conditions: High temperature, light rainfall.
      - Soil: Black soil.
      - Distribution: Maharashtra, Gujarat.
    - Jute:
      - Conditions: Hot and humid.
      - Soil: Alluvial soil.
      - Distribution: West Bengal, Bihar.
  - Beverages
    - Tea:
      - Conditions: Warm climate, heavy rainfall.
      - Soil: Well-drained acidic soil.
      - Distribution: Assam, Darjeeling.
    - Coffee:
      - Conditions: Moderate temperature, high humidity.
      - Soil: Well-drained soil.
      - Distribution: Karnataka, Kerala.


Manufacturing Industries
Importance and Classification
Importance
Economic growth: Contributes to GDP and employment.
Infrastructure development: Supports transportation, energy, and construction.
Self-reliance: Reduces dependency on imports.
Exports: Boosts foreign exchange earnings.
Classification
Agro-based industries: Use agricultural raw materials (e.g., sugar, cotton).
Mineral-based industries: Utilise minerals as raw materials (e.g., iron and steel).
Chemical industries: Focus on petrochemicals and synthetic products.
High-tech industries: Include electronics and IT.
Cottage & small-scale industries : Handloom, pottery, and handicrafts are examples of small-scale industries.
Agro-based Industries
Sugar Industry
Raw material: Sugarcane.
Regions: Uttar Pradesh, Maharashtra, Tamil Nadu.
Uses: Produces sugar, ethanol, and molasses.
Textile Industry
Cotton
Raw material: Cotton.
Regions: Maharashtra, Gujarat, Tamil Nadu.
Uses: Cloth, yarn production.
Silk
Raw material: Silk cocoons.
Regions: Karnataka, West Bengal, Assam.
Uses: Luxury textiles and garments.
Mineral-based Industries
Iron & Steel Industry
Significance: Backbone of industrial development.
TISCO: Jamshedpur, Jharkhand.
Major Plants:
Bhilai: Chhattisgarh.
Rourkela: Odisha.
Visakhapatnam: Andhra Pradesh.
Petrochemical and Electronics Industries
Petrochemical Industry
Products: Plastics, synthetic fibres, rubber, and detergents.
Importance: Versatile applications and economic contribution.
Regions: Gujarat (Vadodara), Maharashtra (Mumbai).
Electronics Industry
Products: Computers, communication devices, and consumer electronics.
Significance: High-tech development and global competitiveness.
Regions: Bengaluru, Hyderabad, Pune.
Government policies
PLI scheme, Make in India


Transport
Importance of Transportation
Economic Growth
Facilitates movement of goods and services.
Boosts trade and commerce.
Connectivity
Links rural and urban areas.
Promotes cultural and social integration.
National Security
Enhances accessibility to border areas.
Modes of Transport
Roadways
Types: Highways, expressways, rural roads.
Examples: Golden Quadrilateral, NH44.
(connects Delhi, Mumbai, Chennai, Kolkata), NH44 (longest highway in India, from Srinagar to Kanyakumari)
Railways
Significance: Bulk transportation of goods and people.
Examples: Konkan Railway, Rajdhani Express.
Airways
Features: Fastest mode of transport, connects remote areas.
Examples: Indira Gandhi International Airport (Delhi), Chhatrapati Shivaji Maharaj International Airport (Mumbai).
Waterways
Types: Inland waterways, international shipping.
Examples: National Waterway-1 (Ganga), ports like Mumbai and Chennai.
Advantages and Disadvantages
Roadways
Advantages: Flexibility, door-to-door service.
Disadvantages: Traffic congestion, pollution.
Railways
Advantages: Cost-effective, eco-friendly for bulk transport.
Disadvantages: Fixed routes, delays due to maintenance.
Airways
Advantages: Fast, efficient for long distances.
Disadvantages: Expensive, weather-dependent.
Waterways
Advantages: Cheapest for heavy and bulky goods.
Disadvantages: Slow, seasonal dependency.


Waste Management
Impact of Waste Accumulation
Spoilage of Landscape
Reduces aesthetic value.
Affects tourism and property values.
Pollution
Air Pollution: Release of harmful gases.
Water Pollution: Contaminates water bodies.
Soil Pollution: Decreases soil fertility.
Health Hazards
Spreads diseases through pests and toxins.
Affects respiratory and skin health.
Effect on Terrestrial and Aquatic Life
Terrestrial Life: Loss of habitat and biodiversity.
Aquatic Life: Death of marine organisms due to plastic and chemicals.
Need for Waste Management
Environmental Sustainability
Protects ecosystems and natural resources.
Public Health
Reduces disease outbreaks and health risks.
Resource Optimisation
Promotes efficient use of materials.
Methods for Reducing, Reusing, and Recycling
Reducing
Minimise waste generation.
Examples: Avoid single-use plastics, practice mindful consumption.
Reusing
Prolong the use of items.
Examples: Reuse glass jars, donate old clothes.
Recycling
Convert waste into reusable materials.
Examples: Recycling paper, plastic, and metals.
Methods of Safe Disposal
Segregation
Separation of biodegradable and non-biodegradable waste.
Facilitates recycling and composting.
Dumping
Controlled Dumping: Scientific landfills to minimise pollution.
Open Dumping: Unhygienic and harmful.
Composting
Converts organic waste into manure.
Eco-friendly and improves soil quality.

Part-I: Map Work

CHAPTER-1

Interpretation of Topographical Maps

Learning Objectives

The students will be able to:

- Locate features with the help of a four figure or a six-figure grid
- Identify landforms marked by contours, with special emphasis on contours and contour intervals
- Interpret colour tints and conventional symbols
- Identify different types of scale and measure distances
- Identify directions (eight cardinal directions)
- Identify settlement patterns, drainage patterns, and natural and man-made features

Revision Notes

Topographical Map

Scan Me!

➢ The term 'topography' is derived from two Greek words: '*topos*' meaning 'a place' and '*grapho*' meaning 'to draw'.

➢ A topographical map is also known as an Ordnance Survey Sheet. They are also called the *topo sheets* or the *topo maps*. The topographical sheets in India are prepared by the government department known as the Survey of India.

➢ A topographic map is a detailed and accurate graphic representation of cultural and natural features that appear on the Earth's surface.

➢ Topographic maps have multiple uses in the present day, such as in geographical planning or large-scale architecture, earth sciences, and various other geographical disciplines. They are used by professionals like soldiers and scientists.

➢ Topographical maps are usually drawn to a large-scale, showing details of natural and man-made features such as mountains, hills, rivers, villages, roads, and bridges.

➢ *Grid lines* are an imaginary network of equidistant lines used on a topographical map. They are drawn and shown in red colour.

➢ A grid reference helps in finding the exact location of a place on a map using a set of lines called the Eastings and the Northings.

➢ Lines extending from north to south on a map are called Eastings. The Easting that lies to the left of the object is to be read. Eastings increase numerically eastward.

➢ Lines extending from east to west on a map are called Northings. The Northing that lies below the object is to be read. Northings increase numerically Northward.

➢ These lines have been named as the Eastings and Northings because they indicate distances eastward and northward, respectively, from the point of origin.

➢ These lines are numbered. The Easting numbers are to be quoted first followed by the Northing numbers.

➢ The points at which the vertical and the horizontal grid lines cross are called coordinates.

➢ The squares enclosed by the network of grid lines are called Grid Squares.

➢ There are two types of grid references—four-figure and six-figure grid references.

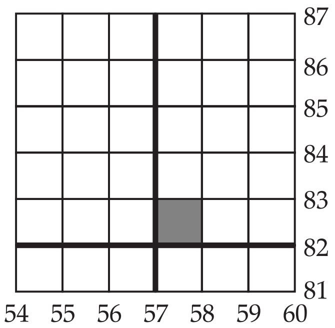
➢ The four-figure grid references are used to locate large areas such as lakes, relief features, etc. while six-figure grid references are used to locate the exact specific places or features such as police station, spot heights, wells, causeways, temples, etc.

Steps to find Grid References:

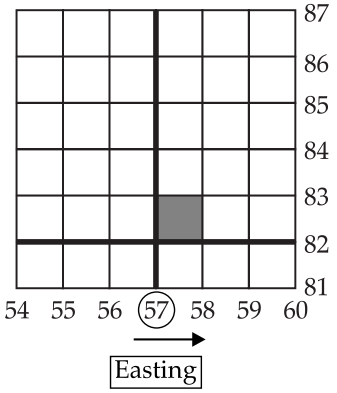
➢ When giving a grid reference, give the Easting number first and the Northing number second, similar to the reading a graph, where the '*x*'-axis coordinates are given first, followed by the '*y*'-axis.

Four-figure Grid Reference:

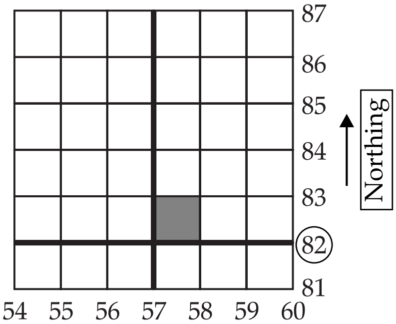
➢ Example: Find the four-figure grid reference of the shaded square in the given grid.



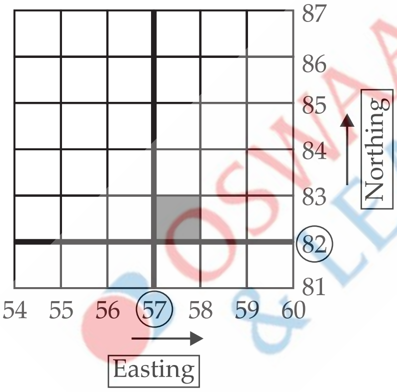
(i) In the given grid square, first locate the Easting number of the shaded square, which is 57.



(ii) Then locate the Northing number of the same shaded square, which is 82.



(iii) Now, write both numbers together by placing the Easting number first and the Northing number second. Thus, the four-figure grid reference of the given grid square is 5782.

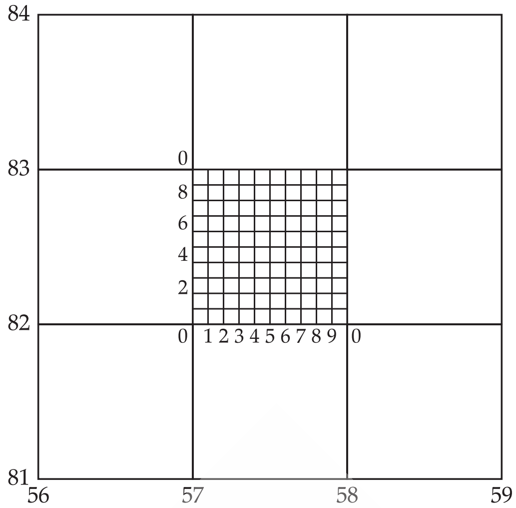


Six-figure Grid Reference: **[2024 Board]**

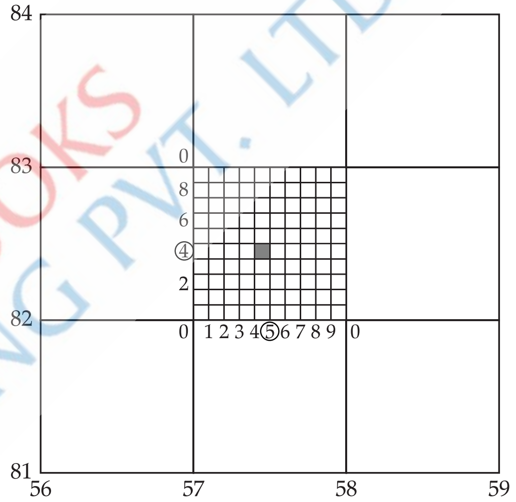
➢ Example: Find the six-figure grid reference of the shaded square in the given grid square.

(i) First, locate the four-figure grid reference, here it is 5782.

(ii) Then divide each side of the grid into ten parts, both vertically and horizontally. Mark the divisions along the Easting and the Northing lines. Here, Eastings are marked from 0 to 9, and Northings from 0 to 8 in a sequence of even numbers, i.e., 0, 2, 4, 6, and 8.



(iii) Find the intersecting point, which indicates the location of the reference point. Here, it is 5 for the Easting and 4 for the Northing.



(iv) Add this Easting number to the Easting value of the four-figure grid reference (here, it is 575), and add this Northing number to the Northing value of the four-figure grid reference (here, it is 824).

(v) Thus, 575824 is the new six-digit number that becomes the six-figure grid reference of the given grid square.

➢ A contour is an imaginary line that is drawn on the map to connect places having equal heights above mean sea level. Such lines are also known as isohypses ('iso' means equal and 'hypse' means height). **[2024 Board]**

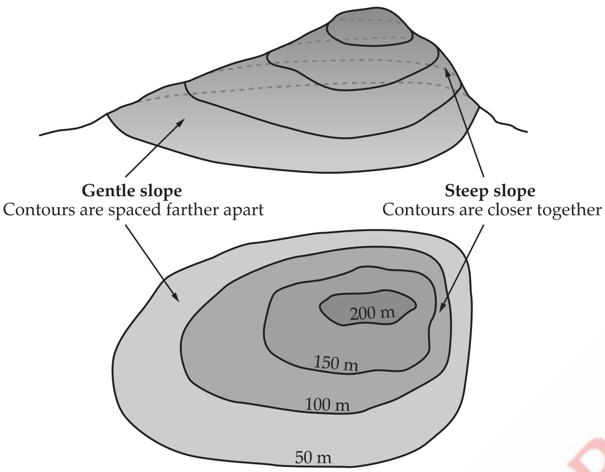
➢ Every point on a contour line has the same elevation, marking the separation between uphill and downhill. Contour lines do not touch or cross each other except at a cliff and every 5th contour line is darker in colour, called index contours.

➢ Contour interval (C.I.) is the vertical difference between two consecutive contour lines, which remains constant throughout the map—for example, 20 metres in the Survey of India Map Sheets No. 45D/7 and 45D/10.

➢ Gradient refers to the steepness of slope. It is the vertical rise of land in relation to horizontal distance.

➢ Different types of slopes are represented through contour lines. These are– gentle slope, steep slope, concave slope, convex slope, and terraced or stepped slope.

➢ **Gentle slope:** When the degree or angle of a feature's slope is very low, the slope is considered gentle. The contour lines representing this type of slope are spaced far apart.



Contours and layered relief

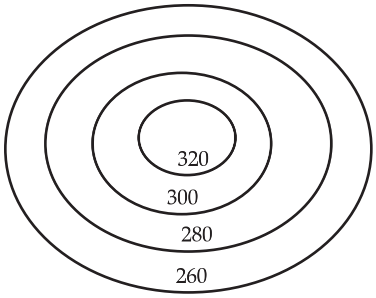
➢ **Steep slope:** When the degree or angle of a feature's slope is high, the slope is steep. The contour lines are drawn close to one another.

➢ **Concave slope:** A slope that has gentle a gradient in the lower part of a relief feature and steep gradient in its upper parts is called a concave slope. In this type, contour lines are widely spaced in the lower parts and are close together in the upper part.

➢ **Convex slope:** A slope that has gentle gradient in the upper part of a relief feature and steep gradient in its lower part is called the convex slope. In this type, contour lines are widely spaced in the upper parts and close together in the lower part.

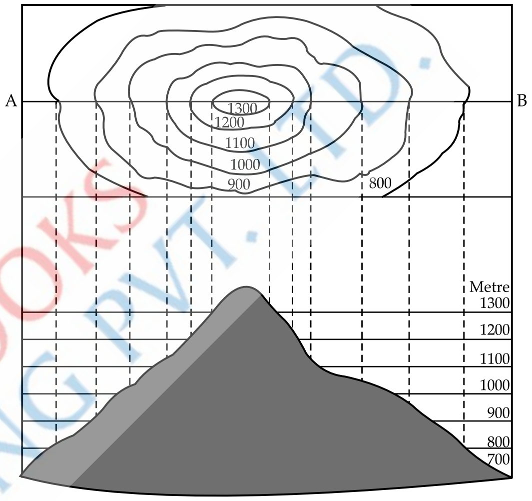
➢ **Terraced slope:** When two contour lines are found alternately found close together and widely spaced—i.e., the slope rises gently, then steeply, then gently again—this pattern is called a terraced or stepped slope.

➢ **Conical hill:** It is a landform with a narrow peak at a high elevation, typically less than 3,000 ft above the surrounding. It is shown by closed, almost circular contour lines and represented by concentric contours. Spacing between the contours is usually uniform.



Conical Hill

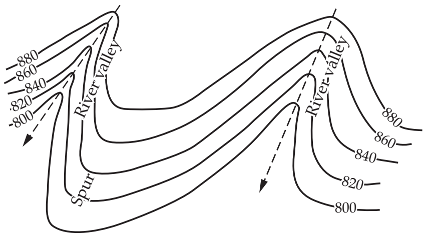
➢ **Mountain:** It is an elevation rising more than 3,000 ft above the surrounding areas. On a map, mountains are represented by closely spaced contour lines with increasing contour values.



Mountain

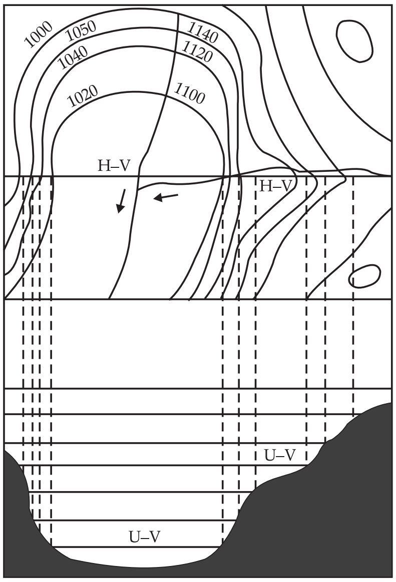
➢ **Valleys:** There are four types of valleys—V-shaped valley, U-shaped valley, hanging valley, and gorge or canyon valley.

● A **V-shaped river valley** is represented by V-Shaped contour. It is a narrow, low-lying area between hills and usually points in the direction of the downward slope.



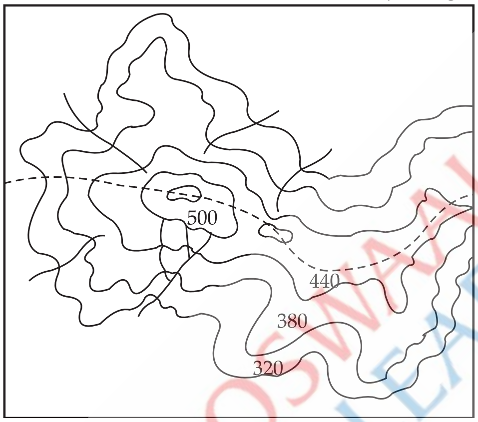
V-shaped River Valley

● **U-shaped valleys** have a flat, wide bottom and steep, parallel sides. The lowest part of the U-shaped valley is shown by the innermost contour line, with a wide gap between the two sides of the stream. The contour values increase at uniform intervals in the outward direction. A hanging valley is a high-lying tributary valley with a steep slope connecting it to the main valley. The slopes of this valley are shown by closely spaced contours, generally forming a V shape.



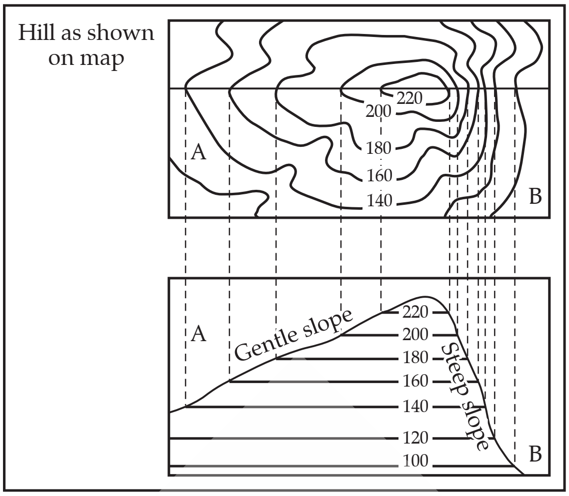
U-shaped and Hanging Valley

➢ Watershed or Water divide: It appears as a ridge or highland dividing two areas drained by different river systems. On one side of a watershed, rivers and streams flow in one direction and on the other side, they flow in opposite direction. It is also called a *water parting*.



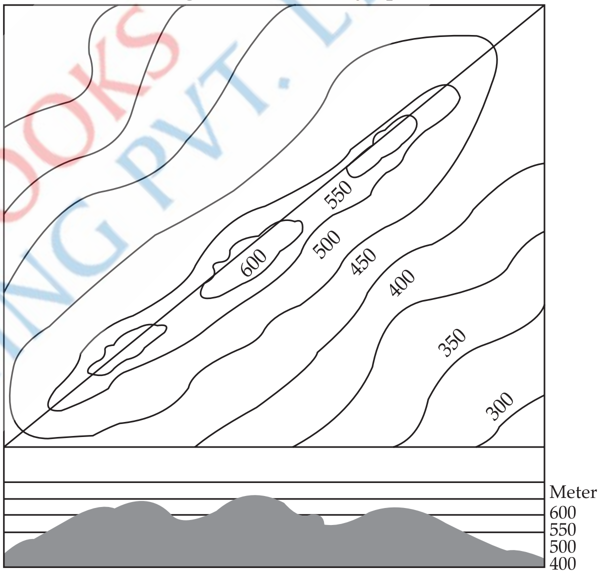
Water divide

➢ **Escarpment (scarp):** It is a landform with sides sloping in opposite directions. It has a steep slope on one side and a gentle slope on the other. The steep slope is called a *scarp*, and the gentle slope is called a *dip slope*. A scarp has closely spaced contours, while a dip slope has widely spaced contours.



Escarpment

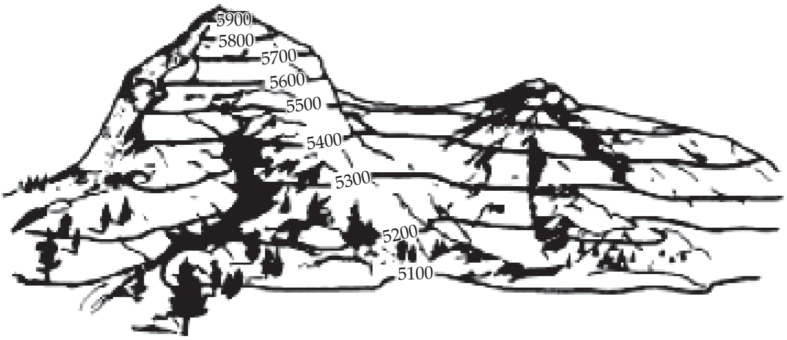
➢ **A ridge:** It is a high, elongated chain of hills with two or more peaks, shown as elliptical contours. The contours are elongated and closely spaced.

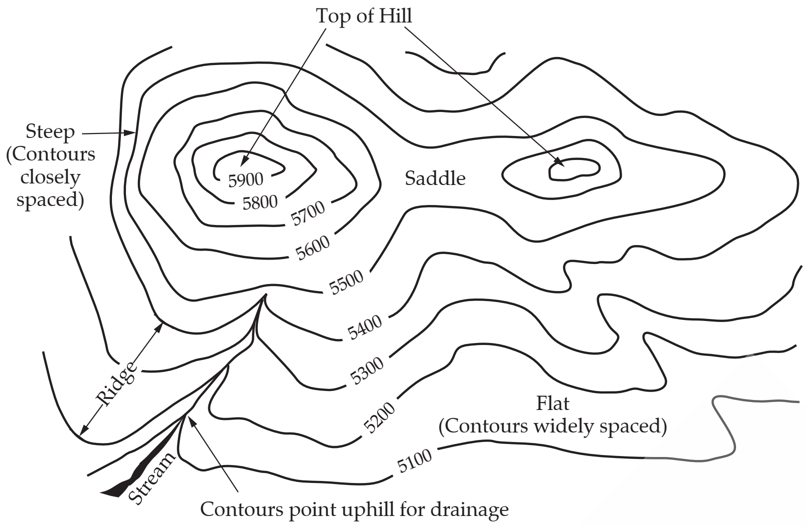


Ridge

➢ **A col:** It is a narrow, high pass through a mountain chain, formed by the meeting of river or glacier valleys from opposite sides of the range.

➢ **A saddle:** It is a pass that slopes gently between two peaks and is shaped like a horse saddle. It is broader than a col.





➢ The **scale** of a map is the ratio between the distance on the map and the corresponding distance on the ground. There are three ways of expressing scale:

(i) The Statement of Scale: It expresses the relationship of map to the ground in words e.g., two centimetres to one kilometre, and is expressed as 2 cm = 1 km.

(ii) The Representative Fraction (R.F.): It is the ratio between the distance on the map and the distance on the corresponding ground, expressed as a fraction. The numerator denotes the length on the map, and the denominator denotes the actual distance on the ground. E.g., 1:50,000.

$$\text{Representative Fraction (R.F.)} = \frac{\text{Distance on the Map}}{\text{Distance on the Ground}}$$

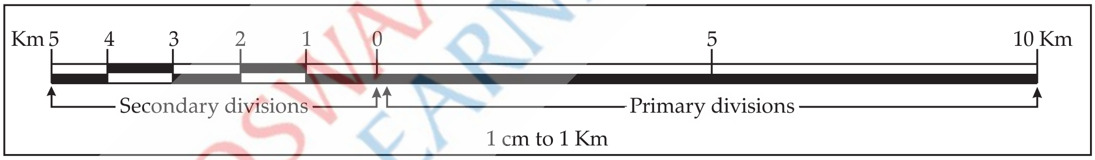
Example: If a scale is 2 cm to 1 km, calculate the R.F.

$$\text{R.F.} = \frac{\text{Distance on the Map}}{\text{Distance on the Ground}}$$

$$\frac{2\text{ cm}}{1\text{ km}} = \frac{2\text{ cm}}{1000 \times 100\text{ cm}}$$

$$\frac{1}{50,000} = 1:50,000$$

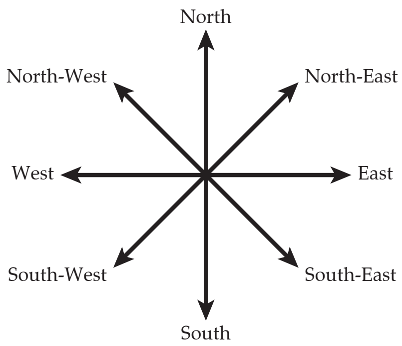
(iii) The Linear or Graphical Scale: It is divided into primary and secondary divisions. It is 10 to 20 centimetres long and divided into 8 divisions of 2 centimetres each, representing one kilometre each from 1 to 7 kilometres.



Linear Scale

➢ There are various methods to measure distance on a map such as: (i) measuring distance along a straight line, (ii) measuring distance along curves and (iii) calculating area using the grid square method.

➢ There are four main **cardinal directions**—North, South, East and West.



Eight Cardinal Points

➢ These directions are further divided into four intermediate directions—North-East, North-West, South-East, and South-West.

➢ The north to which the magnetic compass needle points is called the **Magnetic North**.

➢ The **Magnetic Compass** is an instrument used for finding directions.

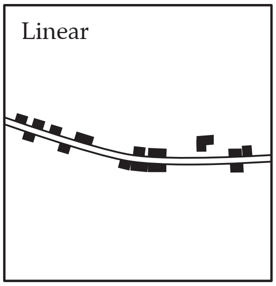
➢ Bearing is the horizontal angle between the direction of an object and the clockwise direction from north.

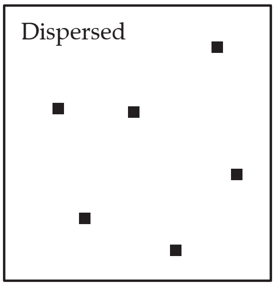
➢ A bearing with reference to the magnetic north-south line is called the **Magnetic Bearing**.

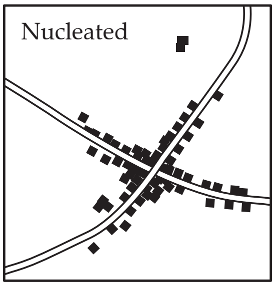
➢ A bearing measured with reference to the geographic north-south line is called **True Bearing**.

➢ Settlement refers to a cluster of inhabited houses–urban and rural. **[2024 Board]**

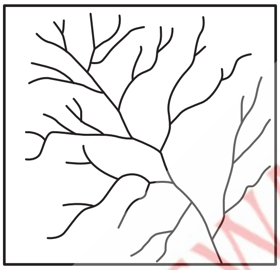
➢ There are different patterns of settlements such as: nucleated or clustered, dispersed or scattered, linear, radial, and rectangular.



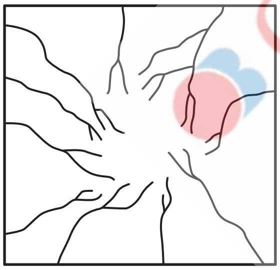




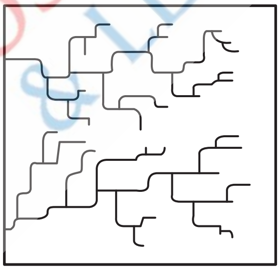
- **Drainage Pattern** – The different forms in which rivers flow, depending upon the factors such as rock structure, topography, climate, and slope of the land.
- There are different drainage patterns, such as dendritic, trellis or rectangular, radial, disappearing or irregular, and parallel.



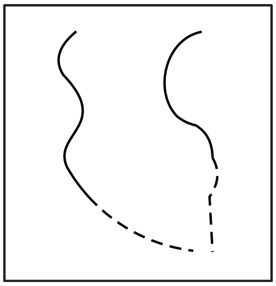
Dendritic Drainage



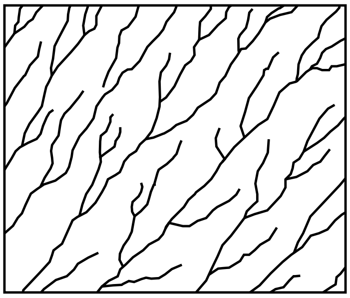
Radial Drainage



Rectangular Drainage /Trellis



Disappearing Drainage/Irregular Drainage

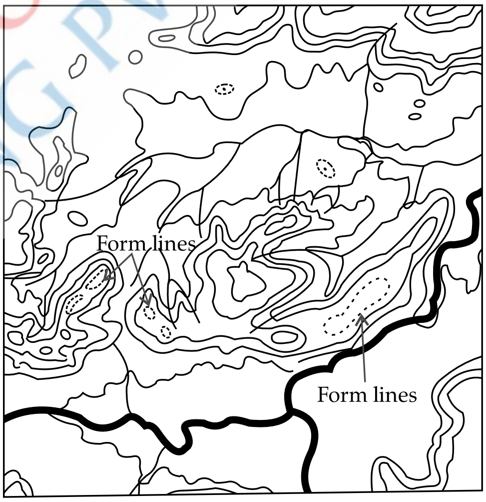


Parallel Drainage

- The height represented by a triangle (Δ) followed by a number such as Δ 364, indicates its elevation above mean sea level and is called a **Triangulated Height**. It is more accurate than a Spot Height because it is calculated using trigonometry.
- **A Spot height** is represented on the map by a black dot in front of the number, such as .275, which indicates the elevation of that specific point above mean sea level.

• 275

- **Form Lines:** These are used in addition to contour lines. They are broken lines running in the direction of contours. They help in indicating minor details not shown by contours with large intervals.



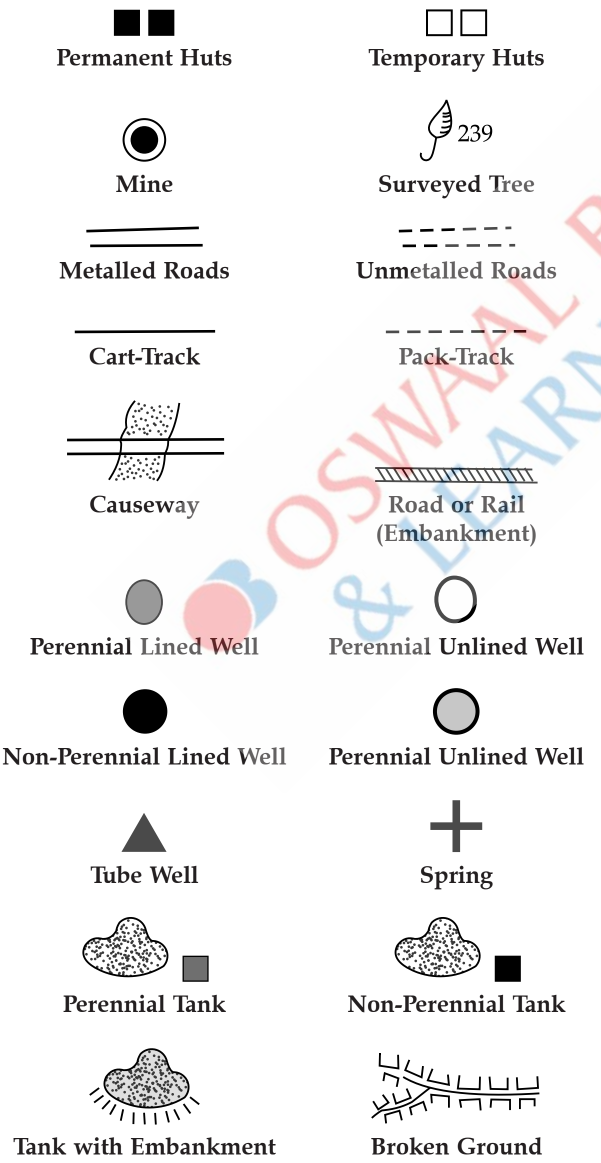
Form lines

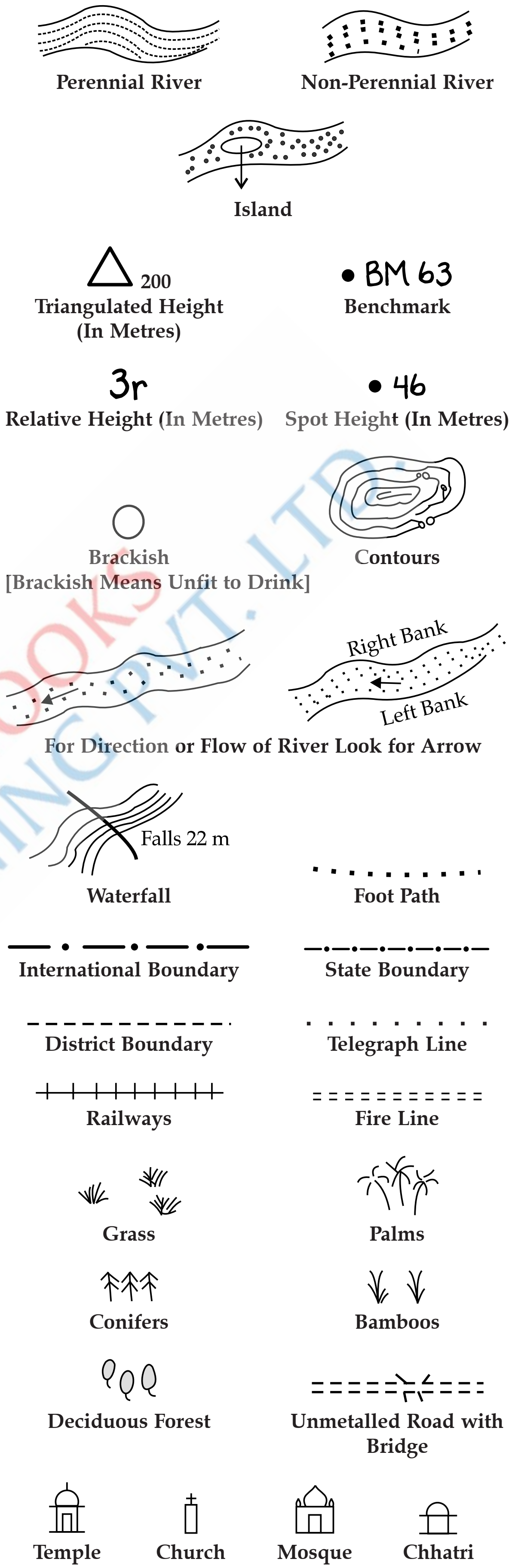
- Surveyors make a permanent cut or mark on a rock, stone, prominent building or place called a **Bench Mark** (BM). It indicates the height in feet or metres of that place above mean sea level, e.g., BM 610.
- **Relative Height/Depth** is represented by a small 'r' written along with a number, such as, 22r. It indicates the relative height or depth of a particular point from the surrounding surface and not from the sea level. It is marked beside well, tank, embankment, peak, dam, sand dune, riverbanks, etc.
- A topographical map uses signs and symbols to represent certain man-made and physical features. These are called conventional signs and symbols. **[2024 Board]**
- **Causeway:** A raised metalled road over a small or non-perennial stream, or marshy area, that serves as a temporary bridge but is not a full bridge.

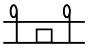
- **Conventional Signs and Symbols:** These are illustrated by a legend or an index given in the lower margin of the map.
- Different colours are used in a toposheet to represent various features. They are as follows:

(i) Brown: Contour lines, their numbering, form lines, and sand features.
(ii) Blue: All water bodies such as wells and lakes, where they contain water.
(iii) Yellow: All cultivated areas.
(iv) Green: All wooded or forested areas, scattered trees and scrubs.
(v) Black: All names, riverbanks broken ground, dry streams, surveyed trees, spot heights and their numbering, railway tracks, telephone and telegraph lines.
(vi) Red: Grid lines and their numbering, roads, cart tracks, settlements, huts, and other buildings.
(vii) White: Uncultivated or barren land.

Conventional Signs and Symbols



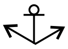



Idgah


Tomb

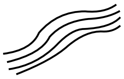

Lighthouse


Fort


Anchorage


Grave

CONTOUR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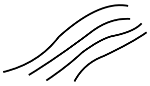
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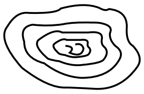
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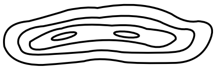
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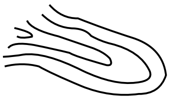
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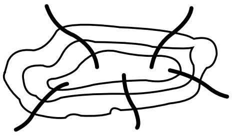
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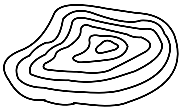
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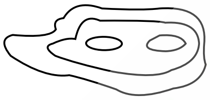
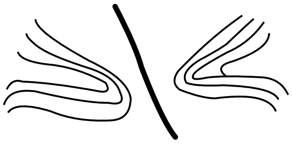
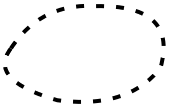

Col or Saddle


A River Gap


Form Line

Key Terms

- **Topographic Maps:** These are large-scale maps that depict both physical and man-made features of the landscape, characterised by contour lines to show in detail ground relief of the land.
- **Grid Reference:** It helps locate the exact location of a place on a map through a set of lines called Eastings and Northings.
- **Coordinates:** The points where the vertical and the horizontal grid lines intersect.
- **Magnetic North:** The north to which the magnetic compass needle points is called the Magnetic North.
- **Scale:** It is the ratio between the distance on the map and the corresponding distance on the ground.
- **Cardinal Directions:** The four principal directions on a compass—North, South, East, and West.
- **Conventional Symbols:** The signs and symbols used on the map that are internationally recognised are called conventional symbols.

CHAPTER-2
Location, Extent and Physical Features: India

Learning Objectives

The students will be able to:

- Locate and identifying the different physical features of India including, mountain passes and plains.
- Identify the rivers of India.
- Understand the direction of winds, and the distribution of minerals and soil.
- Locate the cities and population (dense and sparse).

Revision Notes

Introduction

- India is centrally situated in the southern part of the Asian continent, at the head of the Indian Ocean.
- India extends from 8°4′ N to 37°6′ N latitudes and from 68°7′ E to 97°25′ E longitudes.
- India covers an area of 3.28 million sq. km, which constitutes approximately 2.42% of the world's total land area.
- The longitude 82°30′ E, passing near Mirzapur in Uttar Pradesh, is considered the Standard Meridian of India for calculating Indian Standard Time (IST).
- India lies entirely in the Northern Hemisphere. The Tropic of Cancer (23°30′ N) divides India into nearly two climatic zones.
- India has neighbouring countries: Pakistan and Afghanistan in the northwest; China, Nepal and Bhutan in the north; and Myanmar and Bangladesh in the east. The island of Sri Lanka lies to the south of India, separated by the Palk Strait and the Gulf of Mannar.
- India is a peninsula surrounded by water on three sides. The Indian Ocean lies to the south, the Arabian Sea to the west, and the Bay of Bengal to the east of India.

➢ India's coastline is relatively regular, limiting the number of natural harbours along its shores.

➢ The Andaman and Nicobar Islands lie in the Bay of Bengal between latitudes 6° N and 14° N.

➢ Indira Point, southernmost point of India located at 6° N that was submerged in 2004 by the tsunami.

➢ The Indian subcontinent is separated from the rest of the Asian continent by a chain of high mountains comprising the Himalaya, the Karakoram, the Hindu Kush, the Kunlun and the Tien Shan. It is divided into six physiographic divisions:

Physical Features of India

(i) The Northern Mountain System
(ii) The Northern Plains
(iii) The Great Indian Desert
(iv) The Peninsular Plateau
(v) The Coastal Plains
(vi) The Islands

➢ **The Northern Mountain System consists of:**

(i) The Trans-Himalayan Region or Tibet Himalayan Region
(ii) The Himalayas
(iii) The Eastern Hills or Purvanchal Range

➢ The Northern Mountain Wall of India is also known as the Himalayan Range.

➢ The Himalayas are young fold mountains. They run in a west to east direction from the Indus to Brahmaputra, forming an arc that covers a distance of about 2,400 km. Their width varies from about 400 km in the west to about 150 km in the east.

➢ The Himalayas may be divided into three parallel ranges along their longitudinal extent:

(i) The Himadri Range
(ii) The Himachal Range
(iii) The Shivalik Range

➢ The Himadri is also called the Greater Himalayas or the Inner Himalayas. The Greater Himalayas or Himadri has average height of 6000 metres and width lies between 120 km - 190 km.

➢ The Greater Himalayas has the world's highest peak, Mt. Everest (8,848.86 metres). The world's second-highest peak, Mount K2 (or Godwin Austen), is in the Karakoram Range and, while Kanchenjunga is the highest peak of the Himalaya in India. **[2024]**

➢ It has many **passes**, namely the Karakoram Pass, the Shipki La Pass, the Mana Pass, the Nathu La Pass.

➢ The Himachal is also known as the Himachal Himalayas, or the Middle or Lesser Himalayas.

➢ The Lesser Himalayas have height range of 1,000 m to 4,500 m, and average width is 50 km.

➢ The Himachal Range runs parallel to the Himadri.

➢ Some of the important ranges, such as the Pir Panjal Range, the Dhauladhar Range, and the Nag Tibba Range, are situated here.

➢ Some of the most important hill stations in this range like Mussoorie, Shimla, Nainital, Almora, and Chail.

➢ The Pir Panjal, Bidli, Golabghar, and Banihal are some important passes of the Lesser Himalayas.

➢ The Shivalik Range is also called the Outer Himalayas.

➢ The Shivalik Range is not a continuous; its height varies from 900 metres to 1100 metres, and the width lying between 10 km and 50 km.

➢ The Trans-Himalayas lie to the north of the Great Himalayan Range. It is a 1,600 km long mountain range in Tibet, India, and a region under territorial dispute, extending in a west-east direction parallel to the main Himalayan range, also known as the Tibet Himalayan Region.

➢ The valleys lying between the Shivalik and the Lesser Himalayas (Himachal) are called 'Duns', such as Dehra Dun, Kotli Dun, and Patli Dun.

➢ In the east, the Himalayas bend towards the south, forming a series of hills known as the Eastern Hills or the Purvanchal.

➢ **The Doon, Bhabhar, Terai, and Bhangar** are the other important features of the Himalayas.

➢ The Himalayas are of great significance for the following reasons:

(i) The Himalayas act as a climate divide, as they protects the northern India from severe cold in winter and act as a barrier for the moisture-laden monsoon winds that bring heavy rainfall to the North Indian Plains.

(ii) The Himalayas are the source of many perennial rivers due to the glaciers present there. Rivers such as the Ganga, the Yamuna, the Ghaghra, the Gandak, the Gomti, the Kosi, the Sharda and the Brahmaputra provide water for drinking and irrigation to the entire Northern Plains.

(iii) The rivers originating in the Himalayas are a major source of hydroelectric power.

(iv) The Himalayas act as a physical barrier against invaders.

(v) The region attracts thousands of tourists from India and abroad due to the large number of hill stations, such as Mussoorie, Nainital, Shimla, Srinagar, and Ranikhet, and pilgrimage centres such as the shrines of Badrinath, Kedarnath, Amarnath, Vaishno Devi, and Kailash-Mansarovar.

(vi) The Himalayas also attract adventure seekers, as it provides a lot of opportunities for trekking, hiking, river rafting, skiing, and more.

(vii) The Himalayas are rich in forest resources, providing both hard wood and soft wood and is also a home to a wide variety of wild animals such as yaks, bears, tigers, elephants, red pandas, and snow leopards.

(viii) It is a storehouse of minerals such as copper, lead, nickel, cobalt, tungsten, coal, as well as also gold, silver and precious stones.

(ix) The rivers flowing down from the Himalayas through the Northern Plains carry alluvium and deposit it in the floodplains, which are is very fertile.

➢ The plains of Northern India have three major rivers—the Indus, the Ganga and the Brahmaputra—along with their tributaries.

➢ The River Ganga originates from the Gangotri Glacier in the Himalayas, the River Indus from the Kailash Range, and the Brahmaputra River from the Angsi Glacier in Tibet near Lake Mansarovar.

➢ The Great Plains lie to the south of the Himalayas and to the north of the Indian Peninsula.

➢ The Northern Plains

➢ The Northern plains are about 2,400 km long and their width varies from about 300 km in the west to about 150 km in the east.

➢ They mainly include the states of Punjab, Haryana, Uttar Pradesh, Bihar, West Bengal, and Assam.

➢ They can be sub-divided into the following three areas:

(i) The Punjab Plains

(ii) The Ganga Plains

(iii) The Brahmaputra Plains

➢ The Northern Plains are of great significance of the following reasons:

(i) Due to the presence of a good network of rivers and a favourable climate, these plains support dense population.

(ii) The fertile soil, perennial rivers and favourable climate have made the Northern Plains excellent agricultural land.

(iii) A large number of multipurpose dams have been constructed across the rivers to provide water for irrigation and to generate electricity.

(iv) Socially and religiously, the Northern Plains are of great significance. Many religious and historical cities like Prayagraj, Varanasi, Haridwar, and Mathura, are situated along the rivers.

(v) Due to the flat surface of the Northern Plains, construction of a network of roads and railways is possible. The rivers are also navigable and allow comfortable transportation. This also facilitates easy access to an efficient communication system.

➢ The Great Indian Desert is a sandy plain covered with barchans and sand dunes.

➢ It lies along the western margin of Aravalli Hills and to the south-west of the Gangetic Plain. It is called the Thar Desert.

➢ It is the seventeenth-largest desert in the world and the ninth largest hot, subtropical desert globally.

➢ Covering an area of around 77,000 square miles (about 200,000 square kilometres), it is bounded on the northwest by the Sutlej River, on the east by the Aravalli Range, on the south by salt marshes known as the Rann of Kutch, and on the west by the Indus Valley.

➢ More than 60% of the desert lies in the states of Rajasthan and extends into Gujarat, Punjab, and Haryana. About 85% of the Thar Desert is located within India, with the remaining 15% in Pakistan.

➢ This region experiences semi-arid and arid climatic conditions. It receives less than 150 mm of annual rainfall.

➢ The Luni is the only river in this area. There are a few salt water lakes, such as the Sambhar Lake, the Kuchaman Lake, the Phalodi Lake.

➢ The vegetation cover is sparse, consisting mainly of bushes.

➢ The Great Indian Desert is of considerable significance because of the following reasons-

(i) The grasses in the desert has multi-purpose medicinal benefits.

(ii) The camel, a desert animal, is an important means of transport for carrying both people and goods.

(iii) On 18 May 1974, India conducted its first nuclear weapon test in the Thar Desert.

(iv) Minerals such as limestone, marble, phosphorite, feldspar, gypsum, are obtained from this region.

➢ The **Peninsular Plateau** lies to the south of the Northern Plains. It is a triangular-shaped tableland and is the largest and the oldest of all the physiographic divisions.

➢ It has an elevation of about 600 m to 900 m and is a stable terrain due to the lava tract formed due to the volcanic eruptions.

➢ It is spread the states of Gujarat, Maharashtra, Bihar, Karnataka and Andhra Pradesh.

Plateaus of India

Scan Me!

➢ The plateau can broadly be divided into the following two subdivisions:

(i) The Central Highlands

(ii) The Deccan Plateau

(i) The Central Highlands: The northern part of the Peninsula, to the north of the Vindhyas, is known as the Central Highlands. They are further divided into the Malwa Plateau, Bundelkhand Plateau, Baghelkhand Plateau and the Chhota Nagpur Plateau.

(a) The Malwa Plateau is bounded by the Aravalli Range in the west, Bundelkhand in the east, and Vindhya Range in the south. It stretches across Rajasthan, Madhya Pradesh, and Gujarat.

(b) The Chhota Nagpur Plateau is in eastern India, covering much of Jharkhand and adjacent parts of Odisha, Bihar, and Chhattisgarh.

(c) The Meghalaya Plateau is an extension of the Peninsular Plateau towards the north-east and consists of the Garo, Khasi and Jaintia Hills, along with their outliers formed by the Assam Ranges.

(d) The Bundelkhand Plateau and the Baghelkhand Plateau are eastward extensions of the Central Highlands. Its height ranges from about 550 m to 1,033 m and lies in eastern Madhya Pradesh.

(ii) The Deccan Plateau: It extends from the Vindhyas to the southern tip of the peninsula. The Vindhya Range and its eastern extensions, namely Mahadeo Hills, Kaimur Hills and Maikal Range form its northern edge. The Western Ghats and the Eastern Ghats mark their western and eastern edges.

➢ The Western Ghats, the Eastern Ghats, the Vindhya Range, the Satpura Range, and the Aravalli Hills are the mountains of Peninsular India.

The Thar Desert

Scan Me!

➢ The major Peninsular rivers are divided two categories: the east-flowing rivers and the west-flowing rivers.

➢ The east-flowing rivers are the Mahanadi, the Godavari, the Krishna, the Cauvery or Kaveri.

➢ The west-flowing rivers are – the Narmada, the Tapi, the Luni, the Sabarmati, and the Mahi.

➢ The rivers of Northern India differ from those of Peninsular India in the following ways:

	Rivers of Northern India	Rivers of Peninsular India
1.	They are perennial and snow-fed.	They are non-perennial, seasonal, and rain-fed.
2.	These rivers are longer and have more tributaries.	These rivers are shorter and have fewer tributaries.
3.	These rivers are young and are eroding, transporting and depositing silt and sediments.	These rivers are in their old stage and are only depositing silt and sediments.
4.	They carry more silt as they erode sedimentary rocks.	They have less silt as they erode igneous rocks.

Significance of Peninsular plateau:

➢ Peninsular India is a storehouse of minerals.

➢ The north-western part is rich in iron-rich basaltic lava and the soil derived from it is black soil. This soil is used for growing, cotton, rubber, coffee, millets, and tea.

➢ The Peninsular Plateau is covered with various types of forests—deciduous and evergreen forests.

➢ The plateau rivers have a large number of waterfalls which are used for hydroelectric power generation.

➢ The Western Ghats have some of India's best biodiversity; they are therefore considered as a hotspot region of India.

➢ The coastal plains of India run parallel to the Arabian Sea and the Bay of Bengal, along the Peninsular Plateau.

➢ The Deccan Plateau is flanked by two coastal plains on either side:

(i) The Western Coastal Plains

(ii) The Eastern Coastal Plains

➢ The Western Coastal Plains consist of the Gujarat Plains, the Kutch Peninsula, the Kathiawar Peninsula, the Konkan Coast, the Kanara Coast, and the Malabar Coast.

➢ The Eastern Coastal Plains consist of The Utkal Plain, the Northern Circars, and the Coromandel Coast and have prominent **deltas** on the rivers, namely, the Mahanadi, the Godavari, the Krishna, and the Cauvery (Kaveri).

➢ These coastal plains also include several lakes, such as Chilika Lake in Odisha, and Kolleru and Pulicat lakes in Andhra Pradesh.

➢ The Coastal Plains are quite significant because of the following reasons:

(i) They are a source of precious minerals.

(ii) The Kerala coast has large quantities of monazite, which is used for nuclear power.

(iii) The sedimentary rocks of these coastal plains contain large deposits of mineral oil.

(iv) The ports account for approximately 98% of the international trade.

(v) Fishing is an important activity in these coastal plains.

➢ India has two groups of islands, namely:

(i) The Andaman and Nicobar Islands

(ii) The Lakshadweep Islands

➢ The Andaman and the Nicobar Islands are situated in the Bay of Bengal.

➢ They are a group of 572 islands.

➢ The chief islands are- Andaman Islands and Nicobar Islands.

➢ The Lakshadweep Islands are situated in the Arabian Sea.

➢ They are composed of small coral islands.

➢ The Lakshadweep Islands consist of 36 islands.

➢ The northern subgroup of the Lakshadweep Islands is called Amindivi Islands, while the remaining islands are part of the Laccadive group

The Coastal Plains and Islands of India

Scan Me!

➢ The two groups of islands (The Andaman and the Nicobar Islands) are separated by the **10° Channel**. It is so named because it lies on the 10-degree line of latitude to the north of the equator.

➢ The Indian islands are of great significance for the following reasons:

(i) Both islands are rich in flora and fauna, and are of great strategic importance to India.

(ii) The Andaman and the Nicobar Islands have many beaches, which are an attraction for tourists.

Key Terms

➢ **Pass:** It is a route longitudinal route a mountain range or across a ridge.

➢ **Doon:** Doons/Duns are longitudinal valleys between the Himachal and the Shivalik ranges; e.g., Dehradun.

➢ **Bhabhar:** These are porous, gravelly plains at the foot of the Himalayas.

➢ **Terai:** It is a marshy, swampy area to the south of Bhabhar formed when water from the Bhabhar seeps into the soil and suddenly appears.

➢ **Khadar and Bhangar:** The new alluvium brought down by rivers in low-lying zones is known as Khadar soil while the old alluvium soil found in the riverbeds above the flood plain level is known as Bhangar.

➢ **Perennial Rivers:** The rivers which contain water throughout the year—due to rain in the rainy season, melting ice in summer and freezing snow in the winter—are called the perennial rivers.

➢ **Peninsular Plateau:** It is the oldest landmass, made up of lava flows, and is composed of the old crystalline, metamorphic, and igneous rocks.

➢ **Delta:** A river delta is a triangular-shaped landform that forms by deposition of sediment carried through a river near its mouth, where it meets the sea and gets divided into distributaries.

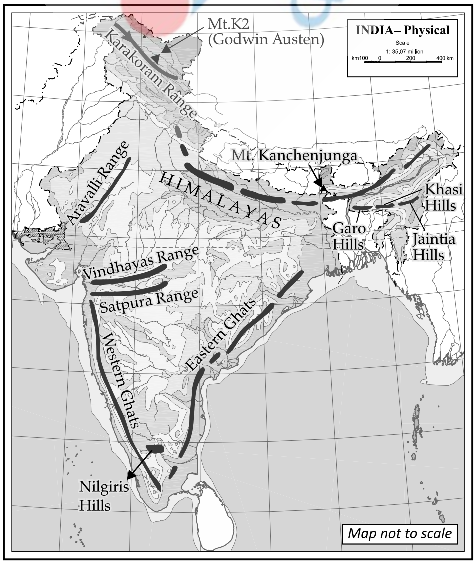
➢ **Ten Degree Channel:** The Ten Degree Channel lies on the 10-degree line of latitude to the north of the equator, and separates the Little Andaman and Nicobar Islands in the Bay of Bengal..

MAP OF INDIA

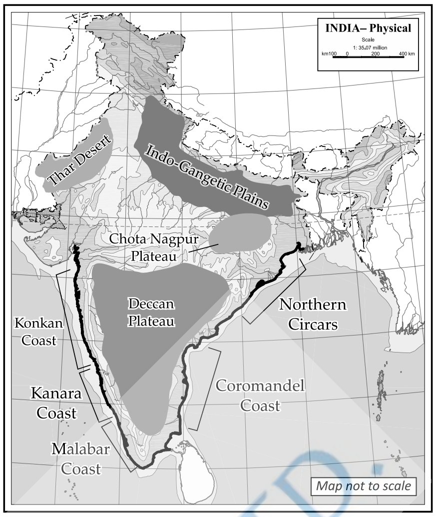
On the outline map of India, candidates will be required to locate, mark, and name the following:

- **Mountains, Peaks and Plateaus:** Himalayas, Karakoram, Aravalli, Vindhyas, Satpura, Western and Eastern Ghats, Nilgiris, Garo, Khasi, Jaintia, Mount Godwin Austin (K2), Mount Kanchenjunga, Deccan Plateau, and Chota Nagpur Plateau. **[2024]**
- **Plains:** Gangetic Plains and Coastal Plains (Konkan, Kanara, Malabar, Coromandel, Northern Circars). **[2024]**
- **Desert:** Thar (the Great Indian Desert). **[2024]**
- **Rivers:** Indus, Ravi, Beas, Chenab, Jhelum, Satluj, Ganga, Yamuna, Ghaghra, Gomti, Gandak, Kosi, Chambal, Betwa, Son, Damodar, Brahmaputra, Narmada, Tapi, Mahanadi, Godavari, Krishna, Cauveri, and Tungabhadra. **[2024]**
- **Water Bodies:** Gulf of Kutch, Gulf of Khambhat, Gulf of Mannar, Palk Strait, Andaman Sea, Chilika Lake, and Wular Lake. **[2024]**
- **Passes:** Karakoram and Nathu-La passes.
- **Latitude and Longitudes:** Tropic of Cancer, Standard Meridian (82°30′E).
- **Direction of Winds:** South-West Monsoon (Arabian Sea and Bay of Bengal Branches), North-East Monsoon, and Western Disturbances. **[2024]**
- **Distribution of Minerals:** Oil - Mumbai High (offshore oil field) and Digboi; iron – Singhbhum, Coal – Jharia. **[2024]**
- **Soil Distribution:** Alluvial, laterite, black, and red soil.
- **Cities:** Delhi, Mumbai, Kolkata, Chennai, Hyderabad, Bengaluru, Kochi, Chandigarh, Srinagar, Vishakhapatnam, Prayagraj. **[2024]**
- **Population:** Distribution of population (Dense and Sparse). **[2024]**

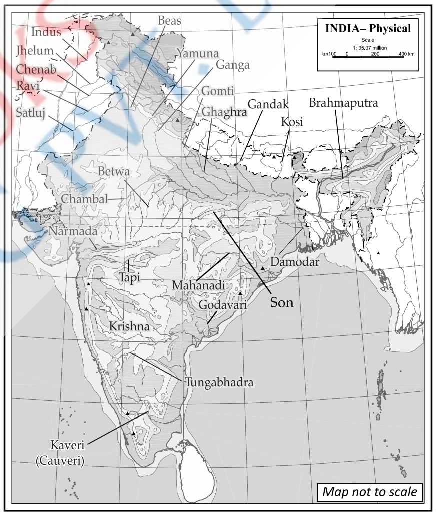
1. The Mountain Ranges, Mountain Peaks and Hill Ranges of India:



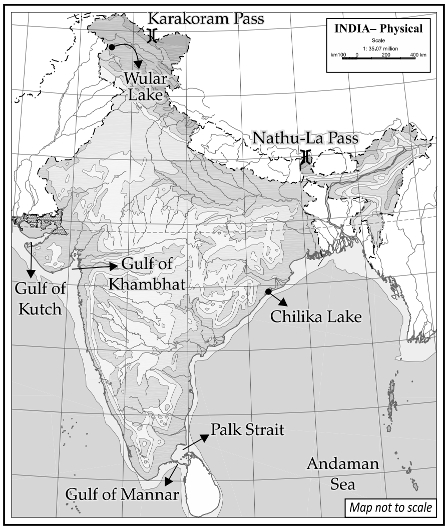
2. The Plains (The Northern Plains and Coastal Plains), Desert (Thar) and Plateau Regions of India:



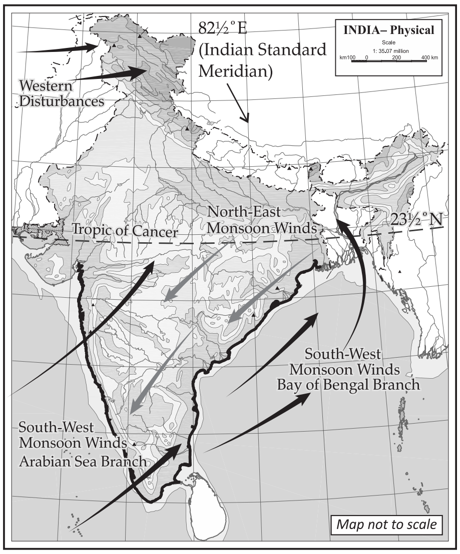
3. The Major Rivers of India:



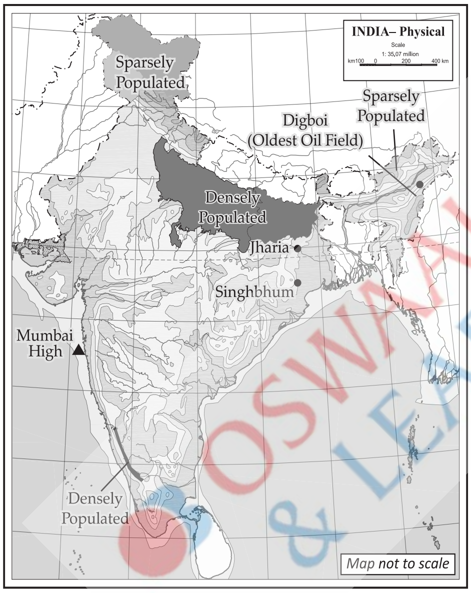
4. Important Passes, Lakes and Gulfs of India:



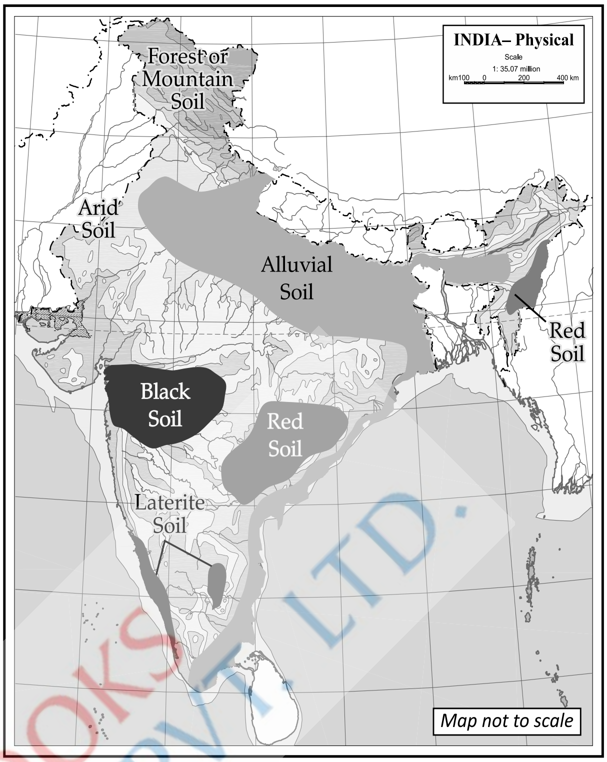
5. Monsoon Winds and Important Latitude and Longitude of India:



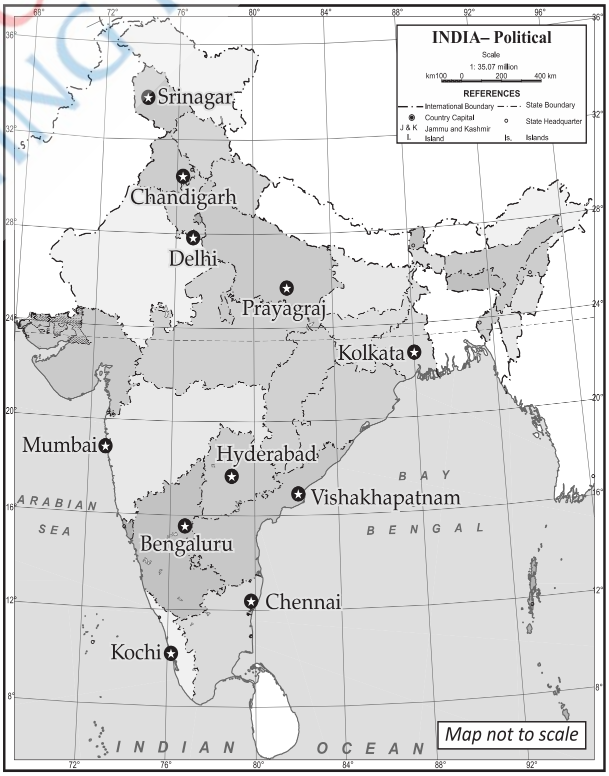
6. Iron-ore Mines, Oil fields, Coalfields and different population density zones in India:



7. Major Soils in India:



8. Major cities in India:



Part-II: Geography of India

CHAPTER-3
Climate

Learning Objectives

The students will be able to:
- Explain the difference between climate and weather.
- Familiarise themselves with the elements of weather and climate.
- Recognise different local winds of the country and their significance.
- Explain the factors that influence the climate of India.
- Highlight the mechanism of the Indian monsoon and identify the types of seasons in India.

TOPIC-1: DISTRIBUTION OF TEMPERATURE, RAINFALL, AND WINDS IN SUMMER AND WINTER

Concepts Covered: Different Types of Climates, Range of Temperatures, Rain-shadow Areas, Loo, Kalbaisakhi, Mango Showers

Revision Notes

Introduction

➢ Climate refers to the sum total of weather conditions and variations over a large area for a long time (more than thirty years). Weather refers to the state of the atmosphere over an area at any given time.

➢ The world is divided into different climatic regions. India, located in Asia, experiences tropical monsoon climate.

➢ Temperature and precipitation vary from region to region and season to season due to India's vast size and diverse topography.

➢ The difference between the highest and lowest temperature in a year is the annual **range of temperature**.

➢ The elements of weather and climate are the same. They are as follows:

(i) Temperature **(ii)** Atmospheric pressure
(iii) Wind **(iv)** Humidity
(v) Precipitation

➢ The elements for a place can be represented using diagrams called climographs or climatographs. Climographs or climatographs show average monthly values of maximum temperature, minimum temperature, and rainfall for a given place.

➢ The northern part of India, which lies beyond the Tropic of Cancer (23½° N) experiences a continental type of climate, characterised by extremely hot summers and cold winters.

Climate of India
Scan Me!

➢ The northern part of the country is situated far from the influence of the sea.

➢ The southern part of India lying in the tropical region between 8°N and 23½°N experiences moderate temperatures. Regions close to the coasts experience an **equable climate.**

➢ Due to the proximity of the sea, the southern coastal regions experience an equable, or maritime, or oceanic type of climate.

➢ There are noticeable both the type and the amount of rainfall received by the region.

➢ In the northern part of India, rainfall mainly brings relief and is orographic in nature, due to the presence of mountain ranges. In the upper parts of the Himalayas, it mostly occurs in solid form as snowfall, while in the rest of the country, it falls as rain.

➢ The moisture-laden winds in summer blow from the sea and to the land and strike the **windward side** of the mountain, causing heavy rainfall, while the other side of the mountain, called the **leeward side**, receives scanty rainfall.

➢ Rainfall is erratic, uncertain, and unpredictable due to its variation time. Thus, it sometimes causes floods or droughts.

➢ In India, rainfall occurs mainly in during summer.

➢ During summer, moisture-laden winds blow from the Arabian Sea and bring heavy rainfall to the western, the central, and the northern regions due to the presence of the Western Ghats and the lofty Himalayas.

➢ The eastern coastal region receives scanty rainfall during summer but heavy rainfall in winters due to the north-east monsoons.

➢ Besides the summer monsoon, the north-western region also receives rain in winter due to **Western Disturbances** that originate over the Mediterranean Sea and blow towards India.

➢ Rainfall is also unevenly distributed. Some regions like Rajasthan receive less than 50 cm of rainfall, the central part and the Deccan Plateau receive low rainfall of about 80 cm while places like Mawsynram near Cherrapunji receive around 2,500 cm of rainfall.

➢ The distribution of rainfall depends on the relief of the land, the direction of the winds from the sea and the path of the cyclonic winds.

➢ Due to the increasing temperatures in summer, the winds become hot and dry and affect local areas severely. Loo is a local wind that blows in the northern regions of the country during May and June and causes heatstroke. It is a gusty, hot, and dry wind.

➢ Similarly, Kalbaisakhi is a local wind that blows in states of West Bengal and Assam in the month of April, which is accompanied by thunderstorms and heavy rainfall.

➢ This rain is beneficial for the growth of tea in Assam and for rice and jute in West Bengal.

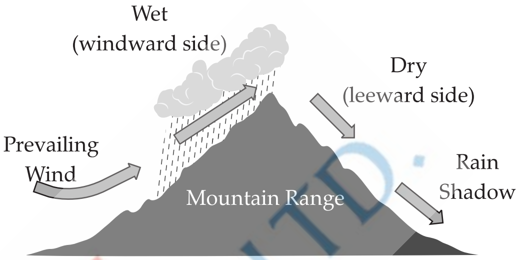
➢ Mango showers (also called cherry blossoms) are local winds that blow in Kerala during the summer months, especially in May and June. This wind helps in the growth of mango, coffee, and tea.

➢ In winter, the winds blow from the land towards the sea in a north-east direction. These winds are cold and dry and move slowly.

Key Terms

➢ **Equable Climate:** It is experienced near and along the coasts due to the influence of the sea on summers and winters which moderates both summer heat and winter chill. There is little difference in the temperature throughout the year. E.g., cities such as Mumbai and Chennai.

➢ **Windward Side:** The side of the mountain facing the moisture-laden winds that receive heavy rainfall after striking the mountain, e.g., the western side of the Western Ghats.



➢ **Leeward Side or Rain Shadow Area:** The leeward side of the mountain which receives very little rainfall. It is also the rain shadow area, e.g., Deccan Plateau.

➢ **Western Disturbances:** The temperate cyclones originating over the Mediterranean Sea, which blow towards India and brings rainfall to its north-western parts.

TOPIC-2: FACTORS AFFECTING CLIMATE AND SEASONS, MONSOON AND ITS MECHANISM

Concepts Covered: October Heat, Western Disturbances, Jet Streams, El Nino, Seasons of India, Features of rainfall in India

Revision Notes

➢ India has diverse climatic conditions.

➢ A number of factors influence the climate of India:

(i) Monsoon winds

(ii) Latitude

(iii) Relief

(iv) Altitude

(v) Western Disturbances

(vi) Distance from the sea

(vii) Air currents

(viii) Ocean currents

➢ The Himalayas act as a climatic barrier. They prevent the south-west, moisture-laden winds from crossing over them, resulting in heavy rainfall in the across the Indian subcontinent. It also protects the Indian region from the cold Siberian winds.

➢ The monsoon winds bring summer rainfall across the whole of South Asia. The south-west monsoon winds move from the Arabian Sea and the Bay of Bengal to the low-pressure areas of north and north-west India. **[2024]**

➢ These winds bring heavy rainfall during summer, from June to September. In October, as it withdraws and retreats, it picks up moisture from the Bay of Bengal and brings considerable rainfall on the eastern coastal plain of India. The winds blowing over the land are cold and dry.

➢ The important line of latitude, the Tropic of Cancer, passes through the middle of India, dividing it into two zones: the **Temperate Zone** (northern part) and the **Tropical Zone** (southern part). The southern tropical zone remains warm throughout the year and experiences virtually no winter season. **[2024]**

➢ The regions north of the Tropic of Cancer never experience the overhead sun, while all the places to its south experience it twice a year.

➢ Relief is also an important factor influencing the climate of India. The Western Ghats act as a barrier in the way of the south-west monsoon winds blowing from the Arabian Sea. This results in heavy rainfall on the Western Coastal Plains, which lie on the windward side.

➢ The Himalayas in the extreme north prevent the moisture-laden winds from crossing them, causing heavy rainfall in the major portion of the Indian subcontinent. The Aravalli Range in the west runs parallel to the southwest monsoon winds and thus does not cause any rainfall in that region.

➢ The higher the altitude, the lower the temperature. This is due to normal lapse rate; for every rise of 166 metres there is a decrease of 1 °C of temperature. Therefore, the mountains are cooler than the plains.

➢ The three major water bodies surrounding the Indian subcontinent: the Indian Ocean, the Arabian Sea and the Bay of Bengal have a significant influence and impact on the climate. They are a major source of rainfall in India. Also, due to its proximity; the coastal regions experience moderate climatic conditions.

➢ During the winter season, due to the Western Disturbances over the Mediterranean Sea, cause westerly cyclonic winds to blow towards India, bringing the north-western part of the country under their influence. These cyclonic winds bring rainfall to the north-western part of India.

➢ Due to their distance from the sea, areas far away from the influence of the sea experience a continental type of climate—characterised by hot summers and cold winters. Coastal areas, on the other hand, experience equable or maritime climate due to their proximity to the sea. Land and sea breezes are caused due to the differential heating and cooling of land and sea.

➢ Air currents that determine the arrival and departure of the monsoons are known as jet streams. The Westerly Jet Stream prevails over the Northern Plains, while the Easterly Jet Stream steers the Tropical Depression over India.

➢ **El Niño** is a warm ocean current in the Pacific Ocean which increases the surface temperature of the sea and affects the Monsoon winds in the Indian Ocean. It often leads to weak monsoon rainfall, causing drought-like conditions in the Indian subcontinent.

➢ Monsoons are periodic seasonal winds caused due to differential heating and cooling of land and sea.

➢ Monsoons are divided into two wind systems:

(i) The Summer Monsoon Wind System

(ii) The Winter Monsoon Wind System **[2024]**

➢ In India, based on monsoon variations, the year may be categorised into four main seasons. They are:

(i) The Hot Summer Season (March to May)

(ii) The Hot and Wet or Rainy Season (June to September)

(iii) The Retreating South-west Monsoon (October–November)

(iv) The Cold and Dry Winter Season (December–February)

➢ The hot summer season begins in March as the sun starts moving northwards and shines vertically over the Tropic of Cancer. During this period, the temperatures rise and goes up to 48 °C.

➢ Due to the moderating influence of the sea, the heat is less intense in the southern part of India. Plateaus and hills also remain cool due to their elevation.

➢ The rainy season begins in June and continues until September. The differential heating and cooling of land and sea develop intense low pressure over the large landmass and high pressure over the sea.

➢ The moisture-laden monsoon winds enter the Indian mainland from the south-west and bring heavy rainfall, often accompanied by thunder and lightning.

➢ The sudden, violent onset of rainfall in the first week of June is termed the burst of the monsoon.

➢ Kerala is the first state to receive the monsoon showers and the last to see them retreat.

➢ The southwest monsoon is divided into two branches:

(i) The Arabian Sea branch

(ii) The Bay of Bengal branch

➢ After the monsoon, in the month of October, the Southwest Monsoon begins to retreat from the northern part of India and is thus called the **Retreating Monsoon**. **[2024]**

➢ During this period, due to the apparent movement of the Sun, the low-pressure trough is gradually replaced by high pressure.

➢ The retreating winds are dry and slow in their process. The combination of high temperature and humidity gives rise to oppressive weather, termed 'October Heat.'

➢ Due to local variations in heat and moisture, tropical depressions originate in the Bay of Bengal, leading to tropical cyclones.

➢ By the end of November, the winter season begins and continues till March. The temperature decreases from south to north. January is the coldest month.

➢ The north-east winds prevail over the country during the winter season and blow from land to sea. When part of these winds blow over the Bay of Bengal, they pick up moisture from there and strike the Eastern Ghats, thereby shedding heavy rainfall along the coastal plains.

Indian Seasons

Scan Me!

➢ A characteristic feature of the winter season is the inflow of cyclonic winds from the west. These cyclonic winds, caused by low-pressure, originate over the Mediterranean Sea and are known as Western Disturbances.

➢ These winds bring rainfall to the north-western plains of India during the winter season.

➢ The distribution of rainfall is influenced by the following factors:

(i) Pressure conditions

(ii) Direction of relief features

(iii) Direction of the moisture bearing wind

(iv) Cyclonic depression

➢ **Main features of rainfall in India are as follows:**

(i) Rainfall occurs mainly during the three months of the rainy season.

(ii) The rains are mainly of the relief type, i.e., the windward side of the mountain receives more rainfall than the leeward side.

(iii) Less rainfall is received from other sources, such as convectional and cyclonic rainfall.

(iv) Rainfall is generally erratic in nature.

(v) India, being an agrarian country, is dependent on rainfall, which significantly affects its economy.

Key Terms

- **Temperate Zone:** The region beyond the Tropic of Cancer (23½°N), where the sun is never overhead.
- **Tropical Zone:** The region between the Equator and the Tropic of Cancer where the overhead sun is experienced twice a year.
- **Jet Streams:** Air currents that determine the arrival and departure of the monsoon.
- **Monsoon:** Derived from the Arabic word 'Mausim' which means season. These are periodic seasonal winds caused by the differential heating and cooling of land and sea.
- **Burst of the Monsoon:** The sudden violent onset of rainfall in the first week of June accompanied by thunder and lightning.
- **Retreating Monsoon:** The withdrawal of monsoon from the Indian subcontinent in October.

CHAPTER-4
Soil Resources

Learning Objectives

The students will be able to:

- Define soil and explain its significance.
- Identify the major types of soil in India.
- Describe the distribution, composition, and characteristics of different types of soil.
- Define soil erosion and its causes.
- Outline methods preventing and conserving soil from erosion.

TOPIC-1: TYPES OF SOILS, DISTRIBUTION, COMPOSITION AND CHARACTERISTICS

Concepts Covered: Alluvial Soil, Black Soil, Red Soil, Laterite Soil

Revision Notes

- Soil is the uppermost thin layer of the Earth's crust, made up of organic matter, minerals, and weathered rocks, covering the Earth's surface.
- The process of soil formation is known as pedogenesis.
- Natural agents such as temperature changes, running water and wind affect the formation of soil.
- Soil is formed through a process called weathering, where parent rock material breaks down into smaller particles due to physical, chemical, and biological processes.
- **Humus** (decomposed plant and animal remains) is the main constituent of soil. Silica, clay and sand are also important constituents of soil.
- Soil fertility refers to the ability of soil to support plant life.
- The important features of soil fertility are that it contains adequate amount of moisture, is rich in nutrients such as nitrogen, phosphorus, and potassium **(NPK)**, contains organic matter and has sufficient depth to enable plants to grow roots.
- Depending on the process of formation, soil can be categorised into residual (or sedentary) soil and transported soil.
- **Residual** soil is formed 'in-situ' that is, in its original position—through the breakdown of parent rocks; for example, black soil, red soil, and laterite soil. **[2024 Board]**
- Soil that is transported by agents of erosion such as wind and running water is called **transported** soil—for example, alluvial soil.
- Soil is classified into the following types: alluvial soil, black soil, red soil, and laterite soil.

SOIL-Different Types and the Importance of Soil

Scan Me!

Name of the Soil	Formation	Crops Grown	Characteristics	Areas Where the Soils Are Found
1. Alluvial Soil (also called Riverine soil)	Transported soil by the deposition of silts and sediments brought down by rivers.	-Wheat -Rice -Sugarcane -Oil Seeds	**(1)** It is of two types: *Bhangar* (Old Alluvium) and *Khadar* (New Alluvium). **(2)** It is porous and is coarse in the upper region; and fine in the lower region. **(3)** Rich in potash and lime, but deficient in nitrogen, phosphoric acid, and humus (except in the Ganga delta, which is rich in humus). **(4)** The colour of the alluvial soil varies from light grey to ash grey, depending on depth, texture, and maturity.	-Uttar Pradesh -Punjab -Haryana -Jharkhand -Bihar -West Bengal
2. Black Soil (also called Regur soil or Black Cotton soil)	In-situ (residual) soil formed by the weathering of volcanic or igneous rocks.	-Cotton -Sugarcane -Jowar -Wheat -Oil seeds	**(1)** It is fine textured and clayey. **(2)** Its moisture retentive and becomes sticky when wet and crack when dry. **(3)** Rich in lime, magnesium and iron. **(4)** Poor in phosphorous, nitrogen and organic matter. **(5)** These soils are black in colour.	-Maharashtra -Gujarat -Madhya Pradesh -Andhra Pradesh -Karnataka
3. Red Soil	In-situ (residual) soil by the weathering of old hard crystalline and metamorphic rocks.	-Rice -Cotton -Pulses -Can be grown by using fertilisers	(1) It is porous, **friable** and coarse. (2) It does not retain moisture. (3) Rich in iron and potash. (4) Deficient in lime, nitrogen, phosphorous and humus. (5) Responds well to manure or fertilisers. (6) Not prone to waterlogging. (7) It is red in colour due to the presence of iron oxides.	-Tamil Nadu -Goa -Karnataka -Odisha -Meghalaya
4. Laterite Soil	Formed in-situ as a result of leaching under typical monsoonal conditions—high temperature and heavy rainfall, with alternating wet and dry spells.	-Tea -Coffee -Rubber -Cashew -Tapioca -Millets	**(1)** Highly acidic in nature. (2) It is porous and coarse. **(3)** Rich in iron. **(4)** Poor in silica, lime, nitrogen and humus. **(5)** Reddish in colour.	-Summits of Eastern and Western Ghats -Andhra Pradesh -Tamil Nadu -West Bengal -Odisha -Assam

Bhangar Soil	Khadar Soil
1. Older alluvium soil.	**1.** Newer alluvium soil.
2. It is found above the flood levels of rivers and is of a terrace-like structure.	**2.** It lies below the flood levels.
3. This soil is dark in colour and composed of lime nodules or *kankar* and has a coarse material.	**3.** It is light in colour and is composed of newer deposits like silt and clay, so it is fine-grained.
4. It is less fertile.	**4.** It is very fertile.

Key Terms

- **Parent Rock:** The original rock from which the soil is formed, mainly used in the context of soil formation.
- **Humus:** Organic matter in the soil formed by the decomposition of plants and animals remains.
- **NPK:** Abbreviation for Nitrogen (N), Phosphorus (P), and Potassium (K)—essential nutrients that enhance soil fertility.
- **Bhangar Soil:** Older alluvium soil found above the flood levels of rivers.

- **Khadar Soil:** Newer alluvium soil found below the flood levels.
- **Friable Soil:** Soil that can be further broken into smaller fragments.
- **Residual Soil:** Formed 'in-situ' meaning in its original position, by the weathering of parent rocks, e.g., black soil, red soil, laterite oil.
- **Transported Soil:** Soil carried from its place of origin by agents of erosion such as water and wind.

TOPIC-2: SOIL EROSION: CAUSES, PREVENTION AND CONSERVATION

Concepts Covered: Types of Erosions: Gully, Sheet, Rill, Stream Bank, Shore and Slip, Leaching, Shelter Belts, Soil Conservation Measures

Revision Notes

- Soil erosion refers to the removal of the top layer of fertile soil caused by the action of wind, water, or human activities.
- Soil erosion caused by running water are as follows: **[2024 Board]**

(i) Sheet erosion: A large quantity of water flows in the form of sheets that removes the thin layer of topsoil along with vegetation covers or due to heavy rainfall. This results in soil erosion over extensive areas.

Causes of Soil Erosion

Scan Me!

(ii) Rill erosion: As a result of prolonged sheet erosion in the second stage, finger-shaped grooves or rills are formed over a large area, which is known as rill erosion.

(iii) Gully erosion: The removal of clayey soil along drainage lines by running water and formed deep channels that erodes soil—mainly on hillsides—after deforestation and overgrazing.

(iv) Stream bank erosion: Streams and rivers change their course by cutting into their banks, and depositing silt.

(v) Shore erosion: Powerful tidal waves erode and damage coastal areas.

(vi) Slip erosion: During heavy rains, when the water is unable to penetrate into the soil due to underlying impervious rocks, causes the heavy moisture-laden soil slides down from the steep land. This process is called slip erosion. Sometimes, it results in a landslide.

- Wind erosion takes place where there is little or no vegetation due to high velocity movement.
- When wind moves soil particles of 0.1–0.5 mm in size in bouncing or hopping manner, the process is called saltation. Particles larger than 0.5 mm that move by rolling are referred to as soil creep.
- In the process of Saltation, particles smaller than 0.1 mm, or the finer ones, gets separated and transported to the long distances until the wind speed decreases. This is called **suspension.**
- Large-scale deforestation by humans has been witnessed in the Outer Himalayas, the Western and the Eastern Ghats for various land uses, such as constructing railway lines, roads, and buildings.
- Uncontrolled grazing of domestic animals is a major factor for contributing to sheet, gully, and rill erosion.
- In India, soil erosion occurs due to the following causes:

(i) Increasing population
(ii) Erratic nature of rainfall
(iii) Overgrazing by domestic animals
(iv) Faulty farming techniques
(v) Topography of the region
(vi) Deforestation

- The Indian states highly prone in soil erosion include Rajasthan, Madhya Pradesh, Maharashtra, Uttar Pradesh, Gujarat, Andhra Pradesh and Karnataka.
- Soil is an essential resource for agriculture. It recycles organic waste, provides nutrients to the vegetation grown. It also influences the global climate. Therefore, **soil conservation** is necessary for the sustainable development of the country.
- Soil conservation refers to the efforts made by humans to prevent loss of soil from erosion or reduced fertility due to overuse.
- Measures to prevent soil erosion:

(i) Afforestation and re-afforestation: Afforestation refers to the establishment of forests or stands of trees in areas where there was no previous tree cover. Re-afforestation involves planting trees to replace those cut down, typically in a 2:1 ratio. The roots of trees and plants bind the soil, slow down running water, enabling water to get absorbed in the soil. Trees also reduce wind force, preventing the blowing away-of soil particles.

(ii) Restricted grazing of animals: Livestock should be rotated across different pastures and fodder crops should be grown in large quantity.

(iii) Construction of dams: In addition to slowing the speed of river water and controlling river floods, construction of dams also saves soil erosion.

(iv) Proper farming techniques:

(a) Crop rotation: This is a farming method in which farmers grow crops cyclically to minimise depletion of specific soil nutrients, thereby maintaining land fertility.

(b) Contour ploughing: Contours act like bunds. Ploughing along them across the slope prevents soil from being washed away by rainwater or by surface run-off.

(c) Terrace farming: Hill slopes are cut into a number of terraces with horizontal top and steep slopes on the back and front. It is one of the oldest and most effective methods of soil conservation.

(d) Strip cropping: In this system, large fields are divided into strips and grass is grown between the crops to reduces wind velocity and protect the topsoil from erosion.

(e) Shelter belts: When trees are planted along sand dunes, these rows are called shelter belts. They help in stabilising sand dunes and prevent the desert from encroaching on land used for farming.

➢ In India, the government has taken specific measures for soil conservation:

(i) Scheme for the reclamation and development of ravine areas.

(ii) Scheme for control of shifting cultivation.

(iii) The National Project on Development and Use of Biofertilisers and National Project on Quality Control implemented.

(iv) Rainwater harvesting

Key Terms

➢ **Gully Erosion:** Removal of clayey soil along drainage lines by running water and forming deep channels, and causing erosion mostly on hillsides after deforestation and overgrazing.

➢ **Leaching:** Due to high rainfall, lime and silica are leached away from soil leaving behind iron oxide and aluminium compound, which make the soil acidic.

➢ **Sheet Erosion:** It is the slow removal of a thin layer of soil over extensive areas by rainwater, especially in areas with sparse or no vegetation.

➢ **Soil Conservation:** It refers to efforts made by humans to prevent soil loss due to erosion or reduced fertility caused by over usage

➢ **Shelter Belts:** Rows of trees are planted to reduce wind speed and prevent soil erosion.

➢ **Reafforestation:** The practice of planting trees to replace those cut down, typically in a 2:1 ratio.

CHAPTER-5
Natural Vegetation

Learning Objectives

Students will be able to:

- Understand the importance of forests.
- Identify different types of vegetation.
- Understand the distribution and correlation with the environment.
- Relate to forest conservation, afforestation and reafforestation.

TOPIC-1: IMPORTANCE OF FORESTS, TYPES OF VEGETATION

Concepts Covered: . Five Major Vegetation, Regions, Canopy, Decoction, Mangrove

Revision Notes

Introduction

➢ Natural vegetation refers to the plant cover that has not been disturbed over a long time and has grown naturally, depending on the climate and soil conditions of that area. On the other hand, vegetation refers to the plants or crops planted or cultivated by humans.

➢ Grasses, shrubs, and trees constitute the natural vegetation of an area.

➢ **Flora** refers to the plants of a particular region or period, listed by species and considered as a whole, e.g., 4,000 species of plants in the Eastern Himalayas.

➢ A forest is defined as an ecosystem or assemblage of ecosystems dominated by trees and other woody vegetation.

➢ It also refers to a large tract of land covered with trees, and accompanied by undergrowth of shrubs, herbs, and sustaining thousands of life forms. In simple terms, a forest is an area with a high density of trees.

➢ Forests are one of the most important natural resources. Natural resources are all the gifts of nature such as air, water, minerals, trees, etc.

➢ In our day-to-day life, forests play an important role. They are as follows:

(i) Forests are highly productive since they provide us with fruits, leaves, roots, tubers of plants, wood for furniture, wood pulp for manufacturing paper.

(ii) Forest animals provide food in the form of meat to the tribals living in the forest.

(iii) Forests also provide products such as herbs, edible plants, fibres, etc.

(iv) Bamboos in the forests are also a means of livelihood for tribal people, as they make mats, ropes, baskets, etc., from them.

(v) Forests protect against flooding by controlling the flow of water and soil erosion, which leads to silting.

(vi) The roots of trees in the forest bind the soil firmly, which prevents soil erosion.

(vii) Trees absorb carbon dioxide and release oxygen, which is used by human beings and animals.

(viii) The trees also help in regulate the water cycle.

(ix) The forests provide an abode and food for the wild animals.

(x) Forests also provide recreation to humans through various wildlife sanctuaries, national parks, etc.

➢ India has a great variety of plant species. India's location in tropical latitudes supports this variety of natural vegetation throughout the country. This diversity is due to the following factors:

(i) Temperature

(ii) Sunlight

(iii) Soil

(iv) Relief, topography, and location

(v) Precipitation or rainfall

➢ India can be divided into five major vegetation regions. They are:

(i) Tropical evergreen forests or rainforests

(ii) Tropical deciduous forests or monsoon forests

(iii) Tropical desert or thorn forests

(iv) Littoral swampy forests

(v) Montane or mountain vegetation

➢ **Tropical Evergreen or Rainforests :** **[2024 Board]**

(i) Temperature between 25°C and 27°C.

(ii) The amount of rainfall is more than 200 cm.

(iii) Found on the western slopes of the Western Ghats, hills of the North-Eastern region and in the island groups of Lakshadweep, the Andaman and Nicobar Islands.

(iv) The characteristic features are:

1. They are dense and have a variety of trees and shrubs.

2. Trees reach a height of 60 m or more.

3. Due to thick the canopy of trees, herbs and grasses cannot grow.

4. These forests do not have any fixed period of time for shedding of leaves.

5. These forests produce hardwood trees. Not all trees shed their leaves at the same time, which is why these forests always appear green.

(v) The main varieties of trees include ebony, rosewood, mahogany, toon, chaplas, gurjan, sissoo, telsur, rubber, shisham, cinchona, etc.

➢ **Tropical Deciduous Forests or Monsoon Forests :** **[2024 Board]**

➢ Tropical deciduous forests cover most of the forested area in India. As these forests depend on the monsoon, they are also known as monsoon forests.

➢ On the basis of water availability, these forests are categorised into two types :

(a) Moist deciduous forests

(b) Dry deciduous forests

(a) Moist deciduous forests:

(i) Rainfall between 100 cm and 200 cm.

(ii) Temperature between 24°C and 27°C.

(iii) Found in regions like Jharkhand, Chhattisgarh, Western Ghats and in some parts of the Andaman and Nicobar Islands.

(iv) The main characteristic features are:

1. The trees shed their leaves for six to eight weeks during spring and early summer.

2. These forests are the most commercially exploited in India.

3. These forests often have trees growing in pure stands.

4. They provide valuable timber and other forest products.

(v) The main varieties of tress found here are sal, teak, arjun, mahua, shisham, palas, mulberry, semul and sandalwood.

(b) Dry deciduous forests:

(i) Rainfall ranges between 70 cm and 100 cm.

(ii) The annual temperature is between 23°C and 27°C.

(iii) Found in the semi-arid areas of the country like the rain-shadow regions of Western Ghats, and the Aravallis.

(iv) The main characteristic features are:

1. These forests adapt to the climatic condition of that particular region.

2. Forests in wetter regions are moist deciduous, while in drier areas, they consist of thorny bushes.

3. Teak and other trees are found in the northern part of India and in the region of higher rainfall in the Peninsular plateau, along with patches of grass.

4. The trees of these forests completely shed their leaves during a definite period.

(v) The main trees found here include teak, sal, tendu, amaltas, rosewood, bel, khair, etc.

(vi) These forests provide timber, fruits and other valuable products. Parts of these forests are also cleared for agricultural activities.

➢ **Tropical Desert Forests :**

(i) Rainfall is less than 50 cm.

(ii) The temperature ranges between 25°C and 27°C.

(iii) These trees are found in semi-arid regions such as the rain-shadow areas of the Western Ghats, parts of Rajasthan, Gujarat, Haryana, Punjab, Madhya Pradesh, and other areas of the Deccan Plateau.

(iv) The main characteristic features are:

1. Due to scanty rainfall, the vegetation is xerophytic, meaning the trees and plants have adapted to survive in dry environments.

2. The trees are stunted and are accompanied by large patches of coarse grasses.

3. The trees have very small leaves due to a lack of moisture.

4. They have scrub vegetation, i.e., thorny bushes.

5. The stems of these plants fleshy, enabling them to conserve water for a longer period.

The Importance of Forests

Scan Me!

(v) The important trees of these forests include Date Palm, Khair, Ber, Babool, Neem, Kanju, Khejri, Cacti, and Acacia.

(vi) The trees are economically valuable. Ber fruit is useful for making pickles or beverages and is also eaten raw. The timber is hard and durable. Date Palm is consumed raw and used as an astringent. Neem leaves and bark are used in various health-related and beauty products.

➢ **Littoral or Swampy Forests :**

(i) Rainfall is more than 200 cm.

(ii) Temperature ranges between 26°C and 29°C.

(iii) These forests grow along the sea coasts, in wet marshy areas, river deltas, tidal zones swampy regions. They are known by different names such as tidal forests, mangroves, and Sundarbans. The forests found in and around deltas are known as mangrove forests, a type of swampy forest. When a tree or a shrub grows in a saline coastal habitat or brackish water, it is known as mangrove.

(iv) Found in areas around the deltas of the major rivers along the eastern coasts, including the saline swamps of the Sunderbans in West Bengal, along the coasts of Odisha and Andhra Pradesh

(v) The main characteristic features are :

1. These forests are dense, evergreen, and of varying heights.

2. The trees have long roots submerged underwater and possess pores that help them to breathe during high tide.

3. The tidal trees yield hardwood that is strong and durable and used for making boats and boxes.

4. The most important tree is the Sundari tree in the Ganga Delta, from which the word 'Sundarbans' is derived.

5. The Indian tidal or mangrove forests are considered the largest mangrove forests in the world.

6. These mangrove forests are among the most productive and bio-diverse wetlands on Earth.

7. These forests have well adapted to the muddy, shifting, and saline conditions, and grow in brackish marshy areas, mudflats, and estuaries along the coast.

(vi) The important trees incude the Sundari, Gurjan, Hintal, Keora, Rhizophora, Amur, Bhara, canes, etc.

(vii) Littoral forests are economically important too. Mangrove trees are used for fuel, while Sundari trees provide hard, durable timber used for making boats and boxes.

➢ **Montane or Mountain Vegetation :**

(i) Rainfall between 100 cm and 300 cm.

(ii) Temperature ranges between 12°C and 13°C and the humidity varies between 56% and 65%.

(iii) These forests are found in the mountain regions having at altitude between 1,000 m and 4,000 m.

(iv) Mountain vegetation is divided into three types according to the altitude. At the highest altitude, seasonally flowering trees are found. In mid-altitude regions, coniferous trees like pine, deodar, fir grow, while at foothills, mixed forests are found.

(v) The main characteristic features are :

1. These forests are mainly found in the higher hills of Tamil Nadu and Kerala, and in the Eastern Himalayan region including the hills of West Bengal, Assam, Arunachal Pradesh, Sikkim and Nagaland.

2. At lower altitudes, these forests are evergreen forests where the trees are mostly branchy, with broad, dense and rounded leaves.

3. At higher altitudes, the slopes are covered with mosses, lichens, ferns and creepers.

4. It provides fine wood, which is widely used for construction and for making railway sleepers.

5. These forests occur in the temperate zone of the Himalayas at altitude between 1,500 and 3,300 metres, where the annual rainfall varies from 150 cm to 250 cm.

6. The Himalayan moist temperate forests cover the entire mountain range in Kashmir, Himachal Pradesh, Uttarakhand, Darjeeling, and Sikkim.

7. In the peninsular region, these forests are found at an altitude of 1,500 m.

(vi) The important trees of the Montane forests are Indian Chestnut, Birch, Plum, Cinchona, Litsea, Magnolia, Pine, Oak, Hemlock, etc.

➢ **Important trees and their uses:**

(i) Ebony: It is used for ornamental carving, making musical instruments, sports goods, piano keys, etc.

(ii) Rosewood: It is used for making furniture, doors, windows, musical instruments, carvings, etc. Due to its durability, it is often used in the martial art weaponry. It also has medicinal uses; for example, it is effective against acne.

(iii) Teak: Its high oil content makes it highly weather-resistant. It is a valuable timber that is heavy, tough, and durable. It is widely used for making furniture, houses, and ships. The bark of the teak tree is considered astringent and is used to treat bronchitis.

(iv) Mahogany: It is very durable and is mainly used for crafting cabinets and furniture, musical instruments. It resists wood rot and is also used in boat construction.

(v) Sal: It is hard, tough, and heavy wood. It is mainly used for making doors, windows, bridge railings, beams, railway sleepers.

(vi) Sandalwood: It is widely used in the cosmetic industry. It is used as a perfume and its oil is used in make aromatic substances. It is also used for making ornamental objects like statues.

(vii) Ber: It is a hard, tough, and durable wood. The fruit is eaten raw as it is rich in Vitamin C and is also used to make pickle and beverage. It is used for making agricultural implements, boat ribs, charcoal, etc.

(viii) Babool: It has high medicinal value. It is a source of gum and is used as an emulsifier. The twigs and barks are chewed to prevent Vitamin C deficiency. Its wood is used as firewood and charcoal, and for boat building.

(ix) Palas: Its leaves are used to rear lac worms.

(x) Neem: The leaves, bark and roots of this tree have medicinal properties. It helps to cure skin infections, ulcers, allergies, etc. Its oil is used to manufacture beauty and health products.

(xi) Cinchona: Cinchona is used to increase appetite. It is also used for blood vessel disorders including haemorrhoids, varicose veins and leg cramps. It contains quinine, a chemical used to treat malaria. Cinchona bark stimulates saliva and gastric juices.

(xii) Deodar: Its wood is strong, durable, and tough. It is mainly used for construction and for making railway sleepers.

(xiii) Sundari: It is a hard, durable, and strong tree. It is mainly used for making boats and for house construction.

Key Terms

➢ **Natural Resources:** Naturally occurring materials that satisfy human needs, provided that the resources are technologically accessible, culturally acceptable and economically affordable. Examples include are sunlight, air, water, coal, minerals, etc.

➢ **Flora:** Refers to the plant life of a particular region or period.

➢ **Forest:** An ecosystem or assemblage of ecosystems dominated by trees and other woody vegetation.

➢ **Canopy:** The uppermost leafy layer of a forest, formed by the crowns of trees and climbers.

➢ **Mangrove :** A Mangrove is a tree or shrub that grows in coastal saline or brackish water.

➢ **Montane:** Refers to areas or organisms inhabiting mountainous regions.

TOPIC-2: DISTRIBUTION AND CORRELATION WITH THEIR ENVIRONMENT

Concepts Covered: .Global Carbon Cycle, Bio-diverse Regions, Ecosystem

Revision Notes

➢ India is a vast and diverse country and is one of the seventeen **mega-biodiverse regions** of the world.

➢ India has a large variety of forest vegetation and includes many species of hardwoods like Sal, Teak, etc.

➢ Indian forest types include Tropical Evergreen, Tropical Deciduous, Tropical Thorn Forests, Coastal Forests, and Montane Vegetation.

➢ These forests support a variety of ecosystems with diverse flora and fauna.

➢ An **ecosystem** is a geographic area where biotic elements such as plants, animals, and other organisms, as well as abiotic elements such as weather and landscape, interact to form a community. Fauna refers to the animal life found in a particular region, period or environment.

➢ Forests are one of the renewable resources. Renewable resources are those that have the potential to be replenished over time by natural processes.

➢ Forests have an interrelationship with the environment.

➢ Forests plays an important role in modifying climate and weather. They help reduce the mean annual temperature.

➢ They help in lowering maximum temperature and raises the minimum temperature.

➢ Forests increase the precipitation of any area and help controls humidity.

➢ By reducing surface runoff, forest vegetation helps increase the amount of water that percolates into a soil.

➢ Forests help prevent and control soil erosion. They contribute to soil firmness and stability by checking the velocity of wind and by reducing surface runoff.

➢ Forests help reduce floods in both hilly regions and plains by reducing the volume of surface runoff.

➢ Forests have the potential to alter the climatic condition of a region through their influence on the global carbon cycle.

➢ Trees and other green plants produce oxygen and absorb carbon dioxide.

➢ Forests also help control major types of pollution such as water, air, and noise pollution.

➢ Forests also influence the lives of many terrestrial animals.

➢ They act as a source of food and shelter for many wild animals.

➢ Forests improve soil nutrition through the addition of organic matter, leaf decomposition, and root penetration.

➢ Forests provide both major and minor products to humans.

➢ Major forest products include both types of woods i.e., hardwoods and softwoods.

➢ Minor forest products include all non-timber items obtainable from the forests, i.e., leaves, bamboos, grasses, fibres, spices, edible products, etc.

➢ Forests also contribute to the economic development of our country. Forest products hold high commercial value serving as raw material for several industries. Various forest-related activities provide employment to millions of people.

Key Terms

➢ **Biodiverse Regions :** These refer to areas containing a wide variety of plant and animal species.

➢ **Ecosystem:** A community of living organisms together with the non-living components of their environment such as air, water and soil, interacting as a system.

➢ **Renewable resource:** A type of resource that has potential to be replaced over a time by natural process.

➢ **Global Carbon Cycle :** It describes the exchanges of carbon between the Earth's atmosphere, oceans,land and fossil fuels..

TOPIC-3: FORENST CONSERVATION

Concepts Covered: Deforestation, Green House Effects, Green House Effects, Joint Forest Management, Social-Forestry, Agro-Forestry

Revision Notes

➢ Man, in order to satisfy his own needs, indiscriminately cleared forested areas, leading to a decline in forest cover.

➢ **Some reasons for the decline in forest cover are:**

(i) Increasing population

(ii) Conversion of forests into pasture lands.

(iii) Overgrazing by animals.

(iv) Increasing demand for timber.

(v) Construction of **multipurpose river valley projects.**

➢ All these factors have adverse effects, such as a decline in forest productivity, soil erosion and floods, reduced rainfall leading to droughts, and an increase in the greenhouse effect in the atmosphere.

➢ **Forest conservation** is the need of the hour to combat issues arising from deforestation. Forest conservation does not mean denial of use, but rather the proper use of forest resources without causing any adverse effects on our economy or environment. **[2024 Board]**

➢ To conserve forests, several practices must be undertaken by the people. Some of these practices are :

(i) Practise **afforestation** under *'Van Mahotsav'* movement or a Forest festival.

(ii) Discourage the cutting of trees and ensure the planting of two saplings in lieu of felling tree, i.e., practising reforestation.

(iii) Ban the practice of shifting cultivation prevalent among tribal communities.

(iv) Grow trees around Iron and Steel industries and other industrial units.

(v) To use non-conventional or renewable sources of energy such as solar and tidal energy.

(vi) Establish corridors forests to facilitate migration of wild animals.

(vii) Enforce strict conservation measures and laws to check **deforestation**.

(viii) Encourage people's participation in planning, decision making and implementation of forest conservation programmes.

➢ The Ministry of Environment, Forest and Climate Change, a nodal agency in the administrative structure of the Central Government, plans, promote, coordinates, and oversee the implementation of environmental and forestry programmes. This ministry is also the nodal agency in the country for the United Nations Environment Programme(UNEP).

➢ In 1988, the Forest Conservation Act of 1980 was amended to implement stricter conservation measures.

➢ India is one of the few countries to have had Forest Policy since 1894. This policy was revised time to time, in 1952 and again in 1988. India launched its National Forest Policy in 1988 and, which led to a programme named **Joint Forest Management (JFM).**

➢ To focus on the protection, conservation, and development of forests, this policy proposed that specific villages, in association with the Forest Department, would manage designated forest blocks.

➢ **The other objectives of this policy are:**

(i) Restoration of ecological balance.

(ii) Maintenance of environmental stability.

(iii) Preserving natural forests with a wide variety of flora and fauna.

(iv) Checking soil erosion and the expansion of sand dunes in the desert areas.

(v) Increasing forest cover through extensive afforestation and Social Forestry Programmes.

(vi) Meeting the basic needs of the rural and tribal people by providing fuelwood, minor forest product, fodder, etc.

(vii) Encouraging the efficient use of forest produce and optimal substitution of wood.

(viii) Organising people's movements in order to achieve these objectives, with special involvement of women.

➢ In 1970, India declared three major objectives for forestry development:

(i) To reduce soil erosion and flooding.

(ii) To meet the growing needs of the domestic wood products industries.

(iii) To meet the needs of the rural population for fuelwood, fodder, small timber, and miscellaneous forest produce.

➢ Socialforestryreferstothemanagementandprotection of forests and afforestation of barren and deforested lands, with the aim of promoting environmental, social and rural development. It is also known as *'forestry of the people, by the people and for the people'*.

➢ It also aims to encourage plantations by the common people to meet the growing demand for timber, fuel wood, fodder, etc., thereby reducing pressure on traditional forest areas.

➢ Since India has a predominantly rural population and largely depends on fuelwood and other biomass for their cooking and heating, the need for a social forestry scheme was felt.

➢ Another scheme under the Social Forestry Programme is involves raising trees on community land and not on private land.

➢ The Social Forestry Scheme can be categorised into groups such as farm forestry, community forestry, extension forestry and agro-forestry.

➢ Through these schemes aims to benefit entire community and rather than individuals and takes the responsibility of providing seedlings and fertilisers. As a result the community takes the responsibility for protecting the trees.

➢ **The objectives of Social Forestry are :**

(i) To improve the environment for to protect farming from adverse climatic conditions.

(ii) To increase the supply of fuelwood, fodder for livestock, small timber for rural housing, minor forest products for local industries.

(iii) To enhance the natural scenic beauty of the area.

(iv) To provide employment opportunities to unskilled workers.

(v) To utilise rehabilitated or restored land for better production.

(vi) To create forests as recreational centres for the public.

(vii) To develop local cottage industries by providing raw materials.

(viii) To encourage afforestation as a means of conserving soil and water.

(ix) Its mission is to convert urban and industrial areas into green belts.

(x) To enhance the quality of life and raise the standard of living of both the urban and the rural populations.

➢ Agro-forestry may be defined as a sustainable land use system that maintains or increases the total yield by combining food crops together with forest tree and livestock ranching on the same unit of land, using management practices that take care of the social and cultural characteristics of the local people and the economic and ecological conditions of the local area.

➢ It is a part of Social Forestry.

➢ In agro-forestry, forest trees are grown along with agricultural crops on the same piece of land.

➢ It is an intermediate stage between forestry and agriculture.

➢ It aims to conserve soil to improve the growth of forest products and agricultural crops together.

➢ **Objectives of Agro-Forestry :**

(i) To discourage deforestation and reduce pressure on forests for obtaining fuelwoods and other forest produce.

(ii) To restore soil fertility by preventing soil erosion.

(iii) To maintain ecological balance through proper utilisation of farm resources.

(iv) To reduce poverty by encouraging increased production of wood and other tree products for self-consumption and sale.

➢ Most forest resources are owned by the Government of India.

➢ For administrative purposes, forests are classified into three types: **Reserved Forests (RF)**, the **Protected Forests (PF)**, and the **Unclassed Forests (UF)**.

➢ Reserved and Protected Forests are collectively referred to 'Permanent Forest Estates'.

Key Terms

➢ **Deforestation :** The clearing or thinning of forests by humans.

➢ **Multipurpose River Valley Projects:** These are major river valley projects designed to meet basic needs such as irrigation for agriculture, electricity for Industries and flood control.

➢ **Greenhouse Effect:** The warming of the Earth's temperature caused by gases in the atmosphere that trap solar energy. These heat-trapping gases are called greenhouse gases. The most common greenhouse gases are water vapour, carbon dioxide, and methane.

➢ **Joint Forest Management (JFM):** The official and widely used term in India for partnerships in Forest movement involving both the State Forest Departments and local communities.

➢ **Afforestation:** The practice of planting trees in an area that previously had no forest cover. It helps improve the environment by reducing Carbon dioxide in atmosphere.

➢ **Reserved Forests (RF):** Permanently earmarked for production of timber or other forest produce. In these forests, all activities are prohibited unless specifically permitted.

➢ **Protected Forests (PF):** Forests where grazing and cultivation rights are allowed, but subject to a few restrictions.

➢ **Unclassed Forests (UF):** Forests and wastelands not classified as Reserved and Protected Forests.

CHAPTER-6
Water Resources

Learning Objectives

The students will be able to:

- Identify various sources of water.
- Analyse the need for water conservation.
- Evaluate conservation practices.
- Explain irrigation: Its importance and methods.

TOPIC-1: SOURCES OF WATER AND NEED FOR CONSERVATION

Concepts Covered: Watershed Management, Aquifer, Percolation Pits, Surface Runoff Harvesting, Rooftop Rainwater Harvesting, Regional Names of Rainwater Harvesting Systems

Revision Notes

Introduction

➢ The Earth is surrounded by water from all sides, but only about 0.03% is available to us as freshwater in rivers, lakes, and streams.

➢ About 97% of the Earth's water supply is founds in oceans and seas, and due to its high salt content, it is unfit for drinking and for agricultural use.

➢ **Sources of Freshwater:**

(i) Surface water

(ii) Groundwater

(iii) Rainwater

➢ **Surface water:** Water that collects on the surface of the Earth, such as lakes, rivers, streams.

➢ **Groundwater:** Water from rainfall that seeps beneath the Earth's surface, filling porous cracks and spaces in soil, sediment, and rocks.

➢ Groundwater constitutes about 0.66% of usable water on the Earth.

➢ Water is the most essential element on Earth. Life is impossible without it.

➢ The increasing population, industrialisation, urbanisation, and agricultural irrigation have reduced the per capita availability of water.

➢ **We need to conserve water due to the following reasons:**

(i) The high demand for water due to increasing population is leading to the lowering of groundwater levels.

(ii) Rainfall in India is seasonal, erratic and unreliable thus, farmers cannot wholly depend on it.

Rain water Harvesting

Scan Me!

(iii) More than 90% of water is utilised for irrigation.

(iv) Industries also used a lot of water and pollute it as well.

(v) A large amount of groundwater along with lakes, rivers, streams, etc., is polluted and cannot be used without proper treatment.

(vi) Scanty rainfall affects the growth of vegetation, which results in drought and a lowering of groundwater levels.

(vii) Water is essential for generating hydroelectric power.

➢ To overcome the shortage of water due to its increasing demand, we need to manage our water resources.

➢ Several practices have been initiated, and measures have been adopted to conserve water effectively.

➢ Some of these effective measures are: rainwater harvesting, **watershed management**, water-saving technologies, recycling of water, and prevention of water pollution.

➢ Rainwater harvesting is a technique through which rainwater is collected from surfaces on which rain falls, filtered, and then stored for future use.

➢ In other words, it is a technique for increasing the recharge of groundwater by capturing and storing rainwater.

➢ **It includes activities that are aimed at:**

(i) Preventing surface and groundwater.

(ii) Preventing of losses through evaporation and seepage.

(iii) Other techniques aimed at the conservation and efficient utilisation of limited water resources.

➢ Rainwater harvesting has become a necessity, and we need to understand its value by making optimum use of rainwater where it falls.

➢ The main objective of rainwater harvesting is to make water available for future use and avoid flooding of roads.

➢ **Water harvesting has many advantages, such as:**

(i) No land is wasted for storage purposes.

(ii) No population displacement is required.

(iii) Groundwater is not directly exposed to evaporation, and pollution increases the productivity of the **aquifer**.

(iv) It reduces soil erosion.

(v) Recharges groundwater resources.

➢ In the past, there were different water harvesting mechanisms carried out in various regions of the country. They included:

(i) Surface Runoff Harvesting

(ii) Rooftop Rainwater Harvesting

➢ In the past, rainwater harvesting mechanisms in different parts of the country in different parts of the country were known by various names. Surface Runoff Harvesting in the Western Himalayas is known as 'Kul". Kul in Western Himalayas, Zing in Ladakh, Baolis in the Gangetic Plains, Johads in Central India and Rajasthan, Surangam in Western Ghats, Korambu in the Eastern Ghats, and Bhandaras in Deccan Plateau.

➢ Watershed management refers to the efficient management and conservation of both the surface and groundwater resources.

➢ It includes the prevention of runoff as well as the storage and the recharge of groundwater by various methods such as **percolation pits**, recharge wells, borewells, dug wells, etc.

➢ **The elements of water harvesting mechanisms are as follows:**

(i) Catchments: The catchment of a water harvesting system is the surface drainage area that collects rainfall.

(ii) Conduits: These are pipelines or drains that carry rainwater from the catchment or rooftop area to the harvesting system.

(iii) Storage Facility: Rainwater can be stored in any available container, such as masonry or plastic tanks, RCC (Reinforced Cement Concrete), etc.

(iv) Recharge Facility: Rainwater can be used to recharge groundwater aquifers through any suitable structures, such as dug wells, borewells, recharge trenches and recharge pits.

Key Terms

➢ **Watershed Management:** It refers to the efficient management and conservation of both surface and groundwater resources.

➢ **Aquifer:** An underground layer of permeable rock, sediment, or soil that yields water.

➢ **Percolation Pits:** These are used to collect rainwater by percolation through the bottom or sides of the pits into surrounding soils.

➢ **Surface Runoff Harvesting:** A system of collecting and conserving rainwater where it falls that includes both natural and man-made surface for its eventual reuse.

➢ **Rooftop rainwater harvesting:** It is the technique through which rainwater is captured from roof catchments and stored in reservoirs

TOPIC-2: IMPORTANCE AND METHODS OF IRRIGATION

Concepts Covered: .Persian-wheel Method, Wells, Tube-Well Irrigation, Inundation Canals, Perennial Canals, Drip Irrigation, Sprinkler Irrigation

Revision Notes

➢ **Irrigation** refers to the process by which a controlled amount of water is artificially supplied to plants at regular intervals to aid in crop production. **[2024 Board]**

➢ In other words, irrigation is the supply of water to plants through artificial means such as wells, tanks, tube wells and canals, sourced from surface water or groundwater.

➢ Agriculture is the backbone of the Indian economy and about 70% of the population are engaged in it.

➢ About 92% of water is utilised for irrigating the agricultural fields.

Wells and Tube wells

Scan Me!

➢ Though rainfall is remains an important source of water for successful farming, artificial means of irrigation support agriculture to a large extent.

➢ Irrigation is an essential requirement for the development of agriculture in India due to the following reasons:

(i) Uncertainty of rainfall.
(ii) Uneven distribution of rainfall.
(iii) Different crops require varying quantities of water for optimal growth.
(iv) High dependence of crops on rainfall.
(v) Effective utilisation of river water.
(vi) To increase and maximise agricultural production.

➢ There are various methods of irrigation – traditional and modern.

➢ Traditional methods of irrigation include wells, tanks and **inundation canals** and while modern methods include tube wells, **perennial canals**, drip irrigation, spray irrigation, furrow irrigation, and sprinkler irrigation. **[2024 Board]**

I. Well irrigation is one of the oldest method of irrigation. A well is a hole dug in the ground to obtain groundwater. It is generally practised in areas where the soil is soft and easy to dig.

➢ **Well irrigation** is practised in the states of Uttar Pradesh, Punjab, Haryana, Bihar, Goa, Gujarat, Rajasthan, Maharashtra, Madhya Pradesh, Andhra Pradesh, Karnataka, and Tamil Nadu.

➢ There are different methods of lifting water from wells for irrigation in India. These include:

(i) Persian wheel method
(ii) Lever method
(iii) Inclined method

➢ Well irrigation has several advantages and disadvantages.

➢ **Advantages of well irrigation:**

(i) Simplest and cheapest method of irrigation.
(ii) Well can be dug anywhere the soil is soft.
(iii) Oxen used for ploughing the fields can also be used to draw water from the wheel at no extra cost.
(iv) With the use of pumps and tube wells, water can be lifted from great depths.

➢ **Disadvantages of well irrigation:**

(i) It is difficult to dig wells in hilly regions and in the rocky areas of the southern Peninsula.

(ii) Due to uneven distribution of groundwater resources, wells do not function effectively in all areas.

(iii) Owing to excessive withdrawal of groundwater and a falling water table, the conventional wells often dry up.

➢ Tube wells are deeper wells from which water is lifted from a depth of 20–30 metres by using power pumps.

➢ **Tube wells can be drilled in areas with the following conditions:**

(i) Availability of abundant groundwater.

(ii) Soft soil, level land, and fertile areas.

(iii) Availability of cheap and regular electricity to operate tube wells.

➢ **Advantages of tubewell irrigation:**

(i) It is able to irrigate a larger area of agricultural land.

(ii) Large amount of groundwater is easily available.

(iii) It is reliable during the dry season when surface water dries up, as the tube well is drilled deep to reach the permanent water table.

(iv) A large quantity of water can be extracted in a short period of time.

(v) Tube wells have played an important role in the Green Revolution.

➢ **Disadvantages of tube well irrigation:**

(i) Irrigation is not possible if the groundwater is brackish.

(ii) It requires cheap and cost-effective power, which is not available in most states.

(iii) Irrigation is not possible if the groundwater level is low.

(iv) Excessive use of tube wells leads to a decline in the groundwater level.

➢ Tube well irrigation is mainly practised in the states of Punjab, Haryana, Uttar Pradesh, Bihar, West Bengal, Rajasthan, Madhya Pradesh and Gujarat.

II. Canal irrigation is an important and effective method of irrigation in India.

➢ **Canal irrigation** is mainly concentrated in Uttar Pradesh, Punjab, and Haryana.

➢ Since digging is difficult in the rocky and uneven terrain, canals are practically absent in the southern peninsular region.

➢ **There are two types of canals in India:**

(i) Inundation canals

(ii) Perennial canals

➢ **Inundation Canals** are canals drawn directly from the rivers without any regulating structures such as a barrage or dam.

➢ This type of canal provides water for irrigation only during the rainy season and times of flood.

➢ Since the water level drops after the rainy season is over, the canal dries up and thus has limited use.

➢ **Perennial canals** are canals diverted from perennial rivers by constructing a dam or barrage across the river. **[2024 Board]**

➢ These canals help irrigate large areas and can draw water throughout the year.

➢ Today, in India, most canals are perennial.

➢ Canal irrigation is practised in Uttar Pradesh, Punjab, Haryana, Bihar, Madhya Pradesh, Rajasthan and Andhra Pradesh.

➢ Uttar Pradesh has constructed a large number of canals to irrigate around 3,091 thousand hectares of land, which is about 30.91% of the total canal-irrigated area of the country.

➢ The major canals of Uttar Pradesh are Upper Ganga Canal, Lower Ganga Canal, Sharda Canal, Eastern Yamuna Canal, Agra Canal and Betwa Canal.

➢ The major canals in Punjab are Upper Bari Doab Canal, Sirhind Canal, Bhakra Canal, and Bist Doab Canal.

➢ In Haryana, the Western Yamuna Canal, Bhakra Canal, Jui Canal, and Gurgaon Canal are the main canals.

➢ The major canals of Rajasthan are Indira Gandhi Canal, Gang or Bikaner Canal, and Chambal Canal Projects.

➢ **Advantages of Canal Irrigation:**

(i) Perennial canals provide constant supply of water and protect crops during drought situations.

(ii) Canal irrigation has proved to be a boon to the sandy areas of Rajasthan, which now yield good agricultural crops.

(iii) Canal irrigation has transformed Punjab and Haryana into 'the Granary of the Country'.

(iv) Canals carry sediment brought down by the rivers, which is deposited in agricultural fields, enhancing soil fertility.

(v) Although the initial investment for constructing a canal is high, it is economical in the long run.

➢ **Disadvantages of Canal Irrigation:**

(i) During the rainy season, many canals overflow and flood already cultivated areas.

(ii) In areas where water flows excessively into fields, it raises the groundwater level and bringing alkaline salts to the surface rendering, the land unfit for agriculture.

(iii) Due to waterlogging, the soil's capacity to absorb water reduces and damaging the standing crops, and stored grains.

(iv) Canal irrigation is mainly suitable in plain areas.

III. Tank Irrigation is an important source of irrigation, mainly in the southern part of the country.

➢ Tank irrigation is one of the oldest irrigation systems in India.

➢ It is mainly practised in the rocky plateau region of South India, where rainfall is highly seasonal and uneven.

➢ Tank irrigation is practiced in Andhra Pradesh, Telangana, Tamil Nadu, Maharashtra, Gujarat, and Madhya Pradesh.

➢ Tank irrigation is mainly practised in South India due to the following reasons:

(i) Due to hard rocks, non-porous and rocky surface water doesn't penetrate through the soil layers, resulting in minimal seepage loss.

(ii) The Deccan terrain is uneven, with many natural depressions that facilitate the construction of tanks.

➢ **Advantages of Tank Irrigation:**

(i) It is inexpensive, as most tanks are built in natural depressions.

(ii) It is highly beneficial in the uneven, rocky Deccan Plateau, where rainfall is seasonal.

(iii) As wells and tube wells cannot be dug in the rocky terrain of the Deccan Plateau, tanks are easily constructed.

➢ **Disadvantages of Tank Irrigation:**

(i) During the dry season, tanks dry up due to lack of rainwater and fail to provide irrigation.

(ii) Due to sediment deposition, tanks silt up quickly, requiring regular de-silting to maintain their storage capacity.

(iii) Tanks occupy large fertile areas that could otherwise be used for agriculture.

(iv) As tanks are extensive and shallow, a large amount of stored water is lost through evaporation or seepage.

➢ **The Conventional System of Irrigation or the Traditional Methods of Irrigation have several disadvantages. They are:**

(i) These methods are unreliable, as rainfall is seasonal.

(ii) Waterlogging occurs in the low-lying fields due to the excess water, damaging crops and reducing yields.

(iii) Due to salt accumulation in arid and semi-arid regions, large tract of lands become barren and unsuitable for cultivation.

➢ Modern irrigation methods are more efficient and supply water more uniformly than traditional methods.

➢ **The Important Modern Methods of Irrigation are:**

(i) **Drip Irrigation:** This is considered the most efficient and advanced method. It delivers water slowly to the roots of plants through pipes, valves, tubing, etc., thereby conserving water. This helps evaporation. **[2024 Board]**

(ii) **Sprinkler Irrigation:** It is a method in which water is supplied to the plants uniformly through a nozzle fitted in a pipe. It is widely used in arid areas as it checks and controls the wastage of water through evaporation and seepage. **[2024 Board]**

One advantage of this type of irrigation is that there is minimal loss of water through evaporation and seepage; however, it is expensive and suitable only for small area.

(iii) **Furrow Irrigation :** It is a type of surface irrigation in which furrows are dug between the rows of crops crops, allowing water to be evenly distributed across the field. It is one of the oldest methods of irrigation and is economical.

(iv) **Spray Irrigation :** In this method, water is sprayed onto the crops from high-pressure sprayers through a long hose. It is expensive, but it utilises water more efficiently. Since the water is sprayed, a considerable amount evaporates.

Key Terms

➢ **Well :** A pit or dug, bored, or drilled into the ground to access groundwater. It is drawn by a pump or by a pulley.

➢ **Tube well :** A deeper well from which water is lifted from a depth of 20–30 metres by using power pumps.

➢ **Inundation Canals:** Canals drawn directly from the rivers without any regulating structures like barrage or dam.

➢ **Perennial Canals:** Canals taken off from perennial rivers by constructing a dam or barrage across the river.

➢ **Persian Wheel Method :** It is a mechanical water lifting device operated by draught animals such as bullocks, buffaloes or camels..

CHAPTER-7
Mineral and Energy Resources

Learning Objectives

The students will be able to:
- Understand the various mineral resources, such as metallic and non-metallic minerals.
- Understand and explain conventional sources of energy.
- Evaluate the economic and environmental implications of hydroelectric power projects.
- Analyse the role of non-conventional energy sources in sustainable development and environmental conservation.

TOPIC-1: MINERAL RESOURCES

Concepts Covered: . Metallic & Non-Metallic Minerals, Iron Ore, Manganese, Copper, Bauxite, Limestone

Revision Notes

➢ **Minerals** are defined as solid, inorganic, naturally occurring substances with a definite chemical, physical properties and a general structure.

➢ Minerals are very important because:

(i) They are a source of raw materials and form the basis for industries.

(ii) They generate employment in the mining sector.

(iii) They are a source of source of energy e.g., coal and petroleum.

(iv) Minerals also help in earning foreign exchange.

➢ **Minerals are classified into two main categories:**

(i) Metallic minerals

(ii) Non-metallic minerals.

(i) Metallic minerals are those that can be melted to obtain new products such as iron, copper, bauxite, tin, gold, and manganese.

➢ These minerals are usually hard, ductile and malleable, and are associated with igneous rocks.

➢ Metallic minerals have a lustre or shine of their own.

➢ **They are further categorised into:**

(a) Ferrous minerals

(b) Non-ferrous minerals

➢ **Ferrous minerals** are those that contain iron content, iron ore, whereas non-ferrous minerals do not contain iron, such as bauxite, copper and gold.

Non-metallic minerals are either organic or inorganic in origin.

➢ They are generally associated with sedimentary rocks.

➢ They do not have a metallic lustre and break easily.

➢ **Non-metallic minerals** are non-malleable.

➢ They are further classified as: (a) Organic non-metallic minerals, and (b) Inorganic non-metallic minerals.

➢ Organic non-metallic minerals include fossil fuels, while inorganic non-metallic minerals include mica, graphite and limestone.

➢ **Some unique characteristics of minerals** are:

(i) They are not evenly distributed.

(ii) They are exhaustible.

(iii) They have an inverse relationship between quality and quantity.

➢ India is rich in mineral deposits, with a wide variety of minerals, such as iron, bauxite, mica, copper, gold, chromite, manganese, and limestone.

➢ **Iron Ore:** India is rich in iron deposits and is the largest producer in Asia.

➢ In India, four varieties of iron ore are found:

(i) Magnetite: It is the finest iron ore and contains more than 70% iron. It has excellent magnetic properties. It is brown to blackish in colour and is also called the black ore.

Minerals in India

Scan Me!

(ii) Haematite: It is the most important industrial iron ore. It contains about 60% to 70% pure iron. It is reddish in colour and is also called red ore.

(iii) Limonite: It contains about 40% to 60% iron. It is a low-quality iron ore. It is yellowish to yellowish-brown in colour.

(iv) Siderite: It contains many impurities and has an iron content of about 40% to 50%. It is a residual ore. It is a carbonate of iron and is found near coalfields.

➢ The primary use of iron ore is in the production of iron and steel. **[2024 Board]**

➢ Iron ore is usually smelted to produce pig iron (metallic iron), which is used to make steel.

➢ It is alloyed with other elements to make it strong and hard, and then utilised for construction, automobile manufacturing, and other purposes.

➢ **Uses of iron ore:**

(i) Used in aeroplanes and beams for construction purposes.

(ii) Used in manufacturing automobiles, trains, and trucks.

(iii) Used in manufacturing metallurgical products, magnets.

(iv) Used in appliances and surgical instruments in the form of steel.

(v) Radioactive iron is used in medicine as a tracer element.

(vi) Iron blue is used in blueprints, inks, paints, enamels, crayons, and linoleum.

➢ Iron ore is found in the mineral-rich areas of Indian states such as the Chhattisgarh, Jharkhand, Karnataka, Goa, Odisha, Andhra Pradesh, Tamil Nadu, Maharashtra and Rajasthan.

➢ Singhbhum, Keonjhar, Mayurbhanj, and Sundargarh districts have large deposits of iron ore.

➢ Iron ore is exported to the countries such as Italy, Iran, China, and Japan. The main exporting ports are Mormugao of Goa and Visakhapatnam in Andhra Pradesh.

➢ Manganese is an important ferro-alloy ore mined in India.

➢ It is hard, black and iron-like metal, an important raw material for the smelting of iron ore and is used in the manufacture of steel.

➢ **Uses of manganese:**

(i) It makes steel tough and hard and resistant to rust.

(ii) It is used in the manufacture of chemical and electrical equipment.

(iii) It is used to manufacture coloured glass.

(iv) It is used in chemical industries to manufacture bleaching powder.

(v) It is used in dry-cell batteries.

(vi) Manganese is also used to manufacture essential enzymes for the metabolism of fats and proteins.

(vii) Manganese is also used to regulate blood sugar levels, support the immune system, and is involved in development and reproduction.

(viii) It is also useful in plant growth, as it helps reduces nitrates in green plants and algae.

(ix) It is an essential trace element in higher animals.

➢ **The important manganese-producing areas in India are Odisha, Karnataka, Madhya Pradesh, Maharashtra, Andhra Pradesh, Goa, Telangana, Jharkhand, and Rajasthan.**

➢ The main buyers of manganese from India are the countries of USA, Japan, France, the Netherlands, the UK, Germany, and Belgium.

➢ Copper is an important non-ferrous metal.

➢ It is a good conductor of electricity and is ductile.

➢ Copper is found in both old as well as young rock formations, and occurs as veins and as bedded deposits.

➢ Copper contains a very small percentage of metal; therefore, copper mining is an expensive and labour-intensive process.

➢ **Uses of Copper:**

(i) Since copper is a good conductor of electricity, it is used for making for electrical wires.

(ii) It is also used in automobiles and the defence industries.

(iii) It is alloyed with nickel and iron to make stainless steel.

(iv) It is alloyed with aluminium to make duralumin.

(v) When alloyed with zinc, it is called brass, and when alloyed with tin, it is called bronze.

(vi) Copper is also used in building construction, plumbing, and in shipbuilding.

➢ India has low-grade copper ore and, therefore, thus imports from the countries such as the USA, Canada, Japan, Mexico, and Zimbabwe.

➢ The major copper mines are located in Khetri in Rajasthan, Singhbhum in Bihar and Malanjkhand in Madhya Pradesh. Besides these, Guntur in Andhra Pradesh and Nagpur in Maharashtra.

➢ Bauxite is an important ore of aluminium. **[2024 Board]**

➢ It is mixed with sand and iron oxide.

➢ It is sticky like clay and is found in the Tertiary deposits of the Peninsular Plateau.

➢ Bauxite contains about 60–70% of aluminium oxide and aluminium is extracted by smelting it.

➢ India has huge deposits of bauxite and ranks fifth in world production.

➢ The largest integrated aluminium plant in India is located at Renukoot in Uttar Pradesh.

➢ It receives its supply from Amarkantak Plateau of Madhya Pradesh and Ranchi of Jharkhand.

➢ **Uses of Bauxite:**

(i) Bauxite is rust-resistant, strong, and lightweight metal.

(ii) It is mainly used in aircraft, automobiles, shipping industry, household appliances, rail wagons, coaches, etc.

(iii) Aluminium extracted from bauxite is a good conductor of electricity and is used in electrical equipment industries.

(iv) It is also used making mirrors, headlight reflectors and in telescopes.

➢ The main bauxite deposits are in Odisha, Goa, Gujarat, Maharashtra, Madhya Pradesh, Chhattisgarh, Jharkhand, Karnataka and Tamil Nadu.

➢ **Limestone:** It is an inorganic non-metallic mineral and a type of sedimentary rock.

➢ **Uses of Limestone:**

(i) It is used as a basic raw material in the cement industry.

(ii) It is used as a flux in the iron and steel industries.

(iii) It is used in the production of chemicals, paper, glass, and fertilisers.

➢ Major limestone-producing states in India include Madhya Pradesh, Rajasthan, Gujarat, Andhra Pradesh, Chhattisgarh, Odisha and, Jharkhand.

Key Terms

➢ **Ore:** Minerals that contain very high percentage of a particular metal and from which the metal can be profitably extracted are called ores. Ores are essentially the raw materials for producing metals. For example, iron ore used to obtain iron and, bauxite to obtain aluminium.

➢ **Metallurgy:** It is the science that deals with the nature, uses, production and purification of metals.

➢ **Exhaustible minerals:** These are minerals that exist in limited quantities and are non-renewable. For example, fossil fuels.

TOPIC-2: CONVENTIONAL SOURCES OF ENERGY AND ITS MECHANISM

Concepts Covered: .Coal-Gondwana Coalfields, Damodar Valley, Types of Coal, Petroleum-Oil Refineries, Natural Gas

Revision Notes

➢ Conventional sources of energy are non-renewable, that cannot be replenished once they are consumed.

➢ Conventional sources of energy include:

(i) Coal

(ii) Petroleum

(iii) Natural Gas

(iv) Hydel Power (Hydroelectricity)

➢ Conventional sources of energy are hazardous to the environment as they cause pollution.

➢ These sources of energy are also expensive.

(i) Coal: **[2024 Board]**

➢ Coal is an organic sedimentary rock, and formed from the accumulation and preservation of plant materials in a swamp environment, deltaic regions, and coastal plains.

➢ Coal is a combustible rock and a significant fossil fuel, though its reserves have depleting due to ever-increasing demand.

➢ Coal is widely used and its primary use is in the generation of electricity.

➢ India ranks third in the world in terms of coal production.

➢ Coal is a bulky material that loses when used, as it is reduced to ash. Therefore, heavy industries and thermal power stations are usually located on or near coalfields to minimise transportation costs.

➢ Raniganj in West Bengal is the oldest coalfield in India while Jharia in Jharkhand is the largest coalfield in India.

➢ In India, coal is classified into two geological ages:

(i) The Gondwana Coalfields

(ii) The Tertiary Coalfields

➢ **The Gondwana Coalfields:** These are a little over 200 million years old and mainly consist of bituminous coal. They make up about 98% of the total coal reserves in India and are available unlike anthracite coal. The extensive resources of Gondwana coal are located in:

➢ **The Damodar Valley (West Bengal–Jharkhand):** Jharia, Raniganj and Bokaro are important coalfields. Apart from these, they are also found in the Godavari Valley, the Mahanadi Valley, the Son Valley and the Wardha Valley.

➢ **The main advantages of coal are:**

(i) It is a source of direct heat and energy for domestic purposes.

(ii) It is one of the cheapest forms of energy.

(iii) It provides numerous raw materials to chemical industries such as benzole, ammonia, coal tar and, coal gas.

(iv) Coal is also a source of many by-products such as coke, tar, ammonium sulphate, phenol, naphthalene, and benzene.

➢ **The main disadvantages of coal are:**

(i) Coal releases carbon dioxide, which and the effect of environment leading to greenhouse gas emissions and the effect of global warming.

(ii) The coal reserves in India are concentrated in the Chhota Nagpur region.

(iii) The transportation cost of coal is significantly high.

(iv) Coal reserves are limited in India.

(v) Coal found in India is of poor quality.

➢ **Uses of Coal :**

(i) It is primarily used as a energy source for heat or electricity.

(ii) It is used to run railway locomotives, machinery,, dynamos, and ship engines.

(iii) It is used to produce electricity. Among the various uses of coal, thermal power generation is the most important.

(iv) Coal is essential for the iron and steel industries.

(v) It is also used in building materials—for firing bricks, potteries, and in iron and brass foundries.

➢ On the basis of carbon content, coal may be classified into four types:

(a) Anthracite

(b) Bituminous

(c) Lignite

(d) Peat

➢ **(a) Anthracite Coal:**

(i) It is the highest quality of coal—a hard and compact variety with the highest carbon content of about 90%.

(ii) It is associated with highly transformed sedimentary rocks subjected to high pressure and temperature inside the Earth.

(iii) It ignites with difficulty but burns for a long duration with a smokeless flame.

(iv) It is lustrous, shiny, and jet black in colour.

(v) It has a high heating value and leaves behind very little ash after burning.

(vi) It has a high calorific value and is suitable for domestic use since it is smokeless.

(vii) Anthracite coal is found only in Jammu and Kashmir.

(b) Bituminous Coal:

(i) It is a black, hard, brittle, and compact coal with 50% to 80% of carbon content.

(ii) Due to high carbon content, its calorific value is very high and it has low moisture content.

(iii) It is used as steam coal, household coal, coking coal, and gas coal.

(iv) Steam coal is the best bituminous coal as it contains 80% carbon.

(v) Bituminous coal is black and lustrous and is commonly used as household coal.

(vi) It is relatively a soft coal and contains tar like substance called bitumen.

(vii) The highest grade bituminous coal is the coking coal, which is an important ingredient in iron and steel industry used for smelting in the blast furnaces.

(viii) Bituminous coal is found in Odisha, Jharkhand, Chhattisgarh, West Bengal, and Madhya Pradesh.

(c) Lignite Coal:

(i) Lignite is also referred to as "brown coal".

(ii) It is a soft, brown, combustible sedimentary rock.

(iii) It is a low-grade coal due to its relatively low heat content.

(iv) It has a carbon content of only around 40%.

(v) It has high moisture content and is less combustible.

(vi) It is found in Tamil Nadu, Rajasthan, Kerala, West Bengal and Puducherry.

(d) Peat Coal:

(i) Peat is an accumulation of partially decayed vegetative or organic matter that has undergone varying degrees of decomposition and carbonisation.

(ii) It contains high moisture and small percentage of **volatile matter**.

(iii) It has lower carbon content and is of inferior grade.

(iv) The formation of peat is the first step in the geological development of other fossil fuels.

(v) It is found in the Nilgiri Hills, the Kashmir Valley, and swampy areas of coastal plains.

(ii) Petroleum: **[2024 Board]**

➢ Petroleum is derived from the Greek words 'Petra' meaning rock and 'oleum' meaning oil.

➢ It is a naturally occurring liquid found beneath the Earth's surface.

➢ It is a mixture of hydrocarbons and other organic compounds.

➢ Petroleum is a fossil fuel which is formed when large quantities of dead organisms are buried beneath the sedimentary rocks such as shale, limestone, sandstone. Therefore, it is also called mineral oil.

➢ It is called 'liquid gold' because it is in liquid form and has a high commercial value, like gold.

➢ Petroleum includes all liquid, gaseous, and solid hydrocarbons.

➢ As a liquid, it is known as crude oil; the gaseous form is called natural gas; and the solid forms include tar, bitumen, and asphalt.

➢ Some petroleum products include are petrol, diesel, lubricants, paraffin wax, slack wax, tar, kerosene, liquefied petroleum gas (LPG).

➢ Byproducts of petroleum include fertiliser, petroleum jelly, insecticide, soap, linoleum, and perfumes.

➢ **Advantages of Petroleum :**

(i) Petroleum has high energy density, as 1 kg of burnt oil can generate approximately 10,000 Kcal of energy.

(ii) Oil extraction is easy and cost-effective due to modern technologies.
(iii) Petroleum in liquid form can be transported over long distances through pipelines or tankers.
(iv) It has been the primary energy resource for many power plants and has broad areas for applications and thus has high energy demand.
(v) It is widely used as a fuel for transportation on land, sea and in the air.
(vi) Petroleum is also used for power generation.
(vii) Its fuel derivatives include ethane, diesel, petrol, kerosene, and LPG.
(viii) Petrochemicals are chemical products derived from petroleum after refining process.
(ix) Examples of petrochemical products include fertilisers, gasoline, synthetic rubber, synthetic fibre, explosives, dyes, crayons, paraffin wax pesticides, perfumes, paints, varnishes, phenol, PVC, lubricating oil, printing ink, photography film, carbon black, polyester, safety glass, herbicides, detergents, and cosmetics.

➢ **Disadvantages of Petroleum:**

(i) Petroleum is an expensive and is in high demand due to its limited supply.

(ii) It is a natural fossil fuel and is non-renewable.

(iii) It is not environmentally friendly, as its burning and extracting generates **greenhouse gases** that lead to pollution and **global warming.**

(iv) It is highly flammable and can cause fires.

(v) It is harmful to marine animals as oil spills during extraction and transportation can be fatal to aquatic animals.

➢ **Oil Refineries :**

➢ Oil refineries are industrial units where crude oil is refined and processed to produce useful products like petrol, petroleum naphtha, LPG and diesel oil, and asphalt.

➢ There are 23 oil refineries in India of which 20 are in the public sector and 3 refineries in the private sector or the joint sector.

➢ In India, the maximum oil production comes from the Mumbai High offshore field.

➢ The main oil deposits in India, in order of their importance are :

(i) Mumbai High

(ii) Oilfields of the Eastern Region

(iii) Oilfields of the Western Region

➢ Mumbai High produces superior quality of crude oil compared to the Middle Eastern countries.

➢ The Oil in Mumbai High is extracted using the drillship named Sagar Samrat, which has a maximum drill depth of about 20,000 feet.

➢ Oil refineries are located close to the oilfields or near ports due to the following reasons:

1. To minimise the cost of transport.

2. To avoid transportation of mineral oil to the interior regions of the country as it is highly flammable.

➢ **Some of the oil refineries in India are:** Mathura Refinery, Mumbai Refinery, Haldia Refinery, Barauni Refinery, Panipat Refinery, Digboi Refinery, Visakhapatnam Refinery, Kochi Refinery, and Jamnagar Refineries.

➢ The oldest oilfield in India is Digboi Oilfield, situated in the Eastern Region of India. The other oilfields in this region include: Moran, Bapapung, Hausanpung and Hugirijang.

➢ The Cambay Basin in Gujarat is the major oilfield in the Western Region of India.

➢ The other important oilfields in Gujarat are Kalol, Koyali, Kosamba, Sanand, Kathana, Ankleshwar and Navagaon.

(iii) Natural Gas:

➢ Natural gas is a mixture of hydrocarbon-rich gases, primarily methane, nitrogen, carbon dioxide.

➢ Natural gas reserves are located beneath the Earth's surface, often near the crude oil deposits.

➢ It is not used in its raw form; it is processed and refined into a cleaner fuel before use.

➢ It is mainly used as a fuel for generating electricity and heat.

➢ When compressed, natural gas is used as vehicle fuel and is known as CNG (Compressed Natural Gas).

➢ Liquefied Petroleum Gas (LPG) is supplied to household for cooking purposes.

➢ It is a by-product obtained during the refining the crude oil.

➢ The main constituents of LPG are butane and propane, which are flammable hydrocarbon gases.

➢ LPG is used as fuel for domestic purposes such as for cooking and in vehicles.

➢ Ethyl mercaptan, which gives off a foul smell, is intentionally added to LPG so that any leakage can be easily detected.

➢ LPG is also being replaced in some areas with PNG (Piped Natural Gas), which is supplied through pipelines instead of storing in the cylinder.

➢ Compressed Natural Gas (CNG) is a fuel used in place of petrol, diesel, and LPG.

➢ In CNG, methane is stored at high pressure.

➢ Three-fourths of India's natural gas comes from Mumbai High and with the rest from Assam, Rajasthan, Tamil Nadu and Tripura.

➢ **Advantages of Natural Gas:**

(i) Natural Gas is considered environmentally friendly as it emits less carbon—about 60–90% fewer smog-producing pollutants.

(ii) It can be stored safely and transported efficiently through pipelines or cylinders.

(iii) It is reliable and convenient for cooking and operating many appliances.

(iv) It is cheaper and cleaner than petrol or diesel.

(v) Natural Gas is colourless, odourless and lighter than air.

(vi) Hydrogen and ammonia are produced from natural gas and are used in fertilisers, paints and plastics.

(vii) There is an abundant supply and a strong reserve of natural gas expected to last for centuries.

➢ **Disadvantages of Natural Gas:**

(i) Though found in abundance, it is non-renewable due to its increasing demand.

(ii) Gas leakage poses a high risk, as it is colourless, odourless, and tasteless.

(iii) It is highly volatile and need to be handled carefully during transporting.

(iv) The infrastructure for production and distribution is quite expensive, including plumbing systems and specialised tanks.

(v) To use it as fuel, all constituents other than methane must be extracted; this processing results in various byproducts such as: hydrocarbons, sulphur, water vapour, carbon dioxide, helium, and nitrogen.

Key Terms

➢ **Volatile Matter:** In coal, it is refers to substances, other than moisture that are released as gas and vapour during combustion.

➢ **Hydrocarbons:** These are compound of hydrogen and carbon, such as those that are the main components of petroleum and natural gas.

➢ **Greenhouse Gas:** A gas that contributes to the greenhouse effect by absorbing infrared radiation, e.g., carbon dioxide and chlorofluorocarbons.

➢ **Global Warming:** It is the increase in Earth's average surface temperature due to effect of greenhouse gases.

TOPIC-3: HYDEL POWER

Concepts Covered: .Hydro power, Multipurpose Projects, Auxiliary Dam, Pisciculture

Revision Notes

➢ Hydel power is a shortened term for hydroelectric power. **Hydropower** is one of the most widely used conventional renewable sources of energy for generating electricity. It uses the gravitational force of falling water.

➢ Hydroelectricity is produced from the energy released by fast flowing water or water falling from a height with a great force.

➢ Tidal energy can be produced by creating a reservoir or basin behind a barrage, then passing tidal water through turbines to generate electricity.

➢ It is one of the best, cleanest and cheapest sources of energy.

➢ It plays an important role in reducing greenhouse gas emissions.

➢ There is very little possibility of causing pollution.

➢ Most of the hydroelectric power plants have a dam and a reservoir and their power generation depends on the head of water and the volume of water flowing towards the turbine.

➢ **Advantages of Hydel Power**

(i) There is less pollution due to absence of fuel combustion.

(ii) It is one of the best, cleanest, and cheapest sources of energy.

(iii) It involves lower maintenance costs.

(iv) It plays a significant role in reducing greenhouse gas emissions.

(v) It is reliable, renewable, and sustainable.

(vi) The reservoirs and dams built to produce hydroelectricity help in conserving and storing water.

➢ **Disadvantages of Hydel Power**

(i) It is expensive due to the high investment required to build dams.

(ii) Construction of dams can cause water access problems as it it may alter the water-table level.

(iii) The construction of large dams can cause serious geological damage, such as triggering earthquakes.

(iv) Displacement of people living in the villages and towns, loss of farmland and businesses, physically and disruption to local communities can cause psychologically disturbed.

➢ **Bhakra Nangal Dam**

(i) It is the largest and most significant **multipurpose Project** built in India on River Sutlej.

(ii) It is a joint venture of the governments of Punjab, Haryana and Rajasthan.

(iii) Its main aim is to harness the water of River Sutlej for the benefit of the states involved in the joint venture.

(iv) The Bhakra Nangal project comprises of :

1. Two dams at Bhakra and Nangal
2. Nangal Hydel Channel
3. Powerhouses
4. Bhakra Canal System
5. Electric transmission lines

1. The Bhakra Dam :

- It is one of the highest gravity dams in the world.
- The reservoir of the Bhakra Dam is called Gobind Sagar Lake.
- It is the third-largest water reservoir in India.

2. The Nangal Dam :

- It has been constructed on the River Sutlej, about 13 km downstream of the Bhakra Dam.
- It is an **auxiliary dam** that serves as a balancing reservoir.
- It is one of the longest cement-lined canals of the World.
- Its main function is to turn the turbines of the powerhouses located below the Nangal Dam.

3. The Power House :

- It has been built to generate hydroelectricity from the water of River Sutlej.
- There are four powerhouses at Ganguwal, Kotla, the Right Bank powerhouse and the Left Bank Powerhouse.
- All these powerhouses have a combined installed capacity of 1204 MW.

4. Bhakra Canal System :

- The main Bhakra Canal is 174 km long.
- It provides irrigation to 27.41 lakh hectares in the states of Haryana, Punjab, and Rajasthan.

➢ **Hirakud Dam :**

(i) The Hirakud Dam is built across the Mahanadi River in Odisha.

(ii) It is one of the first multipurpose river valley projects in India.

(iii) The dam was completed in 1953 but was formally inaugurated in 1957.

(iv) The dam is the longest major earthen dam in Asia.

(v) This project also provides irrigation for kharif and rabi crops in the districts of Sambalpur, Bargarh, Bolangir and Subarnapur.

(vi) Due to the successful irrigation provided by the dam, Sambalpur known as the Rice Bowl of Odisha.

(vii) The dam can generate up to 307.5 MW of electricity through its two power plants at Burla and Chiplima.

(viii) The objectives of this dam are:

1. It helps control floods in the Mahanadi Delta.
2. It irrigates about 75,000 sq km of land.
3. It generates electricity through several hydroelectric plants.

The Main Objective of these Multipurpose Projects :

(i) Provision of irrigation, especially to areas of less rainfall.

(ii) Generation of hydroelectricity to support industrial development and provide other basic facilities.

(iii) Control flooding in the Sutlej and Beas rivers.

(iv) Promotion and develop of river navigation.

(v) Promotion of pisciculture (fish farming).

(vi) Soil conservation through **afforestation** and increased timber productivity.

(vii) Disease control by preventing waterlogging.

(viii) Development of recreational and health resorts.

Key Terms

➢ **Hydropower :** It is power derived from the energy of falling water or fast-flowing water.

➢ **Hydroelectric Power Plant:** A hydropower system that uses a dam to store river water in a reservoir. Water released from the reservoir flows through a turbine, which spins and activates a generator to produce electricity.

➢ **Multipurpose Project:** A large-scale river valley project designed to serve multiple purposes simultaneously such as irrigation, flood control, pisciculture and hydroelectricity.

➢ **Auxiliary Dam:** Constructed to confine the Reservoir created by a primary dam, either to allow higher water elevation and storage or to limit the reservoir's extent for increased efficiency.

➢ **Pisciculture:** The controlled breeding and rearing and harvesting of fish in ponds, tanks, or other enclosures for commercial purposes.

➢ **Afforestation:** The planting of new saplings in barren areas to create forests and improve environment, biodiversity, and climate balance.

TOPIC-4: NON-CONVENTIONAL SOURCES OF ENERGY

Concepts Covered: .Solar Energy, Wind Energy, Tidal Energy, Geothermal Energy, Nuclear Energy, Biogas

Revision Notes

➢ Non-conventional sources of energy are those which are developed in the recent past as an alternative to conventional or non-renewable sources of energy.

➢ **Non-conventional sources of energy include:**

(i) Solar energy

(ii) Wind energy

(iii) Tidal energy

(iv) Geothermal energy **[2024 Board]**

(v) Nuclear energy

(vi) Biogas

➢ They are renewable, inexhaustible, and cause less environmental pollution.

➢ They are inexpensive and easy to maintain. Non-conventional resources are also known as Green Energy—the energy resources of the future.

➢ **Non-conventional sources of energy** are gaining importance due to rising energy demand rapid depletion of conventional sources such as coal, and the fast depleting conventional sources like coal, petroleum, and natural gas.

(i) Solar Energy is the primary source of energy and inexhaustible. **[2024 Board]**

➢ India's location is favourable for solar power generation. India lies between 8°N and 37°N with the Tropic of Cancer running through it receiving abundant sunlight for nearly 300 clear days a year. This makes it ideal for solar power development.

➢ To harness solar energy in India several techniques have been developed, as listed below:

(a) Solar Cells or Photovoltaic Cells :

1. Solar energy can be converted into electrical energy using solar cells or **photovoltaic cells**.
2. A solar cell is a device that converts light energy into electrical energy.
3. These cells are made using a silicon wafers.
4. Light shining on the solar cell produces both current and voltage to generate electric power.
5. Solar cells are regarded as one of the key technologies for a sustainable energy supply.
6. Solar cells are used in calculators, wristwatches, traffic signals, streetlights, and water pumps, etc.

(b) Solar Cooker :

1. Solar cooking, utilising the sun's ultraviolet (UV) rays, involves harnessing solar energy to heat and cook food. This sustainable method relies on the direct or reflected sunlight, often using specially designed solar cookers or ovens, to convert solar radiation into heat energy.

2. The UV rays enter the solar cooker and are converted into infrared light rays.

3. The food in the solar cooker is not cooked by the sun's heat directly. It is the sun's rays, converted heat energy, that cook the food.

4. The heat energy is retained by the lidded utensil.

5. A new design of solar cooker has been developed, using a spherical reflector instead of a plane mirror, as the former provides heating effect and efficiency.

(c) Solar Water Heater :

1. Solar energy is most effectively used for heating of water.

2. A sun-facing collector heats a working fluid that passes into a storage system for later use.

3. The plate is a simple, glass-topped insulated box with a flat solar absorber made of sheet metal, attached to copper heat-exchanger pipes.

➢ **Advantages of Solar Energy:**

(i) It is a renewable and inexhaustible source of energy.

(ii) It is environmentally friendly.

(iii) It can be used for varied purposes such as generating electricity, heating, and drying.

(iv) After the initial installation, its maintenance and repair are minimal and inexpensive.

(v) Solar energy conserves fossil fuels such as coal and petroleum for electricity generation and also helps in reducing electricity bills.

(vi) A solar energy system can be installed anywhere, and solar panels can be easily placed in houses.

(ii) Wind Energy

➢ **Wind energy**, or wind power, is the process by which wind is used to generate electricity.

➢ Wind energy is produced through windmills.

➢ The wind turns the blades of the windmill, spinning a shaft that moves the turbine; turbines are connected to a generator, which produces electricity.

➢ A wind farm is a group of wind turbines or windmills located in the same area used for the production of electricity.

➢ Windmills are installed in open areas, coastal regions, or in hilly terrains.

➢ There are both onshore and offshore wind farms.

➢ An onshore wind farm is an inexpensive source of electric power, while off-shore wind farms are steadier and stronger, although their construction and maintenance costs are higher.

➢ They generate a significant amount of electricity.

➢ The largest wind farm network in India stretches from Nagercoil to Madurai in Tamil Nadu.

➢ Muppandal Wind Farm, situated in the Kanyakumari District of Tamil Nadu, is the second-largest onshore wind farm in the world.

➢ **Advantages of Wind Energy :**

(i) Wind energy is abundant and is renewable.

(ii) It is widely available and geographically distributed.

(iii) It is among the cleanest energy sources and does not produce any greenhouse gas emissions during operation.

(iv) It requires minimal land and consumes no water.

(v) It serves as an alternative to burning fossil fuels.

(vi) The electricity generated from wind energy is used for domestic purposes and is cost-effective.

(iii) Tidal energy is a form of hydropower that generates electricity through significant tidal movements.

➢ Tidal energy can be harnessed in two ways:

(a) By using the change in tidal height (potential energy).

(b) By using the flow of water **(kinetic energy)**.

➢ The four main categories of tidal power technology are:

(i) In-stream device or tidal stream generator

(ii) Tidal barrage

(iii) Dynamic tidal power

(iv) Tidal lagoon

➢ **Tidal stream generator:** It utilises the kinetic energy of moving water to power turbines.

(ii) Tidal barrages: They utilise the potential energy in the difference in height between high and low tides.

➢ Tidal barrages are the oldest method of tidal power generation.

➢ A barrage is built across a bay or a river subject to tidal flow.

➢ During high tide, the seawater flows into the barrage reservoir and turns the turbine, which in turn generates electricity by rotating the generators.

➢ When the tides are low, the sea water stored in the barrage reservoir flows out in the sea and the flow of water turns the turbines in its process.

(iii) Dynamic tidal power: It is a promising technology and proposes building very long dams from the coast straight out into the sea or ocean, without enclosing any area.

(iv) Tidal lagoons: These are independent enclosing barrages constructed on high-level tidal estuarine land, which trap the high water and release it to generate power.

➢ **Advantages of tidal energy:**

(i) Tidal energy is an inexhaustible source of energy.

(ii) It is non-polluting and does not produce carbon emissions like fossil fuels.

(iii) It is predictable since tides rise with great uniformity, and energy can still be produced even when the water flows slowly.

(iv) Once a tidal energy power plant is installed, its maintenance costs remain extremely low.

(v) The energy density of tidal energy is significantly higher than that of other forms of renewable energy, such as wind power.

(iv) Geothermal Energy: 'Geothermal' is derived from the Greek word 'Geo' meaning Earth and 'Thermos' meaning heat. It is the thermal energy generated and stored within the Earth.

➢ Due to extremely high temperatures below the earth's crust, hot water and steam rise up. This heat contains enormous energy and power and is tapped generate geothermal power.

➢ To harness geothermal energy, a power plant must be established at a site with a reliable source of superheated water.

➢ Pipes are installed to reach the underground source, forcing fluids to the surface to produce steam. This steam is used to rotate a turbine, thereby generating electricity or geothermal power.

➢ **Advantages of geothermal energy:**

(i) Geothermal Energy is considered a sustainable and renewable source of energy.

(ii) It is inexpensive, reliable, and easily accessible.

(iii) It is environmentally friendly and emits fewer greenhouse gases.

(iv) A geothermal power plant requires low maintenance costs and the electricity bills are reduced.

➢ In India, geothermal plants have the potential to harness about 12,000 MW of energy.

➢ Geothermal plants are located in Manikaran in, Himachal Pradesh and Puga Valley, Ladakh.

(v) Nuclear Power : It is the energy produced through nuclear reactions.

➢ Inside the nuclear reactor, energy is generated through a chain reaction involving uranium atoms.

➢ When the uranium atom is split it releases heat and energy, which are then converted into electricity.

➢ **Advantages of nuclear power :**

(i) A single uranium atom can generate large amount of energy when split, thus producing abundant electricity.

(ii) It is clean and does not produce significant greenhouse gas emissions.

(iii) It is reliable, unlike solar energy, which depends on weather conditions.

(iv) It is a viable alternative to fossil fuels, which are non-renewable.

(v) Although the initial installation cost is high, nuclear power becomes an inexpensive energy source once operational.

(vi) The waste produced by nuclear power plants can be reprocessed and used to create useful materials, such as those used in aircraft manufacturing.

(vi) Biogas: It is a mixture of gases like methane, carbon dioxide and hydrogen sulphide.

➢ It is produced by processing residual waste from livestock, sewage and organic waste.

➢ Biogas is produced by the breakdown of organic matter in the absence of oxygen a process known as anaerobic digestion.

➢ A biogas plant can convert animal manure, plant matter, agro-industrial waste, and slaughterhouse waste into combustible gas.

➢ Plants that use cattle dung are called gobar gas plants.

➢ The biogas can be used for power generation, cooking, and lighting, among other purposes.

➢ **Advantages of Biogas :**

(i) Biogas is an inexpensive and clean source of energy.

(ii) It is non-polluting and does not emit greenhouse gases.

(iii) It is a renewable source of energy.

(iv) There is no storage issue, as gas is supplied directly from the plant.

(v) The residual sludge is a good fertiliser for pastures and meadows and is ecologically beneficial.

(vi) It generates employment in rural areas.

(vii) It produces enriched organic manure that can supplement or even replace chemical fertilisers

Key Terms

➢ **Photovoltaic Cells:** Devices that convert solar energy directly into electrical energy through the photovoltaic effect. Commonly used in solar panels.

➢ **Wind Energy:** A form of renewable energy generated by converting the kinetic energy of moving air into electrical energy using wind turbines.

➢ **Wind Farm:** It is a group of wind turbines or windmills in the same location used for the production of electricity.

➢ **Potential Energy :** It is the stored energy or energy caused by its position. For example, water stored at height in a reservoir.

➢ **Tidal Energy :** It is a form of hydropower that generates electricity through high tidal movements, caused by the gravitational pull of the moon and the sun.

➢ **Kinetic Energy :** It is the energy possessed by an object in motion. For instance, flowing water, moving vehicles, and wind all have kinetic energy.

➢ **Geothermal :** It is the energy produced from the heat of the Earth. It is harnessed by tapping hot water or steam from underground reservoirs to produce electricity.

➢ **Non-conventional sources of energy :** Sources of energy developed as alternatives to conventional (fossil fuel-based) sources. These include solar, wind, tidal, geothermal, and biogas energy, and are typically renewable and environment-friendly.

CHAPTER-8
Agriculture

Learning Objectives

The students will be able to:
- Understand the significance of agriculture in the Indian economy and its role in providing livelihoods.
- Understand the concept of mixed farming and its advantages in diversifying agricultural activities.
- Examine the cropping patterns and major crops associated with rabi, kharif and zaid seasons.
- Evaluate the climatic conditions and soil requirements for the cultivating food crops and cash crops.

TOPIC-1: INDIAN AGRICULTURE: IMPORTANCE, PROBLEMS AND REFORMS CHARACTERISTICS

Concepts Covered: Salinisation, Pruning, Absentee Landlord, Kisan Call Centres, National Agriculture Policy

Revision Notes

Introduction

➢ Agriculture is the most important occupation in India and the primary activity of the country. **[Board 2024]**

➢ The word "agriculture" is derived from two Latin words—'ager' meaning land and 'culture' meaning cultivation.

➢ Agriculture means the cultivation of the soil to grow crops and rear livestock.

➢ Agriculture is the process of producing food, fodder fibre, fuel and other goods through the systematic raising of plants and animals.

➢ India has a vast expanse of agricultural land due to its rich fertile soil and a good network of perennial rivers.

➢ In addition, India has suitable climatic conditions, a good amount of sunshine throughout the year, long growing seasons, etc.

➢ Agriculture is the backbone of the Indian economy. It occupies a significant position in the overall economy of the country contributing about 17% of total GDP.

➢ It plays an important role in the Indian economy for the following reasons:

(i) Agriculture is essential because it feeds millions of people and the ever-increasing population.

(ii) It also helps raise livestock with suitable environmental conditions and provides fodder to them. Over 60% of India's land is arable, and about 70% of rural families are engaged in this occupation for their livelihood.

(iii) Agriculture helps create job opportunities to millions of people. It is the single largest private sector occupation and provides employment to about 58.4% of country's workforce.

(iv) It supports many important industries by supplying of raw materials like cotton and jute textile industries, sugar industries, vanaspati, food processing, etc. **[Board 2024]**

(v) Various small-scale and cottage industries like handlooms, spinning oil milling, rice thrashing, etc., are dependent on agriculture for their raw material.

(vi) Agriculture also provides a good market for farm inputs like implements, fertilisers, pesticides, machinery, etc.

(vii) India's foreign trade and exports are linked to with agriculture. It accounts for more than 18% of the total export earnings.

➢ India is witnessing slow agricultural growth despite its efforts to achieve high agricultural yield.

➢ This is due to unreliable rainfall, a poor irrigation system, farmers' inaccessibility to markets, lack of proper market infrastructure, poor farming techniques, etc.

➢ Many factors contribute to its low development compared to other developed countries. These factors are categorised into four groups as:

(i) Environmental Factors:

(a) Erratic and unreliable rainfall.

(b) Lack of adequate irrigation facilities and dependence **on the monsoon.**

(c) Soil degradation from erosion and **salinisation** destroys productivity.

(d) The repetition of growing the same crops like rice and wheat lead to soil infertility.

(e) Inadequate use of manure and fertilisers, neglect of **crop rotation**, use of poor-quality seeds, inadequate water supply, etc., leads to low productivity.

(f) The use of simple and old agricultural tools, use of absence of or limited use of machines for ploughing, sowing, irrigating, **pruning**, harvesting and threshing result in low yields.

(g) In recent years, the net sown area has decreased due to the shift in cultivation from food crops to fruits, vegetables, oilseeds, etc.

(ii) Economic Factors:

(a) The Indian farmers chiefly practise subsistence farming, where large amounts of manual labour is employed to work on farms, growing only enough to suffice their family's needs and not much is left for sale in the market.

(b) Farmers use primitive methods and obtain poor yields as they lack scientific and technological knowledge.

(c) The location of the market is an important factor. Markets located at a great distance high transportation costs.

(d) Lack of transportation facilities.

(e) The availability of cheap and efficient labour for the cultivation of crops is important. For example, intensive agriculture requires a large supply of cheap labour.

(f) Agriculture is becoming mechanised and requires huge capital investments to purchase machinery, fertilisers, pesticides, and High Yielding Variety (HYV) seeds. The Indian farmers are too poor to buy all these materials.

(g) Globalisation has posed a great threat to Indian farmers. The international market is a significant challenge to the farmers because new agricultural products are being imported easily to India.

(h) The price of farm products in the international market is declining, while in India the price is increasing.

(i) The reduction in import duties on agricultural products proved detrimental to Indian agriculture.

(iii) Institutional Factors:

(a) The average size of landholdings is very small and subject to fragmentation due to **land ceiling** acts and family disputes.

(b) Landholdings are uneconomic due to their small size, and as such the yields are low.

(c) The small land holdings do not generate good income, which results in sale of a small portion of land by the small farmers to repay their debts.

(d) The land tenure system in India is a problem for farmers, making their life miserable.

(e) Though the tenancy problem has been solved to a certain extent Indian farmers are still suffering from insecurity of tenancy.

(f) Absentee landowners cultivate their land cultivated through tenants and sharecroppers which results in less production due to their limited interest.

(iv) Technological Factors:

(a) A majority of Indian farmers are still dependent on primitive and outdated techniques for crop production.

(b) They use inadequate and obsolete implements and fail to apply modern science and technology to agriculture in India because of their poverty.

➢ **Reforms:**

(i) Agriculture is the backbone of the Indian economy. Due to various factors affecting Indian agriculture, the GDP had been declining in the past, which was a serious concern.

(ii) The Government of India took various measures to overcome the decline Gross Domestic Product (GDP).

(iii) It established the Indian Council of Agricultural Research (ICAR), Agricultural Universities, Veterinary Services, Horticulture Development, Research and Development in Meteorology and Weather Forecasting, and **Kisan Call Centres**.

(iv) The **Green Revolution** and the National Agricultural Policy (NAP), introduced by the Indian Government, are the turning point in the development of Indian agricultural sector.

(a) The Green Revolution:

(i) The Green Revolution is one of the major breakthroughs in the agricultural sector in India.

(ii) It is considered the greatest revolution that brought a transformation from food scarcity to food self-sufficiency.

(iii) The Green Revolution was a technological package comprising material components of improved high-yielding varieties of two staple cereals – rice and wheat – with the help of irrigation, fertilisers, pesticides, and associated management skills.

(iv) The Green Revolution had a great impact on Indian agriculture. These impacts are as follows:

1. It enhanced agricultural production and transformed Indian agriculture from subsistence farming to commercial or market-oriented farming.

2. There was a spectacular increase in the production of wheat. Besides wheat, rice, sugarcane, and oilseeds also showed significant increase in the production.

3. A remarkable improvement was also seen with an increase in yields per hectare.

4. The strategy also benefited associated industries like transportation, marketing, food processing, etc., which helped generate additional job opportunities in both the agricultural and non-agricultural sectors.

5. It has paved the way for latest and modern technology to raise productivity per unit of land.

6. The Green Revolution and the new strategy also made a significant change in the cropping pattern.

7. It improved the economy of farmers and increased rural prosperity.

8. The import of food grains has declined considerably.

9. With the adoption of HYV seeds, chemical fertilisers, irrigation methods, production has been enhanced to a quiet high level.

(v) During the 1960's, India adopted the New Agricultural Strategy. It to replace the traditional agricultural practices with modern technological methods and practices.

(vi) The main elements of the New Agricultural Strategy are:

1. Use of significant capital and technological inputs.

2. Adoption of modern scientific farming methods.

3. Use of HYV (High-Yielding Varieties) seeds.

4. Extension of irrigation facilities, particularly groundwater resources.

5. Proper use of chemical fertilisers.

6. Improvement of marketing and storage facilities.

7. Use of insecticides and pesticides.

8. Consolidation of landholdings.

9. Provision of agricultural credit.

10. Rural electrification.

➢ Besides the Green Revolution, many steps have been taken to improve the agricultural production in India. These are as follows:

(i) Enactment of laws to prevent sub-division and fragmentation of land beyond a certain limit.

(ii) Introduction of various land reforms.

(iii) Rational utilisation of the country's water resources to optimum use of irrigation potential.

(iv) The Government declares support prices for farmers, minimises fluctuations in commodity prices, and monitors international prices.

(v) Setting up of Kisan Call Centres, also known as Farm Tele-Advisors (FTAs).

(vi) The Government of India provides subsidies on fertilisers.

(vii) In order to reduce dependence on chemical fertilisers and to increase organic food production, the Government of India launched a national project on **Organic farming**.

(viii) To avoid excessive use of chemical fertilisers, soil testing laboratories have been established to assess soil health and fertility.

(b) National Agricultural Policy (NAP):

The National Policy on Agriculture seeks to:

(i) Tap the growth potential of Indian agriculture.

(ii) Strengthen rural infrastructure.

(iii) Promote the growth of agri-business.

(iv) Create employment in rural areas.

(v) Address challenges arising from economic liberalisation and globalisation.

The salient features of the National Agricultural Policy are as follows:

(i) Aiming for over four per cent annual growth rate over next two decades.

(ii) Increased private sector participation through contract farming.

(iii) Minimising price fluctuations to protect farmers from risks.

(iv) Launch of the National Agricultural Insurance Scheme.

(v) Removing restrictions on the movement of agricultural commodities across the country.

(vi) Exemption from capital gains tax on compulsory acquisition of agricultural land.

(vii) Progressive institutionalisation of rural and agricultural credit.

(viii) High priority given to rural electrification.

(ix) Protection of plant varieties through legislations.

(x) Monitoring of international agricultural prices.

(xi) Adequate and timely supply of quality agricultural inputs to farmers.

(xii) Establishment of agro-processing units and creation of off-farm employment in rural areas.

(xiii) High priority to be given to the development of animal husbandry, poultry, dairy, and aquaculture.

Key Terms

➢ **GDP:** Gross Domestic Product refers to the monetary value of all finished goods and services produced within a country's borders in a specific time period.

➢ **Salinisation:** The process by which salt concentration increases in the soil negatively affecting crops, pastures and trees.

➢ **Crop Rotation:** The practice of growing different types of crops in succession on the same land to maintain soil fertility.

➢ **Pruning:** The act of cutting off unwanted or excess branches from plants or trees.

➢ **Land Ceiling:** The legal limit on the maximum size of land that an individual is allowed to own.

➢ **Absentee Landowners:** Individuals who lease or rent out land for profit but do not reside on or manage the property themselves.

➢ **Green Revolution:** A major agricultural reform that transformed India from a state of food scarcity to one of food self-sufficiency.

➢ **Kisan Call Centres:** A government initiative launched by the Department of Agriculture to provide agricultural advice and services to farmers.

➢ **Organic Farming:** A farming method that involves growing and nurturing crops without the use of synthetic fertilisers and pesticides, focusing on natural growth and soil health.

➢ **Animal Husbandry:** A branch of agriculture dealing with the breeding and care of animals for meat, milk, eggs, fibre, and other products.

TOPIC-2: TYPES OF FARMING IN INDIA

Concepts Covered: Subsistence Farming, Shifting Agriculture, Plantation Farming, Commercial Farming, Intensive Farming, Extensive Farming, Mixed Farming

Revision Notes

➢ The farming system is based on the nature of land, climatic conditions, technological knowledge and irrigation facilities.

➢ In India different types of farming are practiced in different parts of the country.

➢ The different types of farming practiced in India are as follows:

(i) Subsistence Farming

(ii) Shifting Agriculture

(iii) Plantation Farming
(iv) Commercial Farming
(v) Intensive Farming
(vi) Extensive Farming
(vii) Mixed Farming.

(i) Subsistence Farming:

(a) Subsistence Farming is a self-sufficient farming in which the farmer grows enough food to feed himself and his family. Examples include tapioca, yam and maize.

Subsistence Farming
Scan Me!

(b) The farmers have small lands and do not use fertilisers and thus the yield is low.

(c) The output is mostly for local requirements with little or no surplus trade.

(d) The land holdings are small and scattered.

(e) The farmer uses simple and primitive tools with traditional method of agriculture.

(f) Thus, farming tends to be very intensive and double or triple-cropping is practiced. Such type of farming is called the intensive Subsistence Farming.

(g) A good amount of hand labour is required.

(h) This type of farming is highly dependent on monsoon since there is no irrigation facilities and also depends on the natural fertility of the soil.

(ii) Shifting Agriculture:

(a) Shifting agriculture is known as "Slash and Burn Method".

(b) It is a primitive method of farming in which a patch of forest is cleared by felling or by burning trees.

(c) The patch of land is cultivated using primitive tools like sticks and hoes, with no use of machines.

(d) Cultivation on this land is temporarily, usually for two to three years or until the soil fertility is lost.

(e) The farmer then shifts to another piece of land and repeats the same method.

(f) This type of agriculture poses a serious threat to the environment.

(g) It leads soil erosion and may causes floods.

(h) Crops grown in this type of farming include maize, millets, barley, buckwheat, root crops, rain-fed rice, and vegetables.

(i) Yield per hectare is quite low, as no fertilisers are used.

(j) This method is known different names in various regions in India. It is called Jhum in Assam, Poonam in Kerala, Koman or Bringa in Odisha, Khil in the Himalayan region, Podu in Andhra Pradesh, Kuruwa in Jharkhand, and Bewar, Masha, Penda, and Hera in parts of Madhya Pradesh.

(k) The government discourages shifting agriculture, as frequent movement from one plot to another has adversely affected the ecology of the regions.

(iii) Plantation Farming:

(a) It is a commercial type of farming where a single cash crop is cultivated on a large scale. The farmland is known an estate. This is a monoculture agricultural system.

Planting of Tea
Scan Me!

(b) Crops such as tea, coffee, rubber, spices are grown under plantation farming mainly for trade and profit.

(c) This type of farming is practised on large tracts of land, ranging from a few to thousands of hectares.

(d) Only one type of crop is cultivated throughout the year such as rubber, bananas, tea, coffee and cocoa.

(e) Modern methods, techniques and machinery are used for growing crops.

(f) A large amount of capital is invested in purchasing machinery, fertilisers, pesticides, and in setting up processing factories.

(g) Due to the large size of the plantation, large number of labourers are required to tend the crops and work in the nearby processing factories.

(h) Plantation crops earn a substantial of foreign exchange as they are exported in large quantities.

(iv) Commercial Farming:

(a) It is a type of agriculture where crops are grown for sale in the market for profit.

(b) Crops such as sugarcane, oilseeds, jute and cotton that are sold in the market are also called cash crops.

(c) It largely depends on machinery, uses HYV seeds, chemical fertilisers, pesticides, and insecticides to obtain higher yield.

(d) This type of farming is practised in large farms covering over hundreds of hectares.

(e) The degree of commercialisation of agriculture varies in different parts of the country. For example, rice is grown as a commercial crop in Punjab and Haryana, but it is a subsistence crop in Odisha.

(f) As most states have small landholdings, commercial farming is not widely practised in the country.

(v) Intensive Farming:

(a) It is a system that involves higher input of labour, increased use of fertilisers, pesticides, high quality seeds, etc., and also produces higher output relative to the size of the land area. Examples include Soyabean and tomato.

(b) It is practised in the regions with a high population density.

(c) It requires significant amount of irrigation, as it is characterised by a high incidence of multiple cropping.

(d) It involves greater use of both labour and capital.

(e) This type of farming results in high yield per hectare.

(f) Rice and wheat are the main crops grown in intensive farming.

(g) It is generally practised close to markets.

(vi) Extensive Farming:

(a) Extensive Farming, unlike intensive farming requires less labour to cultivate large areas of land.

(b) It uses machinery and scientific methods to produce large quantities of crops.

(c) Mechanisation is effectively employed over large, and flat areas.

(d) It is a large-scale and cost-effective farming technique practiced in a moderately populated area.

(e) Although the yield per hectare may be low, the total output is very high in this type of farming.

(f) Crop specialisation is a key characteristics of this type of farming.

(g) The main crops grown are rice, wheat, and sugarcane.

(h) One of the advantages of extensive farming is that local environment and soil are not harmed by the overuse of chemicals.

(i) This type of farming is not common. It is practised in parts of the Terai region of sub-himalayas and in parts of north-western India.

(j) It is generally practised in remote locations far from markets.

(vii) Mixed Farming: **[Board 2024]**

(a) It is a type of agriculture in which a farmer undertakes different agricultural practices on a single farm to increase income from multiple sources, while also rearing cattle.

(b) In this type of farming, both food and fodder crops are given equal importance.

(c) Along with farming, other activities carried out include poultry, farming, dairy farming, bee keeping, sericulture, piggery, goat and sheep rearing, and agroforestry.

(d) The main advantage of this type of farming is that it ensures a steady income for farmers, as if one activity fails, others can provide support.

(e) It helps maintains soil fertility, biodiversity, minimises soil erosion, and aids in water conservation.

(f) Farmers can grow crops such as sorghum, Pusa Giant Napier, and berseem as fodder for their cattle alongside food crops.

Key Terms

- **Subsistence Farming:** A self-sufficient farming in which the farmers grow enough food to meet the needs of themselves and their families.
- **Shifting Agriculture:** A primitive method of farming where a patch of forest is cleared—either by cutting down trees or burning them—to cultivate crops temporarily.
- **Plantation farming:** A commercial farming in which single cash crop is cultivated on a large scale, typically on an estate.
- **Commercial Farming:** A type of farming in which crops are grown for sale in the market, rather than for personal consumption.

TOPIC-3: AGRICULTURAL SEASONS AND FOOD CROPS

Concepts Covered: Different Types of Crops on the Basis of Season of Sowing, Methods of Cultivation of Rice and Other Food Crops

Revision Notes

- **Crop:** A crop is a cultivated plant grown either on a large scale for commercial use or on a small scale for personal consumption.
- In India, different crops are grown in different seasons.
- India has three main agricultural seasons:

(i) Kharif Season **[Board 2024]**

(ii) Rabi Season

(iii) Zaid Season

(i) Kharif Season:

- Kharif crops are grown in the months of June and July and harvested in September and October.
- Rice, jowar, sugarcane, bajra, ragi, maize, cotton, and jute are some of the important Kharif crops.

(ii) Rabi Season:

- In Rabi season, the crops are sown in October and November and harvested in March and April.
- Crops such as wheat, barley, rapeseed, linseed, gram, peas, mustard and potatoes are grown as rabi crops.

(iii) Zaid Season:

- Some crops are grown just after the Rabi season during the summer months. These are known as Zaid crops.
- Zaid crops are sown in February and March and harvested in May, such as maize, watermelons, and cucumber.
- In India, agriculture occupies about 60% of the total cropped area.
- With the advent of Green Revolution technology, India focused achieving self-sufficiency in food grain production. The Green Revolution led to an increase in crop production, especially of wheat and rice.
- With all favourable conditions for agriculture available in India, farmers grow almost each and every crop.
- The Green Revolution led to an increase in crop production, especially in wheat and rice.
- Important crops grown in India are rice, wheat, pulses, **millets**, barley, jowar, gram, oats, maize, rye, etc. They fall under the category of food crops called **cereals**. **[Board 2024]**
- The cereals occupy about 54% of total cropped area in India

Some major crops in India:

- **Rice:**

(i) It is the most important staple food crop in India.

(ii) India is one of the world's largest producers of both white and brown rice.

(iii) Rice is an indigenous crop grown in all parts of the country especially in the northeastern and in the coastal parts of southern India.

(iv) Rice is grown in the rain-fed areas where the annual rainfall is heavy, making it a kharif crop.

(v) The climatic conditions required for rice cultivation are as follows:

1. **Temperature:** above 25°C
2. **Rainfall:** above 100 cm
3. **Soil:** Deep, fertile, clayey or loamy soil that can retain water in the field.

➢ In India, the Rice crops are grouped into two categories:

(i) The Upland Rice: It is grown on mountainous regions, sown in March–April and harvested in September–October. It is used locally and depends entirely on rainfall.

(ii) The Lowland Rice: It is grown in low lying areas, sown in June and harvested in October. It requires plenty of water and is locally used and supplied to other regions.

➢ Methods of Cultivation: Rice is cultivated by two methods in India. They are:

(i) The Dry Method of Rice Cultivation

(ii) The Wet Method of Rice Cultivation or Puddling

(i) The Dry Method of Rice Cultivation:

➢ Rice grown by dry methods is confined to rain-fed areas and do not have any irrigation facilities.

➢ In areas of moderate rainfall, seeds are scattered by hand, while in regions with heavy rainfall, they are sown in rows using drills.

➢ Wet method of cultivation is practiced in areas with good supply of water. The field is ploughed and filled with 3 to 5 cm of standing water.

➢ Farmers prepare their fields before the rainy season. Weeds are removed, and the field is ploughed to a depth of few inches. Then, manure and fertilisers are added to the soil. Finally, the whole surface of the farm is covered with water of about 2.5 cm. The field is then ready to receive seedlings from the nursery.

➢ **The steps followed to cultivate rice are:**

(i) Sowing and (ii) Processing

(i) Sowing

➢ In India, rice is sown using the following methods:

(a) Broadcasting Method: After ploughing, the seeds are scattered across the field before the onset of the monsoon.

(b) Drilling Method: In this method, seeds are sown in the furrows using a drill made of bamboo.

(c) Dibbling Method: This refers to sowing seeds at regular intervals in the ploughed furrows.

(d) Transplanting Method: Seedlings are first grown in nurseries, and after four to five weeks, when the saplings reach a height of 25 cm to 30 cm, they are transplanted to prepared rice fields. This is a popular method as it gives a higher yield.

➢ **Advantages of the Transplanting Method:**

(1) It enables the select of only healthy seedlings for the planting.

(2) Reduces wastage of seeds.

(3) It minimises weed growth due to prior nursery sowing.

(4) It results in a higher yield.

(e) Japanese Method: It was introduced in 1953 and is the most popular method. In this method, High Yielding Variety (HYV) seeds called Japonica are used.

➢ Important features of Japanese method of rice cultivation:

(i) Use of High Yielding Variety (HYV) seeds.

(ii) Saplings are sown and raised in the nursery beds for four to five weeks.

(iii) Manure is extensively used to increase the yield.

(ii) Processing of Rice: The harvested crop undergoes several processes before it become consumable. They are:

(a) Harvesting: The cutting and gathering of mature crops is called harvesting. A sickle is used to cut the stalks, which are then left to dry in the fields for 3 to 4 days. It is labour intensive process.

(b) Threshing: It is done by beating the sheaves against wooden bars to separate the grains from the stalks.

(c) Winnowing: It is the process of removing the unwanted husk from the grains.

(d) Milling: This process remove the yellowish husk from the grains. Traditionally done by using a wooden mortar, it is now carried out by machines.

➢ The leading rice-producing states in India are: West Bengal, Tamil Nadu, Punjab, Uttar Pradesh, Andhra Pradesh and northeastern states of India.

➢ **Wheat:**

Wheat Cultivation in India

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(i) It is a staple food for people in the northern and northwestern parts of the country.

(ii) It grows best in a cool, moist climate and ripens in a warm and dry climate.

(iii) It is mostly confined to the cool winter regions.

(iv) In the south, the growing period is shorter than in the north.

(v) Wheat is a Rabi crop.

(vi) The climatic conditions are:

1. Sown in October–November and harvested in January in the south, by March–April in the north.
2. **Temperature:** 10°C to 15°C is suitable for sowing and 20°C to 25°C during harvest.
3. **Rainfall:** 50 cm to 100 cm.
4. **Soil:** It grows best in well-drained loamy and clay loam soil, especially black soil.

➢ **Methods of wheat cultivation:**

1. Sowing:

(i) Seeds can be sown by using either the drilling or the broadcasting method.

(ii) The seeds germinate in about three or four days.

(iii) A low temperature is required during the growing season.

2. Harvesting:

(i) Wheat is harvested in April, when the temperature is around 27°C.

(ii) The crop is harvested by using a sickle.

(iii) States such as Punjab, Haryana, Uttar Pradesh, Rajasthan and Bihar use machines for harvesting.

(iv) Threshers are used to separate the grain from the **husk.**

➢ India has shown a tremendous increase in the wheat production in comparison to other crops grown in the country.

➢ The leading wheat-producing states in the country are Uttar Pradesh, Punjab, Haryana, Rajasthan and Madhya Pradesh.

➢ **Millets:**

(i) Millets refers to coarse grains like jowar, bajra, and ragi, which serve as food grains. They are generally used by people living in rural areas.

(ii) The straw of these grains is a valuable cattle fodder.

(iii) These crops can grow in infertile soil withstand harsh climatic conditions. As millets do not require much rainfall to grow, they are also known as **'dry crops'**.

(iv) They grow over a short period of time, i.e., for 3 to 4 months.

(a) Jowar:

(i) It is both, a kharif and a rabi crop.

The climatic conditions are:

(ii) It grows well in dry areas, even without irrigation.

(iii) Temperature: Between 27°C and 32°C

(iv) Rainfall: Below 45 cm. The crop can grow in arid and semi-arid areas.

(v) Jowar can grow on various types of soil, including from heavy and light alluvium, red, grey soil and yellow loams.

(vi) It is widely grown in Maharashtra, Madhya Pradesh, Karnataka, Andhra Pradesh, Telangana, Tamil Nadu, Uttar Pradesh, Gujarat, and Rajasthan.

(b) Bajra:

(i) It is a rainfed kharif crop, grown both as a pure and mixed crop.

(ii) It is grown along with cotton, jowar, and ragi.

The climatic conditions are:

(iii) It is sown in June–July and harvested in September–October.

(iv) Temperature: 25°C and 30°C.

(v) Rainfall: Less than 50 cm.

(vi) Bajra is grown on red soil or sandy loamy soil.

(vii) It is grown mainly in Rajasthan, Maharashtra, Gujarat, Uttar Pradesh, and Haryana.

(c) Ragi or Buckwheat:

(i) It is grown in drier parts of south India almost throughout the year with the help of irrigation.

The climatic conditions are:

(ii) Temperature: 20°C to 30°C.

(iii) Rainfall: 50 cm to 100 cm.

(iv) Ragi is sown between May and June and harvested between September and October.

(v) It can grow in dry conditions and withstand severe drought.

(vi) It is grown on red, light black, and sandy loam soil in Karnataka and Tamil Nadu, and on alluvial loam soil in Uttarakhand, Jharkhand, and Gujarat.

(vii) It gives a higher yield than jowar and bajra but a lower yield than wheat and rice.

(viii) Karnataka is the leading producer of ragi in India. The other producers of ragi in India are Tamil Nadu, Uttarakhand, Maharashtra, and Andhra Pradesh.

➢ **Pulses:**

(i) Pulses form an important part of the Indian diet because they are rich in protein.

(ii) Pulses are grown as rotation crops, being leguminous plants that fix atmospheric nitrogen and enhance fertility.

(iii) Pulses also serve as good cattle fodder.

(iv) The two most important pulses are *Gram* and *Tur*. Other important pulses include *Urad*, *Moong*, *Masur*, *Kulthi*, *Matar*, *Khesari* and *Moth*.

The climatic conditions are:

(v) Temperature: 20°C to 25°C.

(vi) Rainfall: 50 cm to 75 cm.

(vii) Pulses grow on dry light soil, light loamy and Alluvial, Black and Red soils as well—i.e., they can grow in almost any kind of soil.

(viii) *Tur*, *Urad*, and *Moong* are grown as Kharif crop in most part of India, but khesari, masur, matar and gram are grown as Rabi crops in north India.

(ix) *Gram* is grown as Rabi crop and is sown mixed with Wheat.

(x) India is the largest producer and consumer of pulses in the world.

(xi) The major pulse-producing states in India are Madhya Pradesh, Maharashtra, Uttar Pradesh, Rajasthan, and Andhra Pradesh.

Key Terms

➢ **Crop:** A crop is a plant that is cultivated or grown. It can be grown on a large scale for commercial purposes or on a small scale for self-consumption.

➢ **Kharif Season:** This cropping season is from July to October during the south-west monsoon. The main crops are jute and rice.

➢ **Rabi Season:** This cropping season is from October to March. The main crops are wheat and mustard.

➢ **Zaid:** Agricultural crops that are grown in the short duration between rabi and kharif crop seasons, mainly from March to June.

➢ **Cereals:** It denotes all kinds of grass-like plants which with starchy and edible seeds like, rice, wheat. barley, maize, oats, and millets.

➢ **Upland Rice:** It is grown on mountainous regions, sown in March–April and harvested in September and October.

➢ **Lowland Rice:** It is grown in low-lying areas, sown in June and harvested in October. It requires plenty of water and is locally used and supplied to other regions too.

➢ **Broadcasting Method:** After ploughing, the seeds are scattered all over the field before the onset of the monsoon.

➢ **Drilling Method:** In this method the seeds are sown in the furrows with the help of a drill made of bamboo.

➢ **Dibbling Method:** It refers to the sowing of seeds at regular intervals in the furrows.

➢ **Transplanting Method:** Seedlings are first grown in nurseries, and after four to five weeks, when the saplings attain a height of 25–30 cm, they are transplanted to prepared rice fields.

➢ **Japonica:** The high-yielding variety (HYV) seeds used in the Japanese method of rice cultivation.

➢ **Dry Crops:** As millets do not need much rainfall to grow, they are also known as 'dry crops'.

TOPIC-4 : CASH CROPS: SUGARCANE AND OILSEEDS

Concepts Covered: Types of Cash Crops, Methods of their Cultivation-Sett, Ratooning, Crystallisation, Harrowing

Revision Notes

➢ Cash crops are those crops that are primarily grown for sale and export, rather than for the farmer's own consumption. Examples: Tea, sugarcane, etc.

➢ These crops provide provide raw material to agro-based industries and support the farmers financially to improve their living conditions and their farming practices.

➢ The main cash crops are classified as:

(A) Plantation Crops:

➢ Sugarcane

(B) Oilseed Crops:

➢ Groundnut

➢ Mustard

➢ Soya bean

(C) Fibre Crops:

➢ Cotton

➢ Jute

(D) Beverage Crops:

➢ Tea

➢ Coffee

(A) Plantation Crops: Plantation crops are high-value crops of great economic importance, cultivated on large-scale agricultural unit usually of a single crop. For example, Sugarcane.

➢ It belongs to the grass family and grows to a height of more than 3.5 metres.

➢ The sugar in the sugarcane plant is stored in its stem.

➢ Sugarcane is the main source of sugar, *gur* and *khandsari*.

➢ The sugarcane yield per hectare is higher in the south due to the tropical climate of Peninsular India and long crushing season of about 8 months.

➢ **Climatic conditions required for sugarcane are:**

1. Temperature: Grows best in areas with temperatures between 20°C and 24°C.

2. Frosts are dangerous and harmful to sugarcane crop.

3. A dry and cool winter season is necessary during the ripening and harvesting period.

4. It requires 100–150 cm of rainfall throughout the year.

5. Areas of low rainfall require irrigation.

6. The crop grows well in well-drained, rich alluvial, heavy loam or lava soil.

7. It is also grown on black soil, reddish loam, and laterite soil in the peninsular region.

8. Sugarcane is a soil-exhausting crop and thus needs fertilisers, manure and good irrigation facilities.

➢ **Methods of Cultivation:**

1. Sowing: Sugarcane is a labour-intensive crop, planted by the following methods:

(i) Sett Method:

(i) In this method new canes are planted using cuttings from old sugarcane plants.

(ii) These cuttings, called setts, sprout buds that form new stalks within a few days.

(iii) From each cuttings four to five stalks grow.

(iv) A sugarcane plant takes 10 to 15 months to mature.

(ii) Ratooning Method: **[Board 2024]**

➢ It is a method where, during harvesting, the roots and the lower parts of the plant are left in the field to give the ratoon or the ratoon crop.

➢ The successive crops that grow from the remnants is called the **Ratoon**.

➢ Sugarcane can produce for two to three years, though each successive crop yields less than the previous one.

➢ **Advantages of Ratooning:**

1. It saves labour as the does not need to be planted again.

2. This method is inexpensive, as no preparation of the field is required .

3. Ratoon matures early.

➢ **Disadvantages of Ratooning:**

1. Ratoons produce lower-quality crop, as the canes become thinner with low sucrose content over successive years.

2. There is high risk of pests and diseases.

(iii) By Seeds:

➢ Sugarcane was once grown by sowing seeds but this method is obsolete.

➢ It is planted in furrows and then covered with soil.

Sugarcane Seed Production

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- It is grown as a mixed crop in some states of India.

2. Harvesting:

- Sugarcane is harvested when it matures in 10–12 months.
- Harvesting is done before the cane begins to flower.
- The sugarcane harvesting season begins in October–November and ends in April.
- Harvesting is done manually using hand knives, cutting blades, or hand axes.
- It requires skilled labourers, as the stalks must be cut very close to the ground level because the maximum sucrose content is in the bottom of the stem.

3. Processing:

- After harvesting, the canes are taken to the sugar mills quickly to be processed within 48 hours to preserve the sucrose content.
- In the mills, the canes are crushed between the rollers to extract most of the juice.
- To remove soluble and insoluble impurities, the juice is boiled with lime.
- Non-sugar impurities are removed by continuous filtration.
- The juice is then is concentrated by removing water through vacuum evaporation.
- **Crystallisation** occurs, forming brown sugar.
- The by-products of sugarcane are **bagasse, molasses, and press-mud**.
- In India, two-thirds of the sugar produced is used by the *Gur* and *Khandsari* industries.
- **Problems of Sugarcane Cultivation:**

(i) Sugarcane is a soil-exhausting crop and thus needs good amount of fertilisers, increasing production costs.

(ii) In India, the yield per hectare is very low compared to other countries.

(iii) Sugarcane's crushing season lasts only 4–7 months, leading to financial issues as mills and workers remain idle.

(iv) Sugar mills are located far from fields; a delay of more than 24 hours reduction sucrose content in the canes.

(v) Sugarcane is an annual crop, but less land is available for it, limiting crop rotation options for farmers.

(vi) The production cost of sugarcane in India is the highest in the world due to uneconomic process of production, inefficient technology and heavy excise duty.

(vii) Many mills are small and economically unviable.

(viii) Most Indian sugar mills use old, obsolete machinery that needs rehabilitation.

(ix) Sugar industry is facing competition with *Gur* and *Khandsari*, which are excise duty-free and offer better cane prices.

(x) Sugarcane cultivation requires a large amount of water, but irrigation facilities are inadequate.

(xi) Government-fixed prices for sugarcane are often unprofitable for farmers.

- **Role of Government in Solving Problems Faced by Sugarcane Farmers:**

(i) Establish cooperative societies to support sugarcane farmers.

(ii) Develop irrigation facilities to ensure regular water supply to sugarcane fields.

(iii) Provide adequate and timely loans to farmers for purchasing farm machinery and other agricultural inputs.

(iv) Educate farmers on modern techniques through dedicated radio and television programmes.

(B) Oilseeds: These are raw materials for industries such as vegetable oil, hydrogenated oil, paints, varnish, soap, and pharmaceuticals. Examples include groundnut, mustard, and soybean.

- India produces a wide variety of oilseeds.
- India has the largest under oilseed cultivation globally, and oilseeds are a significant source of foreign exchange.
- The principal oilseeds are:

(a) Groundnut

(b) Mustard

(c) Soyabean

- These oilseeds are used for different purposes—some for cooking and some as industrial raw materials in paints, varnishes, hydrogenated oil, soaps and lubricants.
- Groundnut is the leading oilseed, followed by mustard.
- After oil extraction, the remaining residue is known as oil cake.
- Oil cake is used as animal fodder and also serves as good farm manure.
- India is one of the largest oilseed producers in the world.

(a) Groundnut:

- Groundnut is a kharif crop in most parts of India, but a rabi crop in Odisha and southern states.
- It is mainly used to manufacture of hydrogenated oil, margarine, soap, medicines, and cooking oil.
- It is also known as the peanut and monkey nut.
- It is consumed raw, roasted or salted.
- Its **oil cake** is excellent cattle feed.
- **There are Two Types of Groundnut Plants:**

1. The Runner Type
2. The Bunch Type

- **Climatic Conditions:**

1. Temperature: 20°C to 25°C
2. Rainfall: Between 50 cm–100 cm
3. Black soil, sandy loam, and loamy soil are ideal for the crop.
4. The crop is highly susceptible to frost, prolonged drought, continuous rain and stagnant water.

- **Methods of Cultivation:**

(i) Sowing:

1. After ploughing, seeds are sown by scattering, or broadcasting, or drilling method.
2. In most part of India, the seeds are sown in June or July, but in south they are sown in the month of February and March.
3. Seeds are placed at a depth of 5–6 cm in the soil.

4. Adequate moisture in the top 60 cm layer of soil is essential for high yield and good-quality groundnut seeds.

5. **Weeds** damage the crop, so they are controlled mechanically and chemically.

6. Mature pods have wrinkled shells containing one to four seeds each.

(ii) Harvesting:

1. The crop should be harvested at the right time to obtain higher yields of pods and oil.

2. Pre-harvest irrigation is ideal as it loosens the soil for easier harvesting.

3. The Groundnut plant, along with its roots, is uprooted by hand or machine.

4. Once dried, peanuts are threshed to separate the pods from the rest of the bush.

5. Groundnuts are then packed and sent for processing at mills or to markets for trade.

(viii) India is the second-largest producer of groundnut in the world.

(ix) It is widely grown in peninsular India, especially in Telangana and Tamil Nadu, which are the largest producers in the southern India.

(x) In India, Gujarat is the leading producer of groundnuts followed by Maharashtra, Karnataka, Andhra Pradesh, Rajasthan, Madhya Pradesh, Uttar Pradesh, and Punjab.

(b) Mustard:

➢ Mustard is an edible oil seed and one of the most important grown in India.

➢ Mustard thrives in temperate regions, thus it and widely grown in northern India, especially in Uttar Pradesh, Punjab, and Haryana.

➢ It is a Rabi Crop and is also grown mixed with wheat, gram and barley.

➢ In the north, it is mainly used for cooking and its oil cake is used as animal fodder.

➢ Mustard leaves are eaten as vegetable in the north and are also used as a manure.

➢ Mustard grows well in cool climatic conditions and is widely cultivated along the Ganga–Sutlej Plains.

➢ **Climatic conditions:**

1. **Temperature:** 10°C to 20°C

2. **Rainfall:** 25 cm to 40 cm.

3. **Soil:** Alluvial loam is the best, though mustard also grows in sandy to heavy clay soils.

➢ **Methods of Cultivation:**

1. The crop is grown in the winter season.

2. It is grown in rows with wheat, gram and barley.

3. It is sown using the broadcasting or drilling method.

4. Harvesting should be done when the pods begin to turn yellow and the seeds harden.

5. A sickle is used to cut the mustard plants.

6. The plants are tied and and left to dry for five to six days.

7. Threshing is using a stick, and winnowing separate the grain from the husk.

8. It is extensively cultivated in Uttar Pradesh, Punjab, Haryana, Rajasthan, West Bengal, Assam, Bihar, Odisha, Gujarat, and Jammu and Kashmir.

(c) Soyabean:

➢ Soyabean is grown as a Kharif crop in India.

➢ It is rich in protein and in high demand.

➢ It is considered a substitute for animal protein.

➢ It is consumed as soya milk or tofu (soya cheese).

➢ **Climatic Conditions:**

1. **Temperature:** 13°C to 24°C

2. **Rainfall:** 40 cm to 60 cm

3. **Soil:** Moist alluvial soil and friable loamy acidic soils, grows best on sandy loam rich in organic matter.

➢ **Methods of Cultivation:**

1. Soyabean is sown in 40 cm to 50 cm apart through drilling method.

2. It is a rainfed crop and usually does not require irrigation.

3. One deep ploughing and two **harrowings** is done at sowing to maintain optimum soil moisture.

4. Harvesting is carried out in mid-October, usually using threshing machine.

➢ Soyabean is mainly produced in Madhya Pradesh, Rajasthan and Maharashtra. Madhya Pradesh (M.P.) is the leading producer.

Key Terms

➢ **Setts:** Cuttings taken from high-yielding mother plants. These cuttings are called setts. Buds from these cuttings sprout to form new stalks after a few days.

➢ **Ratoon:** A new shoot that grows from the root or crown of a crop plants after harvest, e.g., sugarcane.

➢ **Crystallisation:** A process that separates a pure solid from a solution in the form of its crystals.

➢ **Bagasse:** The dry fibrous residue remaining after the extraction of juice from the crushed stalks of sugarcane. It is used in manufacturing pulp and as biofuel, and is a byproduct of sugarcane.

➢ **Molasses:** A thick, viscous byproduct of refining sugarcane or sugar beets into sugar, used for making yeast, chemicals and pharmaceuticals.

➢ **Press-mud:** Organic solid waste from sugar mill converted into nutrient-rich and good quality organic manure. It is used in fertilisers, shoe polish and wax.

➢ **Hydrogenated oil:** Oil with trans-fatty acids, chemically altered from liquid to solid at room temperature.

➢ **Oil cake:** The residue that is left after the extraction of oil from the oilseeds.

➢ **Weeds:** Wild or unwanted plants that grow around the crops and absorbs their nutrients.

➢ **Harrowing:** A farming tool with spike like teeth or upright disks, used over ploughed land to level soil and uproot weeds.

TOPIC-5 : CASH CROPS: COTTON, JUTE, TEA AND COFFEE

Concepts Covered: Methods of Cultivation and Irrigation of Different Crops

Revision Notes

➢ **Fibre crops:** Fibre crops are field crops grown for their fibres (strong, thread-like materials), traditionally used in making textiles, rope, twine and similar products. For example, cotton and jute.

(a) Cotton: **[Board 2024]**

➢ Cotton is one of the most important fibre and major cash crops grown in India.

➢ It plays an important role in the textile industry and the agricultural economy of the country.

➢ It is a tropical crop grown as a kharif crop.

➢ Cotton in India provides direct livelihood to about 6 million farmers and around 40 million to 50 million people are employed in the cotton trade.

➢ Cotton is grown in black soil also known as regur soil, black cotton soil or black clay soil.

➢ **Climatic Conditions:**

(i) Temperature: Between 21°C and 27°C.

(ii) Rainfall: Moderate rainfall from 50 cm to 80 cm is ideal.

(iii) A long growing period of at least 200 frost-free days is necessary for the plant to mature.

(iv) Cotton grows on a variety of soils ranging from well-drained deep alluvial soils in the north to deep and medium black clay soils in the Deccan and Malwa Plateau and Gujarat.

(v) The quality of cotton depends on length of fibres, fineness, strength, and structure of its fibre.

➢ **Methods of Cultivation:**

(i) Sowing:

1. The seeds are sown using the broadcasting or drilling method.

2. The crop season lasts from 6 to 8 months.

3. Sowing is ideal before the onset of monsoon, i.e., May to September–October.

4. Drip irrigation is the most effective method of watering in cotton farming.

(ii) Harvesting:

1. Cotton can be picked by hand or by machine.

2. The cotton balls, after ripening burst into white, fluffy and shiny fibres.

3. In Punjab and Haryana, cotton is harvested in December–January, before the winter frost can damage the crop.

4. In the peninsular part of India, it is harvested between January and May, as there is no danger of winter frost in these areas.

(iii) Processing:

After harvesting, the cotton crop goes through the following process:

1. The freshly picked cotton is pressed into large bales.

2. The cotton gin mechanically separates the fibres from the seeds and turns it into ginned cotton, also called **lint**.

3. This process of separating cotton fibre from the cotton seed is called ginning.

4. The cotton lint or fibre, is pressed into large bales and transported to the textile mills.

5. At the mill, the bales are washed, combed, and made into an untwisted rope called a silver.

6. A spinning frame turns these silver directly into cotton yarn.

7. Lastly, the yarn is dyed, and looms are used to weave it into ready-to-use fabrics.

➢ Varieties of Cotton:

There are three varieties of cotton grown in India:

1. Long Staple: 24 mm to 35 mm

2. Medium Staple: 20 mm to 24 mm

3. Short Staple: less than 20 mm

➢ Cotton is extensively produced in Gujarat, Maharashtra, Andhra Pradesh, Telangana, and Punjab.

➢ In India, the chief cotton-growing areas are:

(i) North-western Deccan Region

(ii) Central and southern Deccan of Karnataka

(iii) North-west Region.

Major Cotton Producing States

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(b) Jute:

➢ Jute is obtained from the inner bark of two important species:

(i) The White Jute

(ii) The Tossa Jute

➢ The White Jute is hardy, highly adaptable, and grows well on both lowlands and uplands.

➢ Tossa Jute is grown only on uplands.

➢ Jute is used to manufacture a variety of products such as rugs, clothes, gunny bags, hessian, ropes, carpets, strings, tarpaulins, and upholstery.

➢ Jute is in great demand both internally and globally because of it's eco-friendliness, biodegradable nature, and changing mindset of consumers towards jute products over synthetic alternatives.

➢ Jute is referred to as **'Golden Fibre'** for its colour, silky shine, and high cash value, as it earns good revenue.

➢ It is also called the 'Brown Paper Bag' in wholesale trade, especially as the jute fabric is used to make sackcloth. These sacks are further used to pack grains, cement, and fertilisers.

➢ **Mesta** is an inferior substitute for Jute, can withstand drought conditions and can be grown under a wide range of climatic and soil conditions.

➢ Rough bags are made from Mesta.

➢ Jute is 100% biodegradable, recyclable, and environmental friendly.

➢ **Climatic Conditions:**

1. Temperature: Between 24°C and 35°C.

2. Rainfall: About 150–200 cm is ideal.

3. A warm and wet climate with relative humidity of around 90% is favourable.

4. Jute requires 2–3 inches of rainfall weekly during the sowing period.
5. Newly deposited alluvial soil in the Ganga Delta region is most suitable for jute cultivation.

➢ **Methods of Cultivation:**

(i) Sowing:

1. The land should be ploughed properly before sowing the jute seed.
2. Since the jute seed is small, the land should be prepared to a fine tilth.
3. Seeds are sown using either the drilling or broadcasting methods.
4. The seeds are sown in February on lowlands and in March–May on uplands.

(ii) Harvesting:

1. The jute crop takes 8–10 months to mature.
2. Harvesting begins in July and continues until October.
3. The harvesting is done by hand, either by pulling up the stem or cutting it at ground level, then tying it into bundles.

(iii) Processing:

1. After harvesting, the sheaves of jute bundles are immersed in floodwater or stagnant water for about two–three weeks for **retting**.
2. Retting is a microbiological process that loosens the outer bark and facilitates the removal of the fibre from the stalk.
3. After retting, the bark is peeled from the plant and the fibre is removed.
4. Then, the fibres are stripped, washed, rinsed and cleaned and dried in the sun and pressed into bales.

(xi) West Bengal is the leading producer of jute in the country, followed by Assam, Bihar, Odisha and Uttar Pradesh.

(D) Beverage crops: Crops that give suitable drink other than water, such as tea, coffee and juices.

(a) Tea: **[Board 2024]**

➢ It is an important beverage for the people of India as it acts as a stimulant.

➢ Tea gardens are set up on the hill slopes where shade trees are planted in advance.

➢ It grows well on hill slopes due to the favourable climatic conditions, high altitude and because it prevents from water stagnation.

➢ Though tea requires heavy rainfall, but waterlogging at the roots of the plant is harmful. Even prolonged dry spells are unfavourable for the growth of the tea plantations.

➢ A good amount of iron, humus, nitrogenous fertilisers like ammonium and sulphate are ideal for tea growth, resulting in a higher yield.

➢ High humidity, heavy dew and morning fog are beneficial for the growth of new leaves.

➢ **Climatic Conditions:**

1. Temperature: 20°C to 35°C
2. Rainfall: 150 cm – 250 cm
3. Soil: Well-drained, deep friable loamy soil, rich in humus and iron.

➢ Most of the tea plantations in India are found at the elevations ranging between 600 m to 1800 m.

➢ **Methods of Cultivation:** These are two ways in which the seeds are sown, as follows:

(i) From Seeds: High-quality seeds are sown in nurseries and once they grow into tiny plants called seedlings, they are transplanted into the tea estates at the distance of one metre from all sides. This is the transplantation method of tea cultivation. When the tea saplings reach at height of 20 cm, they are transplanted in the garden.

(ii) From Cuttings:

➢ Tea plants are grown in nurseries from cuttings of high-yielding varieties of crops. This is known as clonal planting.

2. Harvesting:

➢ Tea shrubs take three to five years to mature.

➢ Then they produce shoots called 'Flush'. Flush consists of two tender leaves and a bud.

➢ High-quality tea is obtained from flush.

It is done in two ways:

(i) Plucking of tea leaves

(ii) Pruning of tea leaves

➢ Plucking of leaves is done by women labourers.

➢ Tea is picked every 10 days in the lower altitudes but every 15 days in the higher altitudes.

➢ Tea picking is carried out from early April to mid-November.

➢ Two tender leaves and a bud or shoot are usually plucked from each stem and are considered to be fine plucking.

➢ Pruning tea leaves is an essential part of tea cultivation as it helps in maintain the proper shape of the tea bush to a height of about one metre.

➢ The objective of pruning is to encourage new shoots bearing soft leaves and to facilitate the plucking process.

3. Processing: Tea is classified into four types, as follows:

(a) Black Tea **(b)** Green Tea
(c) Oolong Tea **(d)** Brick Tea

(a) Black Tea: Production involves several key steps:

i. Withering: The tea leaves are spread out on racks and left to dry for 14–18 hours. They are also sun-dried left for a day or two.

ii. Rolling: The leaves are processed using the through CTC (Crushing, Tearing and Curling).

The leaves are rolled mechanically for 30 minutes between steel rollers to break up the cells to squeeze out the juices.

iii. Fermentation: The tea leaves are spread out on large boards in 10–15 cm thick layers in a special room with a room maintained at 40°C for two to three hours. The leaves turn copper-red to brown in colour and begin to release its unique aroma.

iv. Drying: After fermentation, the leaves are transported through tiered dryers on metal conveyor belts and are dried for approximately 20 min, giving the leaves its dark brown to black colour.

v. Blending: Tea tasters and expert blenders mix various grades of tea to flavour consistency, as taste varies with different geographical factors.

(b) Green Tea: These have a good flavour and act as stronger stimulants due to their higher tannin (or tannic acid) content.

i. **Withering:** The tea leaves are spread on laths and placed in the sun to wither, retaining their good qualities.

ii. **Heating:** The tea leaves are heated for 10 minutes at 280°C in cast-iron pans. Then, the leaves are pressed against the hot surface.

iii. **Rolling:** The tea leaves are rolled in a rolling machine for 10–15 minutes between two rotating metal plates.

iv. **Drying:** The leaves are dried at a temperature of 60°C for 20–30 minutes

(c) Oolong Tea: It is a type of tea produced through a process that includes withering the plant under strong sunlight and oxidation before curling and twisting. Withering, rolling, shaping, and firing are similar to black tea, but baking or roasting is exclusive to oolong tea.

(d) Brick Tea: This variety of tea is also called compressed tea. These are blocks of whole or finely ground black tea or green tea leaf dust that have been packed in moulds and pressed into rectangular block form.

➢ India is the world's second-largest producer of tea and the fourth-largest exporter of tea. Tea is widely grown in Assam, West Bengal, Tamil Nadu, Kerala, Tripura, Arunachal Pradesh, Himachal Pradesh, Karnataka, Sikkim, Nagaland, Uttarakhand, and Manipur. India is also the largest consumer of tea in the world.

(b) Coffee:

(i) It is the second-most important beverage crop in India.

(ii) Coffee is one of the oldest plantation crops in India. The first seedlings of coffee were sown in the Bababudan Hills in Karnataka.

(iii) Coffee cultivation requires plenty of inexpensive and skilled labour for various operations like sowing, transplanting, pruning, plucking, drying, grading and packing.

(iv) The three main varieties of coffee grown in India are:

(a) Arabica Coffee

(b) Robusta Coffee

(c) Liberica Coffee

(v) (a) Arabica Coffee: It is a superior quality and most expensive variety.

(b) Robusta Coffee: It is cheaper, as its production cost is lower compared to the yield per acre.

(c) Liberica Coffee: It is used for making instant coffee. The coffee plant cannot tolerate direct sunlight and is grown under shade trees like silver oak, orange, banana, jackfruit, and cardamom.

(vi) Climatic Conditions:

1. Temperature:

(a) Between 14°C and 26°C.

(b) It cannot tolerate frost, snow, high temperatures above 30°C and strong sunshine.

2. Rainfall: Annual rainfall between 125 cm and 250 cm is ideal. Stagnant water is harmful, it is grown on hill slopes at elevations of 600 to 1600 metres.

3. Soil: Well-drained, rich friable loams with humus and minerals like iron and calcium is ideal. Laterite soil rich in iron is also suitable for coffee cultivation.

➢ **Methods of Coffee Cultivation:**

1. Sowing:

(a) Coffee seeds are sown in December–January in beds 1.5 cm–2.5 cm apart.

(b) They are propagated from seeds as seedlings or cuttings in a nursery and, then transplanted into large coffee fields.

(c) Pruning is done regularly to ensure easy picking and heavy bearing of coffee berries.

2. Harvesting:

(a) Arabica coffee is harvested in October to November and Robusta from January to April.

(b) Coffee berries are hand-picked.

(c) The two methods of coffee harvesting are:

(i) Selective Harvesting

(ii) Strip Harvesting

(i) Selective harvesting: Selective harvesting involves hand-picking only ripe coffee berries.

3. Processing:

➢ There are two methods of coffee processing, i.e., obtaining coffee beans from the harvested berries.

They are:

(i) Wet Parchment Method: In this method, the fruit covering is removed before drying, followed by pulping, fermenting, washing and drying.

(ii) Dry Parchment Method: It involve the following steps:

1. Coffee berries are sorted, cleaned; ripe, overripe and damaged cherries are separated.

2. Dirt, soil, twigs and leaves are removed from the cherries.

3. The cherries are dried in the Sun and fermented by further sun-drying for a week.

4. After drying, machines peel off the two inner husks layers.

5. Beans are sorted by size and quality, and then packed in sacks for export.

6. Beans are roasted at 99°C and then ground into coffee powder. Roasting gives coffee its brown colour, aroma and taste.

➢ The traditional coffee-producing states in India are Karnataka, Kerala and Tamil Nadu.

➢ Other coffee-growing states include Andhra Pradesh, Odisha.

Key Terms

➢ **Drip Irrigation:** A type of micro-irrigation where water drips slowly onto plant roots from above or below the surface, conserving water and minimise evaporation.

➢ **Lint:** A short, fine fibre that separates from cloth or yarn during processing.

➢ **Ginning:** The process of separating cotton fibre from the seed.

➤ **Silver:** At the mill, cotton bales are washed combed, and made into an untwisted rope called a silver.

➤ **Mesta:** An inferior substitute for jute that withstand drought and grows in a wide range of climatic and soil conditions.

➤ **Retting:** A microbiological process that loosens the outer bark and allows fibre removal from the stalk.

➤ **Brown Paper Bag:** Jute is also called the 'Brown Paper Bag' in wholesale trades as it is used to make sackcloth for packing grains, cement, fertilisers, etc.

➤ **Golden Fibre:** Jute is known as 'Golden Fibre' for its colour, silky shine and high cash value.

➤ **Clonal Planting:** Growing tea plants in nurseries from cuttings of high-yielding varieties.

➤ **Pruning:** Trimming shrubs to promote growth and remove unwanted branches.

➤ **Withering:** Reducing moisture in tea leaves to soften them and develop flavour compounds.

CHAPTER-9
Manufacturing Industries

Learning Objectives

The students will be able to:

- Explain the importance and classification of agro-based industries—sugar and textile (cotton and silk).
- Identify mineral-based Industries—iron and steel (TISCO, Bhilai, Rourkela, Visakhapatnam), petrochemicals and electronics.

TOPIC-1: IMPORTANCE AND CLASSIFICATION OF INDUSTRIES

Concepts Covered: . Agro-based Industry, Mineral-based Industries, Heavy and Light Industry, Large-scale, Medium-scale and Small-scale Industries, Public Sector and Private Sector Industries

Revision Notes

➤ India is one of the newly industrialised countries and is making significant progress in industrial development.

➤ **Industrialisation:** It is the period of social and economic change that transforms a society from an agrarian society into an industrial one.

➤ Industrialisation contributes to the growth of agriculture, trade, transport, and all other economic activities.

➤ Industrialisation is key to economic development in India, with its vast manpower and varied resources.

➤ It is the process of manufacturing consumer and capital goods, and building infrastructure.

➤ **Industry:** An economic activity involving the production of goods, extraction of minerals, or provision of services.

➤ **Need for Rapid Industrialisation in India:**

(i) India is an agricultural country, and, industrialisation, can support the development of agriculture.

(ii) Agriculture alone cannot generate sufficient employment; hence, the establishment of industries can provide large-scale employment opportunities.

(iii) The development of industries producing capital goods such as machines, and equipment enables a country to produce a various goods in large quantities at low costs, leading to towards technological progress.

(iv) Industrialisation leads to the development of infrastructures such as railways, roadways, dams, which promotes the future growth of Indian economy.

(v) Industrialisation is essential for the country's security as it enables self-reliance in defence through the production of indigenous defence equipment.

(vi) Expansion of industries in the backward regions of India is needed to to reduce the regional imbalance.

(vii) India, through industrialisation, should free itself from the adverse effects of fluctuations in the global prices and should adopt a policy of importing fewer primary products while exporting more manufactured goods.

(viii) Industrialisation plays a major role in India's economic development by increasing national income.

➤ Factors Affecting the Location of Industries:

1. Geographical Factors:

(i) Raw Materials

(ii) Energy

(iii) Labour
(iv) Climate

2. Commercial Factors:
(i) Government Policies
(ii) Availability of Finance and Capital
(iii) Organisational and Managerial Skills
(vi) Water Supply
(v) Transport Facilities
(vi) Market Access

➢ **Classification of Industries:**

1. On the Basis Raw Material Source, Industries can be Classified into:

(a) Agro-Based Industries: These industries use raw materials obtained from agriculture. Examples: cotton, jute, textile, sugar, tea, and coffee.

(b) Mineral-Based Industries: These industries use metallic and non-metallic minerals as raw materials and involve as raw materials and are based on ferrous and non-ferrous metallurgical processes. Examples: iron and steel, heavy engineering, cement, machine tool, basic and light chemicals, and fertilisers.

(c) Forest-Based Industries: These industries use forest resources such as wood, rubber, lac, and resin.

(d) Animal-Based Industries: These industries use raw materials obtained from animals such as woollens, silk, dairy products, hides, leather, and poultry.

2. On the Basis of Nature of Products, Industries can be classified into:

(a) Heavy Industries: These industries use heavy and bulky raw materials to produce capital goods and consumer durables, such as iron and steel industry. They require large investment and a large number of workers.

(b) Light Industries: These industries use light raw materials to produce lightweight goods such as sewing machines, cycles, toys, electronic goods, etc. They require less capital and fewer workers.

3. On the Basis of Size and Investment:

(a) Large-scale Industries: These industries require large capital, large number of skilled and unskilled workers, and produce large goods and machineries. For example, iron and steel, ship building, and automobile industries.

(b) Medium-scale Industries: These industries are neither large nor small and include cycle manufacturing, paper mills, and electronics.

(c) Small-scale Industries: These industries are managed by private individuals with capital investment up to ₹1 crore, and fewer workers. For example, weaving, toy and food processing industries.

4. On the Basis of Ownership:

(a) Public Sector Industries: These industries are owned and managed by the Central or State Government. Government holds more than 50% of the share and the rest may be public shares. These serve public welfare and include public utilities like Post and Telegraph, Railways, Oil Refineries, Heavy Engineering industries, and Defence Establishments. For example: Bharat Heavy Electricals Limited (BHEL), Steel Authority of India Limited (SAIL) and Gas Authority of India Limited (GAIL).

(b) Private Sector Industries: These industries are jointly owned and managed by individuals or group of individuals with 100% private ownership. Examples include Reliance India Limited, Wipro, Infosys, and TCS.

(c) Public-Private Partnership (PPP): These industries are owned, managed and controlled by private **entrepreneurs** and the government (no fixed share of government and private owners). For example, the Automobile Corporation of Goa Ltd. and Ipitata Sponge Iron Ltd. were established with TISCO and TELCO of the Tata Group as private promoters.

(d) Cooperative Sector Industries: These industries are formed by groups of people with similar interests and limited means who pool their physical and material resources. They follow government rules and regulations but are not directly controlled by it. For example, Anand Cooperative Society. Generally, producers of raw materials for the given industry are involved in cooperative sector industries.

5. On the Basis of Location and Market:

(a) Village Industries: These industries meet the basic needs of local market, using raw materials, labour, and other resources available within the village. For example, pottery making, matchbox making, weaving, food processing, and khadi.

(b) Cottage Industries: These industries are organised by individuals using private resources and family skills. For example, weaving, handloom, and carpet making.

6. On the Basis of Finished Product:

(a) Basic Industries: These industries process and refine natural substances and form the basis for other industries. For example: iron and steel industry, petroleum industry.

(b) Secondary or Consumer Industries: These industries use the products of primary industries as raw materials to make goods for direct consumer use. For example: textiles and sugar.

(c) Tertiary Industries: These industries provide public utility-based services such as railways, transport, postal services, and banking.

(d) Ancillary Industies: These industries supply spare parts or components required by large industries such as the heavy electrical, locomotives, and aircraft industries.

➢ The distribution of industries in India is highly uneven due to limited access to raw materials and energy resources in various regions.

➢ Financial resources and major **enterprises** are concentrated in large towns and cities, which also influence the distribution of industries.

➢ Several factors are responsible for the uneven distribution of industries in India. They are:

(i) Agro-based industries like cotton, jute and sugar are located in areas where raw materials are available.

(ii) Forest-based industries like paper, resins, plywood, etc., are located near the forest areas.

(iii) Coastal regions have abundant copra, coir and fish canning, so, the related industries are located there.

(iv) Most oil refineries are located near major ports.

(v) Heavy metallurgical industries are concentrated in the regions rich in metallic reserves such as Karnataka, Jharkhand, Odisha, Madhya Pradesh, Rajasthan, and Tamil Nadu.

➢ **Based on major industries, India can be divided into the following industrial regions:**

(a) The Hooghly Belt
(b) The Mumbai–Pune Belt
(c) The Ahmedabad–Vadodara Region
(d) The Bengaluru–Chennai Region
(e) The Chota Nagpur Region
(f) The Gurugram–Delhi–Meerut Belt
(g) The Visakhapatnam–Guntur Belt
(h) The Kollar–Thiruvananthapuram Belt

Key Terms

➢ **Industry:** An economic activity concerned with the production of goods, extraction of minerals, or provision of services.

➢ **Consumer Durables:** Consumer products that do not need frequent replacement, as they last for a long period. For example: washing machine, automobile, and refrigerators. They are also called durable goods.

➢ **Heavy Industries:** Industries that use heavy and bulky raw materials to produce similar products. For example: iron and steel industry.

➢ **Entrepreneur:** A person who sets up his a business, taking financial risks in the hope of profits.

➢ **Enterprise:** An organisation, especially a business, or a complex and important venture intended to earn money.

TOPIC-2: AGRO-BASED INDUSTRIES (SUGAR, COTTON AND SILK)

Concepts Covered: . Location and Problems of Sugar, Cotton Textile, Silk Textile Industries

Revision Notes

➢ Industries that use the agricultural raw materials are known as agro-based industries.

➢ These industries include:

(A) Sugar industry **[Board 2024]**
(B) Cotton textile industry **[Board 2024]**
(C) Silk industry

(A) Sugar Industry

➢ The sugar industry is India's second largest organised industry. India is the second largest producer of sugarcane in the world.

➢ Sugarcane, an important cash crop, is crushed in the sugar mills to produce sugar *Gur* and *Khandsari*.

➢ **Molasses**, **bagasse**, **and press mud** are the by-products of sugar.

➢ Molasses is thick, dark-brown juice obtained from raw sugar during the refining process.

➢ It is used in the alcohol industry for distilling liquor, in the production of citric acid, chemicals, synthetic rubber, and as fuel for mills.

➢ Bagasse is the dry, pulpy residue left after the extracting juice from sugarcane.

➢ It is used as a biofuel, in the manufacture of pulp and building materials, and for the generating steam and power required to operate the sugar factory.

➢ **Press mud** is a residue from the filtration of sugarcane juice.

➢ It is used to make wax, carbon paper, and shoe polish.

➢ Uttar Pradesh is the leading producer of sugar in India followed by Maharashtra and Karnataka.

➢ Other sugar-producing states include Punjab, Haryana, Madhya Pradesh, and Gujarat.

➢ In Peninsular India, Tamil Nadu is the leading producer due to its higher per-hectare yield, higher sucrose content and long crushing season, and it is emerging as the leading producer in the country.

➢ Besides Tamil Nadu, Karnataka and Andhra Pradesh are also major producers of sugar.

➢ In recent decades, there has been a tendency of sugar industry to grow towards the South.

➢ Factors that led to shift of sugar industries to the South:

(i) The maritime climate of the south, which is free from *loo* and frost.

(ii) The availability of black soil, which is well-drained and more fertile than alluvial soil.

(iii) The sugarcane of the south is of superior quality, with a higher yield as compared to the North.

(iv) Excellent transport facilities in Maharashtra and Tamil Nadu have given them an advantageous position in relation to export markets.

(v) The cooperative societies in South are managing the large sugarcane farms by providing better seeds, fertilisers, and irrigation facility.

(vi) The sugar factories in South are located close to the sugarcane farms, which prevents the loss of sucrose content due to minimal transportation time.

(vii) In the south, farmers have new machinery and crushing devices that ensure high yield.

➢ **Problems of Sugar Industry:**

(i) Sugarcane produced in India is of poor quality and has low sucrose content.

(ii) Due to inefficient and uneconomical production, a short crushing season, low yield and distant locations, the cost of production is high.

Sugar Manufacturing
Scan Me!

(iii) As sugarcane is harvested at the same time, sugar mills face pressure and are unable to crush all of it, leading to wastage.

(iv) The supply of raw materials to sugar factories is irregular.

(v) The Government has fixed the prices of sugarcane, thus if the farmers are not offered fair prices, they tend to switch to over other crops.

(vi) Sugar factories use old and obsolete machineries, which should be replaced with modern technological machinery.

(vii) In rural areas, ***Gur*** and *Khandsari* are in greater demand than sugar.

(B) Cotton Textile Industry:

Cotton Textile Industry

Scan Me!

(i) India is one of the largest manufacturers and exporters of cotton textiles in the world.

(ii) Cotton textile industry is divided into two sectors: **powerloom and handloom**.

(iii) Major powerloom cotton mills are located in Maharashtra, Gujarat and Tamil Nadu, while handloom units are found in Mumbai, Ahmedabad, Kanpur, Coimbatore, Howrah, etc.

(iv) Maharashtra and Gujarat are the two leading cotton textile manufacturing states in India.

(v) Mumbai and Ahmedabad together contribute about 50% of the total installed looms.

(vi) Mumbai is known as the 'Cottonopolis of India' and the 'Lancashire of India'.

(vii) Ahmedabad is known as the 'Manchester of India'.

(viii) Several factors have led Mumbai and Ahmedabad to emerge as major cotton textile centres. The main reasons are:

(a) Regular supply of or proximity to raw materials.

(b) Favourable climatic conditions, especially the humid climate.

Major Cotton Producing States

Scan Me!

(c) Well-developed road and rail network within the country, and sea routes for international trade.

(d) The presence of major ports facilitates both export and import.

(e) Availability of cheap and skilled labour.

(f) Access to capital, as Mumbai and Ahmedabad are financial and commercial centres with many banks and financial institutions that provide loans to manufacturers.

(g) Electricity is supplied by the Tata Hydroelectric System in the Western Ghats to Mumbai and the Ukai and Kakrapara hydroelectric units to Gujarat.

(h) High demand for cotton garments in both northern and southern India, as well as in the foreign markets.

➢ **Problems of the Cotton Textile Industry:**

(i) Long-staple cotton is not adequately grown in India, leading to a shortage.

(ii) Many factories are old, obsolete and non-functional, leading to low productivity.

(iii) Maintenance and replacement of old machineries with the new ones require heavy financial investments.

(iv) The growth of cotton textile industries in countries like China, Japan, and African countries, has led to stiff competition, causing India to lose foreign markets.

(v) The cotton textile industry also faces stiff competition from synthetic fabrics such as polyester, nylon, rayon, which are in increasing demand.

(vi) The industry is also suffers from inadequate production due to irregular power supply.

(vii) Mill owners face great difficulties in obtaining capital for modernisation.

(viii) Acute power shortages and outdated machinery lead to low productivity and poor quality of goods, hindering the growth of cotton textile industry.

➢ **Handloom and Khadi Industry:**

(i) The handloom industry is one of the India's oldest, providing employment to millions.

(ii) The handloom industry is mainly located in small town and rural areas.

(iii) Tamil Nadu, Odisha, Uttar Pradesh, Assam and Andhra Pradesh contribute around 50% of the total production, while Manipur, Maharashtra, West Bengal, Kerala, Rajasthan, Jammu and Kashmir, and Karnataka, are other important handloom centres.

➢ **Problems of the Handloom and Khadi Industries:**

(i) Raw materials are often inadequate, insufficient, and of poor quality.

(ii) Most workers are unskilled and come from economically weaker backgrounds.

(iii) These industries use outdated technology which and keep up with fast-changing modern fashions and designs.

(iv) These industries lack capital, forcing them to adopt cost-saving methods.

(v) They also face stiff competition from mill-made cloth, which is cheaper and of better quality.

(C) Silk Textile Industry:

(i) The Indian silk textile industry is a vital part of the national textile sector. India is one of the largest world's largest silk producers.

(ii) The silk industry in India provides employment to around 60 lakh workers.

(iii) India produces four varieties of silk: mulberry and non-mulberry like *Mulberry*, *Muga*, *Tussar* and *Eri*.

(iv) Assam holds a global monopoly in producing golden-yellow *Muga* silk.

(v) The rearing of silk worms to produce silk is called Sericulture. **[Board 2024]**

(vi) Silkworms are reared on mulberry trees.

(vii) About 90% of India's natural silk come from mulberry silk.

(viii) About 92% of India's mulberry silk is produced in Karnataka, Andhra Pradesh, West Bengal, Tamil Nadu, and Jammu and Kashmir.

(ix) The silk industry is mainly concentrated Karnataka, Assam, Andhra Pradesh, West Bengal, Telangana, Tamil

Nadu, Jammu and Kashmir, Jharkhand, Chhattisgarh, Manipur, and Meghalaya.

➢ **Problems of Silk Industry:**

(i) The industry faces competition from artificial silk and synthetic fibres, which are cheaper and durable.

(ii) Import of cheaper alternative textiles from China and other Asian countries affect the domestic market.

(iii) Outdated manufacturing, along with primitive reeling and weaving methods, affects quality and efficiency.

(iv) Fluctuating raw silk prices adversely affect both weavers and the industry.

(v) High production costs and lack of proper marketing hinder industry growth.

(vi) Farmers have limited knowledge about silkworm diseases.

(vii) Poor and unorganised management, with no proper system for silk testing and grading, unlike in Japan.

(viii) Lack of modern technology and power looms is hampering production growth.

Key Terms

➢ **Molasses:** It is thick, dark brown syrup obtained from raw sugar during the refining process.

➢ **Bagasse:** The dry, fibrous residue left after extracting juice from sugarcane.

➢ **Press mud:** The residue left after filtering of sugarcane juice.

➢ ***Gur* and *Khandsari*:** *Gur* (Jaggery) is an unrefined sugarcane product. *Khandsari* is a type of unrefined raw white sugar made from thickened sugarcane syrup.

➢ **Powerloom:** A loom powered mechanically rather than manually, used for weaving fabric.

➢ **Handloom:** A loom operated by hand, unlike powerlooms which are motorised.

TOPIC-3: IRON AND STEEL INDUSTRIES

Concepts Covered: . TISCO, Bhilai and Rorkela, Visakhapatnam, Steel Plants and Mini Steel Plants

Revision Notes

➢ **(A) Iron and Steel Industry:**

➢ Iron and steel is the mother of all industries and forms the backbone of industrial development, as the production and consumption are key determinants of industrialisation and economic growth.

➢ It lays the foundation for other industries by providing raw material to industries for manufacturing industrial machinery, railway engines, railway tracks, electrical machinery, defence equipment, bridges, dams, shops, automobiles, etc.

➢ India has one of the richest reserves of all the raw materials required for the industry, such as iron ore, coal, and cheap labour.

➢ India is the second largest producer of crude steel in the world.

➢ The main raw materials required for iron and steel are iron ore, manganese, limestone, silica, chromate, feldspar, scrap iron, flux and fuel.

➢ To make steel, impurities in the iron ore such as sulphur, silica, phosphorous, lime, etc., need to be removed. This is done through the following process:

(a) During the iron-making process, a blast furnace is fed with the iron ore, coke, and small quantities of fluxes such as limestone.

(b) The slag floats on the molten iron and is collected at the base of the furnace at regular intervals.

(c) The product obtained is known as pig iron, which can be converted into wrought iron, steel and cast iron.

(d) Through deoxidation, the impurities are removed to convert pig iron into steel.

(e) Steel is cast into **ingots** and rolled into various sizes.

➢ **Integrated Steel Plants:** In integrated steel plants, all the processes, from processing raw materials to rolling, are carried out at one place. **[Board 2024]**

➢ **Tata Iron and Steel Company (TISCO):**

(i) TISCO is situated in Jamshedpur and is the oldest steel plant in the country.

(ii) It was established by Jamshedji Tata in 1907, and production started in 1911.

(iii) The plant obtains its raw materials from the following sources:

Iron and Steel Industry

Scan Me!

(a) Iron ore from the Gorumahisani Mines in Mayurbhanj district of Odisha and the Noamundi Mines in Singhbhum district of Jharkhand.

(b) Manganese from Joda in Keonjhar district.

(c) Limestone, dolomite, and fire-clay from Sundargarh district of Odisha.

(d) Coal from the Jharia and Bokaro coalfields.

(iv) The Kharkai and Subarnarekha are the two perennial rivers that supply water throughout the year.

(v) Labour is employed from the states of Bihar, West Bengal, Jharkhand, Chhattisgarh, and Uttar Pradesh.

(vi) It has good accessibility to the market in Kolkata which is not only a market but also has facilities for the export of finished goods.

(vii) Jamshedpur has a good network of roads and railways which are connected to other parts of the country.

(viii) TISCO produces high-grade carbon steel, acid steel and special alloy steel. Carbon steel is used in structural fittings and tin plates, while acid steel is used for making railway wheels, axels bars, rods, sheets, etc.

➢ **Bhilai Iron and Steel Plant:**

(i) The Bhilai steel plant was established in 1953.

(ii) It was established in collaboration with the former USSR.

(iii) It is located in the Durg district of Chhattisgarh.

(iv) The plant obtains its raw materials from the following sources:

(a) Iron ore is obtained from the Dalli Rajhara Mines.

(b) Limestone is obtained in the Nandini Mines near Bhilai.

(c) Manganese is obtained from neighbouring district of Balaghat.

(d) Coal is obtained from the Bokaro, Kargati and Jharia Fields in Jharkhand, and Korba in Chhattisgarh.

(v) The main source of power is the thermal station at Korba.

(vi) Water is supplied to the plant from a system of reservoirs at Tandula.

(vii) The Bhilai Steel Plant has excellent transport facilities as it lies on the Mumbai–Nagpur–Kolkata Railway line which links to major markets.

(viii) A large number of labourers are employed from the states of Bihar, Jharkhand, and Madhya Pradesh.

(ix) The Bhilai Plant produces heavy rails, structural beams, billets and rolled wire, plates for shipbuilding industry.

(x) The plant also produces by-products such as ammonium sulphate, benzol, coal tar and sulphuric acid.

➢ **Rourkela Steel Plant:** **[Board 2024]**

(i) This steel plant was established in 1959 with technical collaboration from the German firm, Krupp and Demag.

(ii) It is located in the Sundargarh district of Odisha.

(iii) The plant obtains its raw materials from the following sources:

(a) Iron ore is obtained from the reserves of Sundargarh and Keonjhar district of Odisha.

(b) Manganese is obtained from Barajamda.

(c) Limestone is obtained from Biramitrapur.

(d) Dolomite is obtained from Baradwara.

(e) Coal is obtained from Jharia, Talcher and Korba fields.

(f) Electricity is supplied by the Hirakud Project.

(iv) Water is obtained from the Mandira Dam across the Sankh River and Mahanadi rivers.

(v) Good transportation on the Kolkata–Nagpur railway line provides easy access to the raw material producing areas and to markets.

(vi) Labourers are employed from Bihar, West Bengal, Jharkhand and Odisha.

(vii) Rourkela Steel Plant produces hot-rolled sheets, cold-rolled sheets, **galvanised sheets,** and electrical steel plates.

(viii) It also produces large quantity of nitrogen, used in manufacturing of fertilisers and various chemicals.

➢ **Visakhapatnam Steel Plant:** **[Board 2024]**

(i) It is an integrated steel plant located in the port city of Visakhapatnam in Andhra Pradesh.

(ii) It is the first shore-based steel plant in India.

(iii) The plant obtains its raw materials from the following places:

(a) **Iron ore** is obtained from Bailadila in Chhattisgarh.

(b) **Limestone**, **dolomite** and **manganese** is obtained from mines of Andhra Pradesh and Odisha.

(iv) It receives power from coalfields of the Damodar Valley.

(v) The plant produces liquid steel and finished saleable steel.

(vi) It is a major export-oriented plant with the advantage of coastal location.

➢ **Mini Steel Plants:** **[Board 2024]**

Mini Steel Plant

Scan Me!

(i) Mini steel plants are smaller units that operate using electric furnaces and mainly use steel scrap and sponge iron or pig iron as their raw material.

(ii) Mini steel plants are located throughout the country away from the integrated steel plants.

(iii) Mini steel plants have the following advantages:

(a) Scrap iron is used as raw material which is cheap and easily available.

(b) These plants require less capital investment and can be set up on a smaller size in industrial towns.

(c) These plants cause pollution as they it runs on electric power.

(d) These plants cater to the needs of the local market, and thereby reducing the burden on large steel plants.

(e) Their construction and gestation periods is short.

➢ **Problems of Iron and Steel Industry:**

(i) The iron and steel industry is a capital-intensive. Many integrated steel plants set up by the public sector have been established with foreign aid.

(ii) The industry lags behind in using advanced technology. Moreover, low labour productivity adversely affects the output of finished products.

(iii) The availability of high-grade coking coal used for smelting iron ore is limited.

(iv) Many small iron and steel plants have shut down due to inadequate supply and rising cost of raw materials.

(v) Government control over iron and steel prices leaves only marginal profit for manufacturers.

(vi) Inefficient management and poor functioning of public sector steel plants affect potential utilisation.

(vii) Rising demand for iron and steel has led to increased imports.

Key Terms

- **Scrap Iron:** Discarded or waste pieces of iron to be recast or reworked.
- **Pig Iron:** Crude iron, a direct product of the blast furnace, poured into moulds and refined to produce steel, wrought iron, or ingot iron.
- **Ingots:** A piece of pure material, usually metal, cast into a shape suitable for further processing.
- **Galvanised Sheets:** A sheet, strip or other steel item coated with a thin layer of zinc to prevent rusting.
- **Integrated Steel Plants:** A unified steel mill where all primary functions of steel production are carried out, such as iron making, steel making, casting, roughing rolling, and product rolling.

TOPIC-4 : PETROCHEMICALS AND ELECTRONICS

Concepts Covered: Organic Molecules, Consumer Electronics, Space Technology, Policy Framework for Technology Sector

Revision Notes

Other Mineral-Based Industries

(A) Petrochemical Industry:

- Chemicals obtained either directly or indirectly from chemical processing, of petroleum oil or natural gas is called **petrochemicals.**
- India is amongst the fastest-growing petrochemical markets in the world.
- Major petrochemicals include acetylene, benzene, ethane, ethylene, methane, propane, and hydrogen, from which hundreds of other chemicals are derived.
- This industry is typically located near an oil refinery that supplies the basic requirements of naphtha or ethylene and benzene. As transportation of petroleum and its products is dangerous, a shorter distance is maintained between the source of raw material and the industrial plants to avoid accidents. Hence, for safety reasons, petrochemical industries are located close to refineries.
- These chemicals are used to manufacture products like synthetic fibres, synthetic rubber, ferrous and non-ferrous metals, plastics, drugs, and pharmaceuticals.
- These products are widely used in the domestic, industrial and agricultural sectors.
- The petrochemical industry produces the following products:

The petrochemical industry produces various products, including adhesives, fertilisers, dyes, detergents, insecticides, pesticides, resins, crayon, plastic sheets, printing inks, paints, and carbon paper.

- **Advantages of Petrochemical Products:**

(a) Petrochemicals are cost-effective, economical, and cheap.

(b) The raw material is easily available and not dependent upon agricultural raw materials.

(c) It is a highly compact, portable source of energy.

(d) It is an excellent source of **organic molecules** for producing plastics, medicines, rubber, fibres, etc.

- There are several petrochemical production units in India. Some of them are as follows:

1. Herdillia Chemicals Ltd. in Chennai.
2. National Organic Chemical Industries Ltd. near Mumbai.
3. Petrofils Cooperative Limited (PCL) in Vadodara.
4. Indian Petrochemical Corporation Ltd. near Vadodara.
5. Reliance Industries in Gujarat.
6. Haldia Petrochemicals Ltd. in West Bengal.
7. Bongaigaon Petrochemicals Ltd. in Assam.
8. Indian Oil Corporation in Gujarat and Panipat.

(B) Electronic Industry: **[Board 2024]**

- The electronic industry in India began around 1965 with the objective of developing space and defence technologies.
- India gradually developed in **consumer electronics,** starting with transistor radios, black-and-white TVs, calculators, and other audio products. Later, colour televisions were introduced.
- The period between 1984 and 1990 is considered the golden period for electronics, during which India's industry witnessed continuous and rapid growth.
- India also exports a wide range of electronic components and products, such as display technologies, entertainment electronics, telecom equipment, semiconductor design, and electronic manufacturing services (EMS).
- The Indian electronic industry has several advantages that support its growth.

These include:

(i) Manpower (skilled and unskilled)
(ii) Market demand
(iii) Supportive government policies

- India has experienced strong growth in the demand for consumer products and durables in recent years, which has facilitated the growth of the electronics sector both directly and indirectly.
- Important requirements of the electronics industry are:

(i) A cool and moderate climate
(ii) Trained manpower
(iii) Affordable manpower

The electronics industry in India is largely concentrated in the southern states of India. Some of the major units are:

1. Indian Telephone Industries (ITI) in Bengaluru.

2. Electronics Corporation of India Ltd. (ECIL) in Hyderabad.

3. Bharat Electronics Ltd. (BEL) in Bengaluru.

(i) Space Technology in India:

➢ Indian Space Research Organisation (ISRO) was established in the 1960s by the Department of Atomic Energy.

➢ ISRO in Bengaluru, Satellite Launching Station in Sriharikota, and the National Remote Agency in Hyderabad were established, giving further impetus to space research in India.

➢ India made remarkable achievement by launching Aryabhata, its first satellite, aimed at developing advanced technology for agriculture, weather forecasting, and more.

➢ Bhaskara-I was later launched, followed by the development of Polar Satellite Launch Vehicles (PSLV) and Geosynchronous Satellite Launch Vehicle (GSLV).

➢ In 2008, India's first scientific mission, **Chandrayaan-I**, was launched to the Moon.

➢ India made further progress by launching indigenously built satellites such as APPLE and the INSAT series.

(ii) Software Industry:

➢ The software industry is the fastest growing electronics industry in India.

➢ The software industry is a leading destination for the IT and IT-enabled services worldwide.

➢ The Department of Electronics has played an important role in enhancing India's competitiveness in IT.

➢ The department has initiated several programmes for manpower development, quality improvement, and promotion of software engineering and research.

➢ The Business Process Outsourcing (BPO) industry is continuously growing.

➢ Software giants such as Infosys, Wipro, and TCS, are providing software solutions to overseas clients.

➢ Bengaluru and Hyderabad are the leading centres of the software industry in India.

➢ Bengaluru is known as the Silicon Valley and IT capital of India. Many MNCs, such as Capgemini and Yahoo, have entered the Indian market.

Key Terms

➢ **Petrochemical:** A chemical obtained directly or indirectly from chemical processing of petroleum oil or natural gas.

➢ **Organic Molecules:** Usually composed of carbon atoms in rings or long chains, bonded with elements such as hydrogen, oxygen and nitrogen.

➢ **Consumer Electronics:** Electronic device or digital equipment used daily by people.

➢ **Space Technology:** Technology developed by space science or the aerospace industry for use in spaceflight, satellites or space exploration.

➢ **Chandrayaan-I:** India's first scientific mission to the Moon, launched in 2008.

CHAPTER-10
Transport

Learning Objectives

The students will be able to:

- Explain the importance of transport.
- List the modes of transport: roadways, railways, airways, and waterways
- Understand the advantage and disadvantages of different modes of transport.

TOPIC-1: IMPORTANCE OF TRANSPORT, MODES OF TRANSPORT: ROADWAYS AND RAILWAYS

Concepts Covered: . National Highways, Expressways, State Highways, District Roads, Rural or Village Roads, Border Roads and Other Roads, Broad Gauge, Metre Gauge and Narrow Gauge

Revision Notes

➢ Transport is the lifeline of people all over the world. Transport reflects the growth and development of a country.

➢ Transport refers to the activity that enables the movement of goods and people from one place to another.

➢ Transport in India has grown significantly over the years in terms of network and in its system.

➢ **Importance of Transport**

(i) Transport facilitates access to natural resources lying unutilised in the hills, forests, and mines and increases economic productivity.

(ii) It links backward areas to urban centres and reduces regional industrial disparity.

(iii) The transport system helps in transporting raw materials and essential machinery to industries.

(iv) Transport protects people during war, natural calamities, and other crisis.

(v) It helps in the process of industrialisation and urbanisation.

(vi) The transport system helps to strengthen unity and brotherhood among people.

(vii) It has shortened distances between various places.

(viii) It helps in supplying finished goods to consumer markets.

(ix) It connects remote areas and villages with cities.

➢ The present transport system of India has developed over the three domains: land, air, and water. This transport system in India comprises four main means of transport as follows:

(a) Road transport (roadways)

(b) Rail transport (railways)

(c) Air transport (airways)

(d) Water transport (waterways)

(A) Road Transport: **[Board 2024]**

➢ India has an excellent network of roads connecting all towns and cities.

➢ According to the National Highways Authority of India, India has the second-largest road network in the world. In India roads are constructed rapidly, yet the density of roads remains low. Jammu and Kashmir has the lowest road density, while Kerala has the highest. The length of roads per 100 sq. km of area is known as density of roads.

➢ Road transport is significant to India's economy and contribute about 4.7% to the **Gross Domestic Product** (GDP).

➢ The road network is important for the country's growth, **social integration**, and for the security of its citizens.

➢ Indian roads can also be classified based on the material used in their construction, i.e., metalled and unmetalled roads.

➢ Indian roads are categorised as follows:

1. National Highways
2. Expressways
3. State Highways
4. District Roads
5. Rural or Village Roads
6. Border Roads
7. Other Roads

1. National Highways:

➢ National Highways are major roads constructed and maintained by the Central Government.

➢ These highways facilitate interstate transport, and movement of defence personnel, and materials in strategic areas, and connect manufacturing centres state capitals, major cities, and important ports.

➢ The National Highways Authority of India (NHAI), constituted in 1988, and is responsible for the development, maintenance and management of National Highways.

➢ NHAI is currently undertaking developmental activities under the National Highways Development Project (NHDP).

➢ To reduce time and distance between India's major cities two major projects have been undertaken by NHAI. They are

(I) Golden Quadrilateral (GQ) :

(a) The largest superhighway project in India.

(b) It connects India's four largest metropolitan cities—Delhi, Mumbai, Kolkata and Chennai by six-lane superhighways.

(c) The main economic benefits of this project are:

(i) Establishing faster transport networks between major cities and ports.

(ii) Providing fast and smooth movement of goods and people within India.

(iii) Developing industries and creating job opportunities in smaller towns better market access.

(iv) It is also providing opportunities for farmers, through better transportation so that the agricultural produce to reach to major cities and ports for export.

(v) Driving economic growth directly through construction, and indirectly through demand for cement, steel and other materials.

(vi) Provides a boost to truck transport across India.

(II) North–South and East–West Corridors (NS–EW):

(a) These are a part of highway project in India.

(b) Jhansi is the junction of the North–South and East–West Corridors.

(c) The North-South Corridor connects Srinagar in Jammu and Kashmir to Kanyakumari in Tamil Nadu, and while the East–West Corridor connects Silchar in Assam to Porbandar in Gujarat.

2. Expressways: These highways are six-lane roads designed for high-speed movement of vehicles without obstacles.

The major expressways are:

(i) Yamuna Expressway

(ii) Ahmedabad–Vadodara Expressway

(iii) Delhi–Gurgaon Expressway

(iv) Mumbai–Pune Expressway

(v) Noida–Greater Noida Expressway

(vi) Delhi–Noida Direct Flyway

(vii) Panipat Expressway

(viii) Bengaluru–Mysuru Infrastructure Corridor.

National Highways in India

Scan Me!

3. State Highways:

➢ State Highways are constructed and maintained by the state governments.

➢ State Highways usually link important cities, towns, and district headquarters within the state, connecting them to National Highways or highways of neighbouring states.

➢ These highways also connect the industries or tourist destinations to important areas in the state.

➢ They also contribute to the development of states.

4. **District Roads:**

➢ These are important roads that connect Talukas, rural areas, and key towns in the districts to the district headquarters.

➢ These roads lie within districts, connecting production areas with markets.

➢ They are maintained by the Zila Parishads.

5. **Rural or Village Roads:**

➢ Village roads are interconnected with one another and with the nearby towns and cities.

➢ Most of the village roads are un-metalled and become muddy and sticky during the rainy season.

➢ It constitutes over 80% of the total road network.

➢ The rural roads sector suffered from lack of systematic planning, quality and sustained maintenance.

➢ These roads are in poor condition, affecting the quality of life in rural areas and farmers' ability to transport produce to market after harvest.

6. **The Border Roads Organisation (BRO):**

➢ It was established on 7th May 1960 to secure India's borders and develop infrastructure in remote regions of the northern and northeastern states.

➢ This organisation constructed the world's highest road from Manali to Leh, at an average height of 4,270 metres.

➢ Besides constructing roads in strategic areas, the BRO clears snow in high-altitude regions and builds airfields, buildings and permanent bridges.

➢ **Advantages of Roadways:**

(i) Roadways are ideal for short distances, as they provide access to every village.

(ii) It is cost-effective compared to other means of transport.

(iii) It transports people and goods quickly and easily.

(iv) It provides door-to-door service.

(v) Roads can be constructed easily in hilly terrain where building railway line is not feasible and air services are inaccessible.

(vi) The movement of goods by road is safer and more convenient.

(vii) Road transport is best suited for carrying perishable goods and people to and from rural areas not served by rail, water, or air transport.

➢ **Disadvantages of Roadways:**

(i) Road transport is not as reliable as rail transport. During rainy or flood season, poor road quality makes them unsafe and unusable.

(ii) Road accidents occur frequently , making this mode of transport unsafe.

(iii) This mode of transport is unsuitable and costly for transporting cheap, bulky goods over long distances.

(iv) It often faces delayed due to heavy traffic, high concentration of vehicles check posts, tolls and octroi collection points.

(v) Road transport is comparatively less organised, irregular, and unreliable. Transport charges are also unstable and unequal.

(B) Rail Transport: **[Board 2024]**

(i) It is an important mode of transport for both freight and passengers.

(ii) It is the fourth-largest and busiest railway network in the world and largest commercial utility employer, with over 1.4 million employees.

(iii) Kolkata, Delhi, Mumbai, and Chennai have their own metro networks.

(iv) Suburban train that handles commuter traffic are mostly Electric Multiple Units (EMUs).

(v) Railways transport all kinds of goods, including mineral ores, fertilisers, agricultural produce, and iron and steel.

(vi) The railway system is divided into 18 zones.

(vii) Rail Gauge is the distance between the inner sides of the two parallel rails that forms a single railway line.

(viii) Based on track width, Indian Railways are classified into three categories:

1. **Broad Gauge**
2. **Metre Gauge**
3. **Narrow Gauge**

➢ **Advantages of Railways:**

(i) Railways allow easy movement of people and bulky goods from one place to another.

(ii) It reduces long distances in terms of time.

(iii) It links industries with markets.

(iv) It has developed and commercialised agriculture It links industries with markets.

(v) Railways are a major source of employment in India , providing jobs to lakhs of skilled and unskilled workers.

(vi) It helps during famines or other crises by carrying foodgrains and essential materials to affected areas.

(vii) Railways play a key role in internal security and in efficient transporting defence equipment to strategic areas.

(viii) Railways have boosted tourism.

(ix) Railways have made travel safer and more comfortable.

(x) Railways act as an integrating force, overcoming social barriers and uniting people across the nation.

(xi) Railways have bridged the gap between the villages and cities , encouraging rural development through new ideas.

(xii) Railways are more affordable for the masses than air transport.

(xiii) Indian Railways introduced Kisan Rail to facilitate the farmers of Maharashtra to increase their income by seamless supply of perishable and essential commodities during the pandemic.

➢ **Disadvantages of Railways:**

(i) Railway lines cannot be built in all areas, especially hilly and dense forested regions.

(ii) Railways require heavy capital investment, with high costs for construction, maintenance, and operations.

(iii) Railways cannot offer door-to-door service.

(iv) Railway are unsuitable and costly for short distance and small goods traffic.

(v) Railways involve more time and effort in booking and collecting goods compared to road transport.

(vi) Long-distance train travel can be tiring and uncomfortable.

(vii) Some industrial regions lack rail access and rely on other transport to move cargo from stations.

(viii) Train travel is limited to land and cannot cross oceans.

Key Terms

➢ **Social Integration:** The peaceful unification of social groups through coexistence, collaboration and cohesion.

➢ **Gross Domestic Product:** The total value of goods and services produced in a country during one year.

➢ **Golden Quadrilateral (GQ):** The largest super highway project connecting four biggest metropolitan cities–Delhi, Mumbai, Kolkata and Chennai.

➢ **Broad Gauge:** A railway track with a gauge wider than the standard 1,676 mm.

➢ **Metre Gauge:** Railways with a track gauge of 1,000 mm (3 ft 3 3/8 in).

➢ **Narrow Gauge:** Railways with a track gauge of either 762 mm or 610 mm.

TOPIC-2: AIR TRANSPORT AND WATER TRANSPORT

Concepts Covered: Inland Waterways and Oceanic Waterways, Major Parts of India

Revision Notes

➢ Air transport is the most modern means of transport, as it is fast and time-saving. **[Board 2024]**

➢ India has both domestic and international airlines that carry passengers, freight, and mail.

➢ The Airports Authority of India (AAI) came into existence on 1 April 1995 after merging two existing authorities the National Airports Authority and the International Airports Authority.:

➢ Air transport in India was managed by two corporations: Air India and Indian Airlines.

➢ In 2007, Air India and Indian Airlines merged into one, which is now called Air India.

➢ Air India is the fourth-largest airline in Asia by passengers carried, after China Southern Airlines, IndiGo, and Air China.

➢ There are also several private airlines, such as SpiceJet, Jet Airways (India) Ltd., Inter Globe Aviation Ltd. (Indigo), Go Airlines (India) Pvt. Ltd., etc.

➢ There are three cargo airlines which are operating and providing cargo services in the country. They are:

(i) Blue Aviation Pvt. Ltd.

(ii) Deccan Cargo

(iii) Express Logistics Pvt. Ltd.

➢ Pawan Hans Helicopters Ltd., established in 1985, aims to provide helicopter services to the oil sector in offshore exploration, hilly regions, remote areas, and promotion of tourism.

➢ **Advantages of Air Transport**

(i) Air transport is the fastest and most comfortable mode of transport.

(ii) It can easily reach remote and inaccessible areas such as mountains, forests, deserts.

(iii) It is very useful during the times of war and natural calamities like floods, earthquakes, famines, epidemics, civil unrest.

➢ **Disadvantages of Air Transport**

(i) Air transport is expensive.

(ii) It connects only major cities.

(iii) It is dependent on weather conditions and may be delayed or cancelled, causing inconvenience to passengers.

(iv) It causes pollution, as it runs on petroleum, which is a non-renewable source of energy.

(v) It provides limited and restricted services between destinations.

(vi) It carries a small tonnage but has high freight charges.

(vii) Its maintenance and overhead costs are very high.

➢ Water Transport:

➢ Water transport is the easiest and cheapest mode of transport.

➢ Unlike road and rail transport, no infrastructure needs to be built, as water bodies occur naturally.

➢ It has the largest carrying capacity and is best suited for transporting bulky goods over long distances.

➢ It plays a significant role in connecting different parts of the world and is indispensable to foreign trade.

➢ Water transportation is more efficient and environmentally friendly than other modes of transport.

➢ Water transport can be divided into two categories:

(i) Inland Waterways

(ii) Oceanic Waterways

(i) Inland Waterways: **[Board 2024]**

➢ India has an extensive network of inland waterways in the form of rivers, canals, backwaters, and creeks.

➢ The Inland Waterways Authority of India (IWAI) was established on 27 October 1986 for the development and regulation of inland waterways for shipping and navigation.

➢ **Inland Waterways** should be free of barriers such as waterfalls and rapids.

➢ Rivers of Peninsular India are not suitable for navigation due to the following reasons:

1. The rivers are rain-fed and seasonal.
2. They are shorter in length.
3. These rivers have many waterfalls.

➢ The Inland Waterways Authority has declared five major rivers as National Waterways:

1. **National Waterway No.1 (NW-1):** Ganga-Bhagirathi-Hooghly river system connecting Haldia–Kolkata–Farakka–Munger–Patna–Varanasi–Prayagraj. Total length: 1,620 km – longest inland waterway in India.
2. **National Waterway No.2 (NW-2):** The River Brahmaputra connects Dhubri–Pandu–Tezpur–Neamati–Dibrugarh–Sadiya and connects the Northeast region with Kolkata and Haldia ports.
3. **National Waterway No.3 (NW-3):** The West Coast Canal connects Kollam to Kottapuram. It is the most navigable and has great tourism potential.
4. **National Waterway No.4 (NW-4):** It connects the states of Andhra Pradesh, Tamil Nadu and the Union Territory of Puducherry through canals interlinking Godavari and Krishna rivers.
5. **National Waterway No. 5 (NW-5):** It comprises Talcher–Dhamra stretch of river Brahmani, Geonkhali–Charbatia stretch of east coast canal.
6. **National Waterway No.6. (NW-6):** It is a proposed waterway between Lakhipur and Bhanga of the Barak River. It would facilitate cargo transport through Assam, Nagaland, Mizoram, Manipur, Tripura, and Arunachal Pradesh.

(ii) Oceanic Waterways:

➢ **Oceanic waterways** play an important role in India's transport sector.

➢ India has a vast coastline of approximate 7, 517 km, including islands.

➢ Twelve major ports, along with 185 minor and intermediate ports, support these routes.

➢ These routes are also used for transport between the islands and the rest of the country.

➢ About 80% of cargo traffic is handled by the major ports.

➢ The Indian ports are classified into major, ports, minor ports and intermediate ports.

➢ **The major ports of India are:**

(i) Kolkata (West Bengal)
(ii) Haldia (West Bengal)
(iii) Paradip (Odisha)
(iv) Visakhapatnam (Andhra Pradesh)
(v) Chennai (Tamil Nadu)
(vi) Tuticorin (Tamil Nadu)
(vii) Kandla (Gujarat)
(viii) Mumbai (Maharashtra)
(ix) Jawaharlal Nehru (Nava Sheva) Port, near Mumbai
(x) Mormugao (Goa)
(xi) New Mangalore (Karnataka)
(xii) Kochi (Kerala)

➢ Ennore Port, near Chennai, is India's first public company port and functions as a corporate entity, not a port trust.

➢ **Advantages of Waterways :**

(i) Water transport is cheap with low maintenance costs.
(ii) Heavy and bulky goods can be transported easily.
(iii) It is very useful during natural calamities like floods and rain, when relief operations are conducted via waterways.
(iv) Shipping development is also essential for national defence.
(v) It is a fuel-efficient and eco-friendly mode of transport.
(vi) It is a safe mode of transport.
(vii) Water transport plays an important role in foreign trade.

➢ **Disadvantages of Waterways :** **[Board 2024]**

(i) It is a slow mode of transport.
(ii) It depends on weather conditions.
(iii) Long hours of travel can cause sea-sickness.
(iv) The area covered by water transport is limited, as unlike roads or railways, waterways cannot be easily constructed.
(v) It is less safe due to the risk of boats and ships sinking.

Key Terms

➢ **Inland Waterways:** Rivers, canals, backwaters and creeks that help move goods and people within the country.

➢ **Oceanic Waterways:** Routes used to transport goods and people between islands and the mainland.

CHAPTER-11
Waste Management

Learning Objectives

The students will be able to:
- Explain how waste harms land, water, and health.
- Understand why waste management is needed.
- State methods of safe waste disposal.
- Classify ways to reduce, reuse, and recycle waste.

TOPIC-1: IMPACT OF WASTE ACCUMULATION

Concepts Covered: Domestic Waste, Commercial Waste, Industrial Waste, Municipal Waste, Food Processing Waste, Agricultural Waste, Eutrophication, Sanitary Landfills, Biomagnification, Bioaccumulation

Revision Notes

➢ Waste refers to materials discarded after use, causing both environmental problems and economic loss.

➢ There are three types of waste:

(i) Solid Waste
(ii) Liquid Waste
(iii) Gaseous Waste

➢ Solid waste often accumulates in places like house backyards, street corners, outside hospitals and schools, near water bodies.

➢ Some waste is biodegradable and while some is non-biodegradable.

➢ Some waste is toxic and some are non-toxic.

➢ Waste originates from different sources, such as:

(i) **Domestic waste**
(ii) Commercial waste
(iii) **Industrial waste**
(iv) Municipal waste
(v) Food processing waste
(vi) Agricultural waste

➢ **Impact of Waste Accumulation:**

➢ When these accumulated solid waste is dumped and left unattended, it creates the following problems:

(i) Decomposing waste leads to the growth of **pathogenic** bacteria, virus and fungi.

(ii) Mosquitoes, flies, rodents, insects, etc., also spread diseases due to accumulation of wastes around the houses or nearby places.

(iii) Rainwater carries decomposed waste and pathogens into water bodies, spreading disease.

(iv) Accumulated solid waste spoils the landscape when left to decompose.

Introduction To Waste

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(v) Burning coal, fuelwood, or petroleum releases sulphur and nitrogen gases, which react with oxygen to form sulphur oxide and nitrogen dioxide.

(vi) Sulphur dioxide (SO_2) and nitrogen oxides (NO_x) react with water in the atmosphere to form sulphuric acid (H_2SO_4) and nitric acid (HNO_3). These acids mix with rain and form acid rain.

➢ When acid rain falls on buildings, it causes chemical reactions that corrode surfaces, leaving damage on monuments and statues. Monuments affected by acid rain include the Taj Mahal (Agra), the Houses of Parliament (London), and the Parthenon (Athens).

➢ Pollution is the addition of contaminants (solid, liquid, or gas) or energy (such as heat or sound) into the environment causing harmful changes. Waste accumulation mainly causes three types of pollution:

(i) Air pollution
(ii) Water pollution
(iii) Soil pollution

➢ Major causes of pollution include the burning of fossil fuels, such as coal, oil, natural gas and petroleum. These fuels are used to generate thermal electricity and power run vehicles.

➢ Decomposing waste produces large amounts of methane gas, which is highly explosive if not properly managed.

➢ Waste dumped into water bodies causes **Eutrophication**. Eutrophication is an ecological problem caused by human activities. It leads to dense growth of floating plants in water bodies due to excess phosphorus and nitrogen.

➢ Algae, plankton and other micro-organisms use carbon dioxide, inorganic nitrogen, and phosphate from the water as nutrients.

➢ When aquatic systems have excess nutrients, they suffer from eutrophication, making them unlivable and causing the death of many aquatic organisms.

➢ Unattended domestic, dumped spreads diseases through contamination.

➢ Solid waste chokes drains and gully pits, by the solid wastes causing waterlogging, which leads to mosquito breeding and diseases like malaria and chikungunya.

➢ Hazardous wastes includes toxic substances like lead, which can harm brain development in children.

➢ Harmful **toxins** such as asbestos cause chest and lung cancer, mercury causes brain damage and death, and arsenic causes cancer.

➢ Radioactive waste produced by nuclear reactors and weapons factories causes serious environmental problems.

➢ Accumulated waste also affects terrestrial life which includes human beings, plants and animals.

➢ **Effects of Accumulated Waste on Humans:**

(i) Accumulated solid waste looks filthy, smells bad, and attracts insects and rodents that spread diseases.

(ii) **Sanitary landfills** are unfit for human settlements, as harmful gases like methane and carbon dioxide start to escape within one or two years. **[Board 2024]**

➢ **Effects of Accumulated Waste on Plants:**

(i) Plants are affected directly by toxins from waste or indirectly through the soil.

(ii) Toxins cause of leaf injuries, premature leaf fall, reduced photosynthesis and transpiration, block stomata, due to dust, and stunted root and shoot growth from smoke.

➢ **Effects of Accumulated Waste on Animals and Birds:**

(i) Stray animals and scavengers like dogs, pigs, cows, and rats are directly affected from the wastes as they feed on waste.

(ii) These animals may eat toxic or non-degradable substances from the waste and die due to choking.

➢ Waste accumulation also from the waste affects aquatic organisms both in freshwater and marine environments.

➢ Pesticides and industrial and domestic wastes are major sources of that causes harm to aquatic life.

➢ Biomagnification is the accumulation of toxic chemicals by an organism from water and food through the food chain. Example: mercury in predatory fish.

➢ The concentration of toxins at higher trophic levels is known as bioaccumulation.

➢ **Methyl mercury** can cross the barrier between blood and nerve cells, reach the brain and cause progressive irreversible damage. This occurred in Minamata Bay, where people fell ill after consuming mercury-contaminated fish.

➢ Mercury contamination also occurs from the waste of industries such as paper and pulp chlorine industry, and pesticide industries.

Key Terms

➢ **Domestic waste:** Waste generated from household activities, such as fruit and vegetable peels, paper, polythene, and discarded clothes.

➢ **Industrial waste:** Waste material produced during manufacturing processes.

➢ **Pathogenic:** Refers to organisms such as viruses and bacteria that cause disease.

➢ **Eutrophication:** The enrichment of a water body with nutrients, leading to excessive plant growth and reduced dissolved oxygen.

➢ **Sanitary Landfills:** Waste disposal sites where waste is safely buried to prevent environmental contamination.

➢ **Toxins:** Organic poisons produced by certain bacteria.

➢ **Methyl Mercury:** A highly toxic organic form of mercury responsible for mercury poisoning.

TOPIC-2: NEED FOR WASTE MANAGEMENT

Concepts Covered: Dermatophytosis, Global Warming, Greenhouse Effect, Ozone Layer, Chlorofluorocarbons (CFCs), Acid Rain

Revision Notes

➢ Waste is not only an environmental problem but also an economic loss. It causes different types of pollution.

➢ The rotting garbage produces harmful gases and that mix with the air causing breathing problems for people.

➢ The transmission of diseases due to accumulation of wastes is a major threat to both people and environment.

➢ Due to waste accumulation on land and in water bodies, diseases are spread through flies, mosquitoes, rodents, and pet animals.

Water Pollution

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➢ Various diseases are spread by the following flies, rodents, and pet animals:

(i) **Housefly:** Typhoid, Diarrhoea, Dysentery, Cholera, etc.

(ii) **Sandfly:** Kala-azar, Sandfly fever, etc.

(iii) **Tsetse fly:** Sleeping sickness.

(iv) **Mosquitoes:** Malaria, Filarial, Chikungunya, Dengue, Yellow fever, etc.

(v) **Rodents:** Plague, Salmonellosis, etc.

(vi) Dog: Rabies, etc.

(vii) Cat: Dermatophytosis, Anthrax, etc.

➢ Water is polluted due to industrialisation and urbanisation because:

(i) Sewage contains organic matter that cannot be decomposed.

(ii) Industrial and commercial waste contain toxic agents.

(iii) Fertilisers and pesticides produce pollutants.

(iv) All of these reach the water bodies either directly or indirectly.

Greenhouse gases and Effect

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➢ People are also affected by pollution because they drink contaminated water and also use it for personal purposes.

➢ Viral, bacterial, protozoan, and helminthic are some of the common waterborne diseases.

➢ The greenhouse effect is a natural process that warms the Earth's surface.

➢ When the sun's energy reaches the Earth's atmosphere, some of it is reflected back to space and the rest is absorbed and re-radiated by the greenhouse gases.

➢ The warming up of the atmosphere is due to the greenhouse effect that retains terrestrial radiation. This increase in Earth's temperature is called global warming.

➢ Five primary gases contribute to the greenhouse effect and global warming. These gases are known as greenhouse gases.

They are:

(i) Carbon Dioxide (CO_2)

(ii) Methane (CH_4)

(iii) Nitrogen Oxide (N_2O)

(iv) **Chlorofluorocarbons** (CFCs)

(v) Water Vapour

➢ Global warming has several effects, which are:

(i) It is projected that the global temperature is likely to rise by 2°C to 5°C over the next century.

(ii) Due to the rise of temperature, there is a possibility of melting of ice caps at the Earth's poles.

(iii) There will be widespread changes in the climate in the wind and rainfall patterns, due to increase in temperature across the Earth's surface.

(iv) Higher temperature will cause an increase in transpiration, which in turn will affect the groundwater table.

(v) Due to climatic changes, pathogenic diseases will increase, and there will be rise in the insects and pests.

➢ Due to the emission of nitrogen oxides and CFCs, the ozone layer has been depleting.

➢ The chemical reaction that is caused by the contact of oxides of nitrogen with ozone destroys the ozone layer.

➢ The **ozone layer** is also called ozonosphere (90% occurs in stratosphere, extending from 10 to 18 km) containing high concentrations of ozone molecules (O_3).

➢ The ozone layer absorbs ultraviolet and infrared rays coming from the sun and protects the life on the Earth from its harmful effects.

➢ Without the ozone layer, the direct exposure to ultraviolet rays from the sun that reaching the Earth's surface can cause diseases like skin cancer and cataracts.

➢ Ultraviolet rays cause genetic disorders that affect heredity.

➢ These rays also disturb the ecological balance in marine ecosystems, affecting organisms like algae and fish.

➢ Ultraviolet rays also damage the physical and chemical properties of complex chemical substances.

➢ **Acid Rain:**

(i) Rain containing large amounts of harmful chemicals due to the burning of substances such as coal and oil.

(ii) It refers to the deposition of a mixture of wet (snow, fog, dew, sleet, etc.) and dry (acidifying particles and gases, etc.) acidic components.

(iii) It is caused by emissions of sulphur dioxide and nitrogen oxides, which react with the water molecules in the atmosphere to form acids.

(iv) **Effects of Acid Rain:**

Acid rain has had adverse effects on:

(a) Forests

(b) Freshwater bodies

(c) Soils – acid rain increases soil acidity, killing insects and affecting soil organisms.

(d) Buildings, monuments, and statutes.

(e) Human health – affects the nervous system.

(f) Aquatic species and plant growth.

(v) Acid rain and dry deposition of pollutants on land cause soil pollution.

➢ Polluted soils reduce mineralisation and decomposition processes.

➢ Toxic chemicals in the soil destroy earthworms, nematodes, and other soil organisms.

➢ To protect the biosphere and prevent its destruction, waste accumulation must be controlled.

➢ **Waste management** is essential as improperly stored waste can cause health, safety and economic problems. It involves the collection, transportation and disposal of garbage, sewage and other waste products.

Key Terms

➢ **Waste Management:** Waste management involves the activities and actions to handle waste from its inception to its disposal. This includes collection, transport, and disposal of garbage, sewage, and other waste products, along with the monitoring and regulation of waste management process.

➢ **Dermatophytosis:** A fungal infection of the skin, especially the feet.

- **Ozone layer:** A layer in the stratosphere that prevents harmful radiation from the sun from reaching the surface. Ozone gas absorbs ultraviolet and infrared rays.
- **Chlorofluorocarbons (CFCs):** Organic compound containing only carbon, chlorine and fluorine, produced as volatile derivatives of methane, ethane and propane.
- **Acid Rain:** Rain containing harmful chemicals resulting from the burning of substances such as coal and oil. When sulphur oxides and nitrogen dioxides react with water vapour present in the atmosphere, they form acids like sulphuric acid and nitric acid, which then mix with rain to produce acid rain.

TOPIC-3: SAFE DISPOSAL OF WASTE

Concepts Covered: Segregation of Waste, Dumping of Waste, Composting of Waste

Revision Notes

- Waste management is essential to handle waste from its inception to its disposal effectively.
- Waste disposal is the most crucial step in waste management. This step is essential to prevent harm to the environment, avoid injury or long term progressive health damage.
- Waste disposal refers to the method used to destroy or recycle unused, old or unwanted domestic, agricultural, medical, or industrial waste.
- There are various methods of safe disposal of waste. The methods include:

(i) **Segregation of Waste**
(ii) Dumping of Waste
(iii) Composting of Waste

Waste Management
Scan Me!

(i) Segregation of Waste: It is the first and most important step in safe disposal of waste for an effective waste management.

- Different coloured dustbins are used to segregate garbage like like glass, paper, cloth, metal, wet and dry food waste.
- Two types of colour-coded dustbins are used to collect the waste:

(i) Green-coloured dustbins for biodegradable wastes.
(ii) Blue-coloured dustbins for non-biodegradable waste.

- Biodegradable waste is converted into useful products like compost or gobar gas.
- Non-biodegradable waste can be recycled.

(ii) Dumping of Waste: After segregation, dumping is the next step in waste disposal.

- Waste materials are dumped in open low-lying areas far from the city. Though not environment friendly, it is the cheapest method.
- Dumping grounds are open pits that become breeding grounds for mosquitoes, flies and insects.
- Burning of these waste materials pollute the air and releases foul odour.
- **Sanitary landfill** is another method where waste is dumped daily.
- These are sites where waste is isolated from the environment until it is safe. It is considered safe when it has completely degraded biologically, chemically, and physically.
- The sanitary landfill system of disposing wastes is a biological method with minimal environmental damage.
- Five phases are followed for the disposal of waste:

(i) In the first phase, aerobic bacteria deplete the available oxygen resulting in an increase of temperature.
(ii) In the second phase, anaerobic conditions are established, and hydrogen and carbon dioxide are released.
(iii) In third phase, a large number of bacteria and methanogenic activity, (i.e., methane production) are established.
(iv) In the fourth phase, methanogenic activity stabilises.
(v) In the fifth phase, organic matter depletes, and the system returns to aerobic state.

- Sanitary landfills have several advantages over open dumping as listed below:

(i) Waste from landfills can be used directly as fuel for combustion.
(ii) The location of waste deposition in the landfills are monitored.
(iii) After the landfills are filled, they can be converted into parks or farmland.
(iv) Landfills help prevent pollution and open burning.

- Landfill sites should be covered with non-edible drought-resistant perennial plants.
- The selected plants should thrive on low nutrient soil.
- **Municipal Waste Management:** Municipal authorities take the following steps to manage wastes. They are:

(i) Collection of municipal solid waste.
(ii) Storage of municipal solid waste.
(iii) Transportation of municipal solid waste.
(iv) Segregation of municipal solid waste.

Proper Waste Disposal
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(iii) Composting of Waste: It is a method of **waste disposal** where organic matter decomposes naturally in the presence of oxygen. **[Board 2024]**

➢ A mixture of decomposed organic matter is used to enrich soil fertility.

➢ Compost is made from leaves, grass, vegetable peels, which decompose due to the action of aerobic bacteria, fungi, and other organisms.

➢ **Advantages of Composting:**

(i) Composting provides nutrient rich fertiliser, reducing the use of synthetic ones in farming.

(ii) Compost reduces soil erosion and improves soil structure.

(iii) It prevents plant diseases and and limits the spread of pathogen.

Composting
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(iv) Microorganisms like bacteria, fungi, aerate the soil, speeds up composting, and convert nitrogen into usable form.

(v) Compost increases water retention in sandy soil.

(vi) It prevents pollution by stopping pollutants in surface runoff from entering water bodies.

(vii) Compost revitalises farm soil by restoring trace minerals and organic matter.

(viii) Composting effectively reduces greenhouse gas emissions.

Key Terms

➢ **Waste Disposal:** Waste disposal is the management of waste to prevent environmental harm, injury or long-term health damage.

➢ **Segregation of Waste:** It refers to separating waste using different types of dustbins for glass, paper, cloth, metal, and wet and dry food waste.

➢ **Dumping of Waste Disposal:** Waste is dumped in open pits, which become the breeding ground for mosquitoes, flies, insects, etc.

➢ **Sanitary Landfill:** It is a waste disposal system that uses a biological method with minimal environmental damage.

➢ **Composting:** It is a waste disposal method where organic waste decomposes naturally under oxygen-rich conditions.